

Differential Splicing from Equivalence-Class Counts

Overview

Questions and unit of analysis

The analysis asks two distinct questions:

1. Does local splice-path usage differ by experimental condition within a cell type?
2. Does local splice-path usage differ across cell types within the same subjects?

The primary analysis aggregates cells into *subject-by-condition-by-cell-type pseudobulks* and estimates local splice-path compositions from their observed paired-primer equivalence-class (EC) counts. Biological replication is represented by subject; cells are never treated as independent condition replicates.

A splice block is a local region between constitutive exonic anchors. Transcripts that take the same path through that region are collapsed to one subisoform. This targets local splicing choices, which can be identifiable even when the corresponding full-length isoforms are not.

For condition effects, each hypothesis is indexed by a splice block and a cell type. It is an omnibus test of

$$H_0 : \text{mean path usage is equal across the included conditions.}$$

Only conditions represented by at least three subject pseudobulks are included. With two conditions and two paths, this is a subject-level differential path-log-ratio test. With more conditions or paths, all condition coefficients and path log-ratios are tested jointly.

For cell-type effects, the primary hypothesis is indexed by a nonredundant splice-path partition and jointly tests whether path usage differs among the represented cell types after blocking by subject. Pairwise subject-matched tests are secondary analyses used to localize omnibus signals and to match the hypotheses available from external comparators. A comparator pair must be represented in at least four subjects. Condition tests require at least 10 modeled gene UMIs per retained pseudobulk; the comparator benchmark uses 25 UMIs for alignment with the external-method screens. The cell-type and condition hypothesis families are reported separately.

Workflow

The primary analysis proceeds in five steps:

1. Construct local splice blocks from the isoform annotation and map every supported isoform to a local path.
2. Aggregate paired oligo(dT) and random-hexamer EC counts within each subject, condition, and cell type while retaining separate primer maps.

3. For each block, construct a label-independent pooled isoform baseline and estimate only its $d_b = S_b - 1$ local path-log-ratio perturbations while preserving within-path isoform mixtures and nuisance isoforms.
4. Stabilize local path estimates with two pseudo-observations per path, then fit a subject-blocked omnibus cell-type model or a condition model within one cell type.
5. Calibrate the resulting statistics using R_{null} design-matched Monte Carlo null replicates with a leave-one-event-out empirical null CDF, then apply BH separately to cell-type and condition hypotheses. The default is $R_{\text{null}} = 32$.

The same Tealeaf local-path estimator is used for the comparator benchmark and full-data analyses. It retains both primer-specific EC observations and estimates the identifiable local path coordinates directly. Supporting and deprecated analyses appear in the appendices.

Notation and Dimensions

An *equivalence class* (EC) is a set of isoforms compatible with one UMI. An EC count is therefore observed, whereas its allocation among compatible isoforms is latent. A *splice path* is an ordered local exon chain between two constitutive anchors. A *subisoform* is the set of full-length isoforms taking the same path through one block. PSI means percent spliced in; for two paths it is one component of the path-composition vector. ILR means isometric log-ratio, an orthonormal coordinate system for a composition.

Let $g = g(b)$ denote the gene containing block b , and let $\mathcal{I}_g$ be the set of isoforms retained by the paired-primer model and assigned to gene g by the isoform-to-gene table. Define $T_g = |\mathcal{I}_g| \in \mathbb{N}$. This is a gene-level set, not the set of isoforms whose exons overlap the block, and it is therefore shared by every block of the gene. Let $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$ be the retained, annotated spliced isoforms represented in the block’s GTF-derived path map. Isoforms in $\mathcal{I}_g \setminus \mathcal{I}_b^{\text{sp}}$, including modeled unspliced precursors, remain nuisance isoforms in the EC likelihood.

Let $I_b, S_b \in \mathbb{N}$ be the numbers of qualifying pseudobulks and represented spliced paths for block b , with $S_b \geq 2$. Then $d_b = S_b - 1 \in \mathbb{N}$ is its number of independent path contrasts. There are $P = 2$ primer protocols, and primer p has $K_{gp} \in \mathbb{N}$ ECs for gene g . The principal symbols in the primary model are listed below.

symbol	dimension or domain	meaning
b	$\{1, \dots, \mathcal{B}\}$	splice-block index
i	$\{1, \dots, I_b\}$	pseudobulk index within block b
u	$\{1, \dots, U_b\}$	subject index within block b
c	$\{1, \dots, C_b^{\text{cell}}\}$	cell-type index within block b
p	$\{1, \dots, P\}$	primer-protocol index
t	$\{1, \dots, T_g\}$	isoform index within gene $g(b)$
s	$\{1, \dots, S_b\}$	splice-path index
c_{bucp}	$\mathbb{N}_0^{K_{gp}}$	observed subject–cell-type EC counts
A_{gp}	$\mathbb{R}_+^{K_{gp} \times T_g}$	weighted EC-by-isoform map
$\bar{\theta}_g$	Δ^{T_g-1}	label-independent pooled isoform baseline
δ_{bi}	$\mathbb{R}^{d_b}$	local path-log-ratio perturbation
ψ_{bi}	Δ^{S_b-1}	represented local path composition
y_{bi}	$\mathbb{R}^{d_b}$	ILR coordinates of ψ_{bi}
Q_b	$\mathbb{R}^{S_b \times d_b}$	Helmert ILR basis

Here $\mathcal{B} \in \mathbb{N}$ is the number of blocks, $\mathbb{N}_0 = \{0, 1, \dots\}$, and $\Delta^{S_b-1} = \{\mathbf{v} \in \mathbb{R}_+^{S_b} : \mathbf{1}'\mathbf{v} = 1\}$. Bold lower-case symbols are vectors, upper-case symbols are matrices, $\mathbf{1}_k \in \mathbb{R}^k$ is a vector of ones, $\mathbf{0}_k \in \mathbb{R}^k$ is a vector of zeros, and $I_k \in \mathbb{R}^{k \times k}$ is the identity.

Helmert basis and ILR coordinates

The simplex has only $d_b = S_b - 1$ free directions because path proportions sum to one. The Helmert matrix Q_b is a fixed orthonormal basis for the sum-zero subspace:

$$Q_b' Q_b = I_{d_b}, \quad Q_b' \mathbf{1}_{S_b} = \mathbf{0}_{d_b}.$$

Using one-based row $r \in \{1, \dots, S_b\}$ and column $j \in \{1, \dots, d_b\}$, its entries are

$$(Q_b)_{rj} = \begin{cases} 1/\sqrt{j(j+1)}, & r \leq j, \\ -j/\sqrt{j(j+1)}, & r = j+1, \\ 0, & r > j+1. \end{cases}$$

For any strictly positive $\psi_{bi} \in \Delta^{S_b-1}$, the ILR coordinate is $\mathbf{y}_{bi} = Q_b' \log \psi_{bi} \in \mathbb{R}^{d_b}$, where the logarithm is elementwise. For $S_b = 2$, $Q_b = (1, -1)'/\sqrt{2}$ and $y_{bi} = \log(\psi_{bi1}/\psi_{bi2})/\sqrt{2}$. Thus a two-path ILR test is a scaled log-odds, or differential-PSI, test.

Methods

Tealeaf

Local-path estimation

Fix a gene g and one of its splice blocks b . For subject u , cell-type level c , and primer protocol $p \in \{1, 2\}$, let $\mathbf{c}_{bucp} \in \mathbb{N}_0^{K_{gp}}$ be the observed EC-count vector and let $A_{gp} \in \mathbb{R}_+^{K_{gp} \times T_g}$ be the fixed EC-by-isoform compatibility or conditional-weight matrix. Oligo(dT) and random-hexamer observations have separate count vectors and matrices, so primer-specific positional coverage enters the likelihood without changing the underlying isoform composition. Counts from the same subject and cell type are summed within primer. Conditioning on each observed primer total removes library size from the composition likelihood. An EC may be compatible with multiple isoforms; its count is not first allocated to a transcript or path. Unsupported isoforms and ECs not assigned to g are excluded, but every supported isoform of g , including an unspliced nuisance isoform, remains in every block likelihood for that gene.

After defining the observations eligible for a test, the baseline is estimated from their EC counts and compatibility matrices without using the labels within that set. Let $\mathcal{I}_b$ index the subject-cell-type observations included for block b , and define the primer-specific pooled count vector $\bar{\mathbf{c}}_{bp} = \sum_{(u,c) \in \mathcal{I}_b} \mathbf{c}_{bucp} \in \mathbb{N}_0^{K_{gp}}$. The label-independent pooled isoform composition $\bar{\boldsymbol{\theta}}_g \in \Delta^{T_g-1}$ is the joint conditional-multinomial maximum-likelihood estimate

$$\bar{\boldsymbol{\theta}}_g = \arg \max_{\boldsymbol{\theta} \in \Delta^{T_g-1}} \sum_{p=1}^2 \bar{\mathbf{c}}_{bp}^\top \log \left(\frac{A_{gp} \boldsymbol{\theta}}{\mathbf{1}^\top A_{gp} \boldsymbol{\theta}} \right).$$

Dependence on $\mathcal{I}_b$ is suppressed in the notation. The implementation uses reference-logit coordinates, starts from equal isoform weights, and optimizes this low-dimensional objective by L-BFGS. Primer counts are pooled separately, so their distinct compatibility and positional-coverage maps

remain in the objective, while conditioning on each primer total removes library-size differences. The estimate is computed once for each gene and included observation set and reused across blocks sharing both; because neither observed nor permuted labels enter its estimation, it is unchanged in the matched null analyses.

Let $Q_b \in \mathbb{R}^{S_b \times d_b}$ be the Helmert basis defined above. Let $r_b(t) \in \{1, \dots, S_b\} \cup \{\emptyset\}$ map isoform t to its represented path or mark it as a nuisance isoform, let $\mathcal{R}_b = \{t : r_b(t) \neq \emptyset\}$, and let $\bar{\rho}_b = \sum_{t \in \mathcal{R}_b} \bar{\theta}_{gt} \in (0, 1]$ be the pooled mass of represented isoforms. Thus $\bar{\theta}_g$ supplies the fixed within-path isoform ratios, nuisance-isoform abundances, and represented mass around which each subject–cell-type local path fit is perturbed.

For any $\delta \in \mathbb{R}^{d_b}$, define the perturbed isoform composition $\theta_g(\delta) \in \Delta^{T_g-1}$ componentwise by

$$\theta_{gt}(\delta) = \begin{cases} \frac{\bar{\theta}_{gt} \exp\{Q_{b,r_b(t)}^\top \delta\}}{\sum_{v \in \mathcal{R}_b} \bar{\theta}_{gv} \exp\{Q_{b,r_b(v)}^\top \delta\}}, & t \in \mathcal{R}_b, \\ \bar{\theta}_{gt}, & t \notin \mathcal{R}_b. \end{cases}$$

Define its represented path composition $\psi_b(\delta) \in \Delta^{S_b-1}$ by

$$\psi_{bs}(\delta) = \frac{1}{\bar{\rho}_b} \sum_{t \in \mathcal{R}_b: r_b(t)=s} \theta_{gt}(\delta), \quad s \in \{1, \dots, S_b\}.$$

Thus δ changes only local path usage, while within-path isoform ratios, nuisance isoforms, and total represented mass remain fixed.

For subject–cell-type combination (u, c) , Tealeaf maximizes the two-primer EC log likelihood with a uniform fixed-total path concentration $\zeta \geq 0$,

$$\hat{\delta}_{buc}(\zeta) = \arg \max_{\delta \in \mathbb{R}^{d_b}} \left\{ \sum_{p=1}^2 c_{bucp}^\top \log \left(\frac{A_{gp} \theta_g(\delta)}{1^\top A_{gp} \theta_g(\delta)} \right) + \frac{\zeta}{S_b} \sum_{s=1}^{S_b} \log \psi_{bs}(\delta) \right\}.$$

Each of the S_b paths therefore receives ζ/S_b pseudo-observations, so the total concentration does not depend on the number of represented paths. The penalty stabilizes boundary estimates without shrinking subject–cell-type compositions toward a shared biological baseline. The global value of ζ is selected by the testing-level empirical-Bayes procedure below. Junction, exonic, and ambiguous reads contribute through their compatibility patterns without first assigning reads to global isoforms.

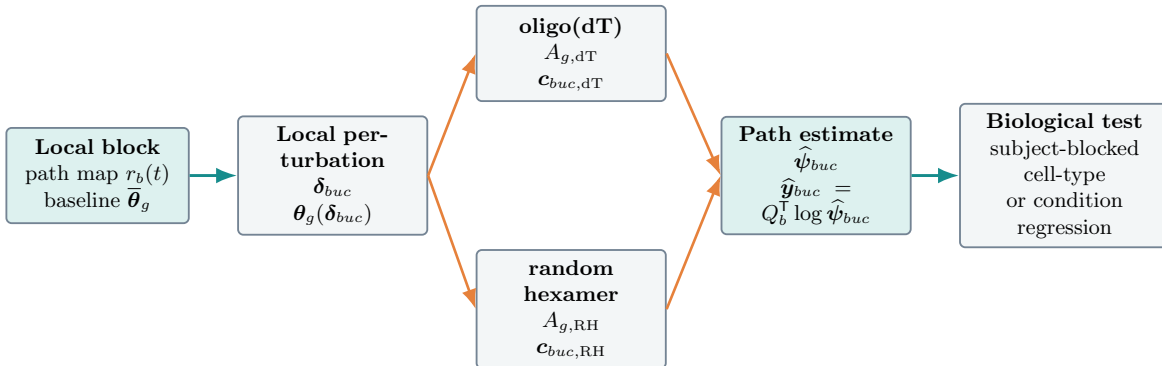


Figure 1: Tealeaf local-path workflow for block b . The pooled baseline and block path map define a low-dimensional perturbation shared by the two primer likelihoods. Their EC counts jointly estimate one path-ILR vector for each subject–cell-type combination, and biological replication enters in the second-stage multivariate regression.

Statistical testing

Let $\hat{\mathbf{y}}_{buc} = Q_b^\top \log \hat{\boldsymbol{\psi}}_{buc} \in \mathbb{R}^{d_b}$ be the estimated splice path usage ratios (make non-redundant via Q_b). The primary cell-type analysis uses every fitted subject-cell-type vector for a block in one multivariate regression. Let $N_b^{\text{obs}} \in \mathbb{N}$, $U_b \in \mathbb{N}$, and $C_b^{\text{cell}} \in \mathbb{N}$ be its numbers of subject-cell-type observations, subjects, and cell types, and stack the responses as $Y_b \in \mathbb{R}^{N_b^{\text{obs}} \times d_b}$. The design $X_b \in \mathbb{R}^{N_b^{\text{obs}} \times (U_b + C_b^{\text{cell}} - 1)}$ contains an intercept, $U_b - 1$ subject indicators, and $C_b^{\text{cell}} - 1$ cell-type indicators. The equal-weight model is

$$Y_b = X_b \Gamma_b + E_b, \quad E_{bn} \stackrel{\text{iid}}{\sim} N(\mathbf{0}_{d_b}, \tau_b^2 I_{d_b}),$$

where $\Gamma_b \in \mathbb{R}^{(U_b + C_b^{\text{cell}} - 1) \times d_b}$ and $\tau_b^2 \geq 0$. The joint Wald test sets the $(C_b^{\text{cell}} - 1) \times d_b$ cell-type coefficient block to zero and therefore has at most $(C_b^{\text{cell}} - 1)d_b$ degrees of freedom. Subject indicators preserve the repeated-measures design while producing one cell-type hypothesis per block. With zero conditional measurement covariance, the REML scalar-variance generalized least-squares model reduces exactly to multivariate ordinary least squares, and the implementation uses this closed form.

For a condition test within one cell type, the subject-level vectors enter an equal-weight multivariate regression with an intercept and $C_b^{\text{cond}} - 1$ condition columns, where $C_b^{\text{cond}} \in \mathbb{N}$ is the number of represented conditions. An isotropic between-subject variance is fitted by REML under the null, and all condition-by-path coefficients are tested jointly.

The secondary pairwise cell-type analysis localizes differences and supplies Tealeaf hypotheses matched to junction-based comparators. For cell types 1 and 2, define the paired subject contrast $\hat{\mathbf{d}}_{bu} = \hat{\mathbf{y}}_{bu2} - \hat{\mathbf{y}}_{bu1} \in \mathbb{R}^{d_b}$. A paired t test is used when $d_b = 1$, and Hotelling's test is used when $d_b > 1$. Scalar sampling variances are moderated across blocks with a robust scaled-inverse-chi-squared prior, refitted inside every null family.

Let $R_{\text{null}} \in \mathbb{N}$ be the number of Monte Carlo null replicates, with default $R_{\text{null}} = 32$. For every block and replicate, the null transformation is a within-subject cell-type-label permutation for the primary omnibus regression, a condition-label permutation for a condition contrast, or a paired subject sign flip for a pairwise cell-type contrast. Collecting replicate $r \in \{1, \dots, R_{\text{null}}\}$ across all blocks gives one null p-value family on which the complete BH procedure can be audited. Calibration pools the null reference p-values across blocks only within matched tested-dimension and subject-count strata. If $\tilde{p}_b \in [0, 1]$ is block b 's uncalibrated reference p-value, $\tilde{p}_{jr}^{(0)} \in [0, 1]$ is block j 's null reference p-value in replicate r , and $\mathcal{S}_b$ is the set of blocks in its calibration stratum, the leave-one-block-out calibrated value is

$$p_b = \frac{1 + \sum_{j \in \mathcal{S}_b \setminus \{b\}} \sum_{r=1}^{R_{\text{null}}} \mathbf{1}\{\tilde{p}_{jr}^{(0)} \leq \tilde{p}_b\}}{1 + R_{\text{null}}(|\mathcal{S}_b| - 1)}.$$

Thus the calibration CDF contains approximately $R_{\text{null}}|\mathcal{S}_b|$ values rather than only R_{null} , while the whole-family BH audit still contains only R_{null} realizations. The default 32 is a computational choice rather than a theoretically distinguished value; sensitivity to 64 and 128 replicates is reported below. BH is applied separately to the cell-type and condition families. Supporting analyses of path-estimate uncertainty, alternative smoothing, and the deprecated global EC GLMM are reported in the appendix.

Tealeaf selects one global fixed-total concentration from $\mathcal{Z} = \{2, 4, 8, 16, 32, 64, 128, 256\}$ by empirical Bayes at the level of the calibrated tests. Let $p_j(\zeta) \in [0, 1]$ be test j 's calibrated p-value after fitting with candidate $\zeta \in \mathcal{Z}$. For each candidate, the two-groups marginal density is $f_\zeta(p) = \pi_{0,\zeta} + (1 - \pi_{0,\zeta})a_\zeta p^{a_\zeta - 1}$, where $\pi_{0,\zeta} \in (0, 1)$ is the null weight and $a_\zeta \in (0, 1)$ is the shape

of a decreasing beta alternative. The parameters maximize log evidence, averaged first over the $n_g \in \mathbb{N}$ tests from gene g and then over the $G \in \mathbb{N}$ genes, and the selected concentration is

$$\hat{\zeta} = \arg \max_{\zeta \in \mathcal{Z}} \frac{1}{G} \sum_{g=1}^G \frac{1}{n_g} \sum_{j:g(j)=g} \log f_{\zeta}\{p_j(\zeta)\},$$

where $g(j)$ maps test j to its gene. Gene-level averaging prevents genes with more blocks or cell-type pairs from dominating selection. This target chooses smoothing for distinguishing the calibrated null/non-null mixture, uses the same uniform-centered estimator in selection and testing, and requires no fitting-data split. The whole selector, including calibration and estimation of $(\pi_{0,\zeta}, a_{\zeta})$, is rerun within each matched null family so the FDR audit includes hyperparameter selection. One concentration is selected per analysis family rather than imposed across distinct hypothesis families. The subject-split comparator benchmark selects $\hat{\zeta} = 32$, while the full-data omnibus cell-type, pairwise cell-type, and condition families each independently select $\hat{\zeta} = 64$. Supporting analyses of alternative smoothing targets, path-estimate uncertainty, and the deprecated global EC GLMM are reported in the appendix.

The selected $\hat{\zeta}$ is used for test statistics, null calibration, p-values, and discovery selection. Descriptive path proportions and effect sizes are refitted with a separate fixed reporting concentration $\zeta_{\text{report}} = 1$ after the discovery set is fixed. This weak barrier stabilizes otherwise nonconvergent boundary fits while avoiding the compression induced by a concentration optimized for testing-level mixture evidence. The reporting refit cannot alter a test statistic, p-value, or discovery. Both concentrations are stated whenever fitted path usage is shown.

Independent long-read validation

We evaluated directional replication in the P56 brain ScISOr-Seq2 data from Joglekar et al., generated independently by the Tilgner laboratory. We used the authors' UMI-filtered annotated-transcript count matrix rather than treating individual reads as biological replicates. The source matrix contains two biological replicates within each of five brain regions. Eight Tealeaf cell types have a prespecified correspondence to its annotations: astrocytes, mature oligodendrocytes, cerebellar granule neurons, cortical excitatory neurons, dentate-gyrus excitatory neurons, CA excitatory neurons, thalamic excitatory neurons, and medium spiny neurons. Source transcript names are mapped to stable GENCODE transcript identifiers and then collapsed with the same block-to-path map used by Tealeaf. For astrocytes and oligodendrocytes, region-specific samples are pooled within replicate index; a sensitivity excluding both cross-region glial mappings is reported below.

Directional validation uses the secondary pairwise BH discoveries because an omnibus hypothesis has no single direction. For selected test j with $S_j \in \mathbb{N}$ paths, let $\Delta_j^T \in \mathbb{R}^{S_j}$ be the difference between the two Tealeaf mean path-composition vectors from the $\zeta_{\text{report}} = 1$ refit, and let $\Delta_{jr}^L \in \mathbb{R}^{S_j}$ be the corresponding difference in long-read transcript-UMI proportions for source replicate $r \in \{1, 2\}$. The pooled long-read effect is Δ_j^L , computed after summing the two replicate count vectors. A test replicates in pooled direction when $(\Delta_j^T)^T \Delta_j^L > 0$, and it replicates in both biological replicates when $(\Delta_j^T)^T \Delta_{jr}^L > 0$ for both r . The pooled sign-flip null rate is $1/2$. For the stricter endpoint, the conditional null is half the observed fraction for which the two source-replicate dot products have the same sign; this preserves their between-replicate concordance and randomizes only the orientation of the Tealeaf contrast. The primary coverage rule requires at least 20 mapped long-read UMIs in each pooled cell type; the stricter endpoint additionally requires at least 10 in each cell-type-replicate combination.

We apply the same directional validation to full-data BH discoveries from LeafCutter, MAJIQ Heterogen, and scQuint, and to a paired junction-CLR ablation refitted over every candidate intron group with BH applied within each cell-type contrast. For junction group j containing $K_j \in \mathbb{N}$ junctions, its short-read effect $\Delta_j^J \in \mathbb{R}^{K_j}$ is the difference between equal-pseudobulk mean junction-composition vectors. This common post-selection reporting estimator avoids comparing method-specific direction estimates. A source junction count is the sum of annotated-transcript UMIs over source transcripts containing that junction, and the long-read junction effect $\Delta_j^{L,J} \in \mathbb{R}^{K_j}$ is formed from the corresponding proportions. The pooled criterion is $(\Delta_j^J)^\top \Delta_j^{L,J} > 0$, with the analogous replicate-resolved criterion. A junction group is eligible only if every tested junction is represented by at least one source annotated transcript and it passes the same depth rules used for Tealeaf. The MAJIQ build requires support in at least three pseudobulks and its PSI-coverage filter requires at least three reads in at least three bins; Heterogen then requires coverage in at least half the pseudobulks within each compared group. MAJIQ edges are restored to their local splicing variations before composition estimation.

Comparators

In addition to the pooled SUPPA2 comparator, we run a primer-aware SUPPA2 comparator using Tealeaf’s paired-primer EC preparation and compatibility blocks. For event e , sample i , and primer p , expected included and excluded counts are obtained by applying the production EC allocations to the standard SUPPA2 transcript sets. A shared event logit is estimated with a primer offset and event-specific beta-binomial overdispersion, with both refit separately for full data and each subject fold. The resulting event PSI values use the same paired Wilcoxon approximation and within-contrast BH correction as pooled SUPPA2. For the matched comparison, event p-values are combined by Simes within each gene–cell-type pair and intersected with Tealeaf’s eligible gene–pair keys in both folds, so the table reports Tealeaf and primer-aware SUPPA2 on the same hypothesis universe. The unrestricted primer-aware audit remains a separate sensitivity analysis.

The same biological subjects are divided into two condition-balanced halves for Tealeaf and every comparator. The benchmark retains the 25-modeled-gene-UMI candidate screen and requires at least four subjects for each cell-type pair; Tealeaf tests 7,671 and 5,140 candidate block-by-cell-type-pair hypotheses in the two halves. Spliced alignments supplied to the external methods are restricted to unique genomic mappings and deduplicated by cell barcode, UMI, genomic start, and CIGAR. LeafCutter tests intron clusters, MAJIQ tests edges within local splicing variations, and scQuint tests intron groups. We additionally run rMATS 4.1.2 on JCEC event counts and SUPPA2 v2.4. SUPPA2 generates the standard SE, A3SS, A5SS, MXE, and RI event definitions and transcript-TPM PSI values; its classical paired Wilcoxon statistic is evaluated with a vectorized normal approximation and corrected within each contrast. rMATS is rerun in paired mode with the PAIRADISE model on JCEC counts. The pseudobulk BAMs are single-end-like, so the count pass uses single-end read handling, while the statistical model pairs matched biological subjects; its IncLevelDifference is converted to the common level-b-minus-level-a orientation before direction checks. For each Tealeaf–comparator pair, scoring retains only genes and cell-type pairs eligible for both methods in both halves. Feature p-values are combined by Simes within each gene–cell-type pair and then across the shared pairs within a gene. For method m , gene g , and half $h \in \{1, 2\}$, let $p_{mgh} \in [0, 1]$ be the resulting gene p-value. The conjunction p-value $p_{mg}^{\text{rep}} = \max(p_{mg1}, p_{mg2})$ tests whether the gene has signal in both halves, and BH at 0.05 is applied on the pairwise shared gene universe. Secondary endpoints are held-half nominal replication and Spearman correlation of gene–pair $-\log_{10} p$ across halves.

Results

Tealeaf reproducibility against comparator methods

Table 1 gives the calibrated direct local-path comparison using the testing-level EB choice $\hat{\zeta} = 32$. Tealeaf has more independently replicated genes than every comparator on the matched shared universes. The matched-split SUPPA2 analysis uses the same subject-fold manifests as Tealeaf, with SUPPA2 event tests aggregated by Simes within each gene–cell-type pair. The advantage is consistent with the intended information gain from combining junction, exonic, and ambiguous EC reads while testing the $S_b - 1$ local path coordinates rather than a larger set of junction coefficients. Because the global concentration was estimated on this benchmark dataset, the numerical advantage is internally assessed and requires confirmation in an independent dataset.

The rows in Table 1 separate two related comparisons. LeafCutter, MAJIQ Heterogen, scQuint, and matched-split SUPPA2 use the matched shared gene–cell-type-pair split benchmark and are the direct reproducibility comparisons to Tealeaf. For SUPPA2, the shared unit is a gene–cell-type pair after Simes aggregation of its event p-values, rather than a Tealeaf block. The rMATS row is explicitly the historical unpaired split audit. The corrected paired-JCEC rMATS rerun is integrated below as a full-data event comparator, rather than being presented as a paired split result that was not rerun. Thus the main split conclusion is based on directly matched gene–pair hypotheses, while SUPPA2 and paired rMATS provide complementary event-level sensitivity and independent long-read validation.

The fold-specific Tealeaf analyses make 1,471 and 1,047 BH calls. The pooled EB evidence rises from 0.190 at $\zeta = 2$ to 0.333 at $\zeta = 32$, then falls to 0.309 at $\zeta = 256$; fitting the halves separately selects 32 and 64. The complete selector was rerun in each of the 32 matched sign-flip null families. Every null family selected $\zeta = 2$, mean fold rejection was 0.04995 at nominal 0.05, and no fold-level null family made a BH call. Propagating each selected null family through the same within-pair Simes aggregation, across-pair gene aggregation, two-fold conjunction, and BH procedure produced zero replicated genes on every comparator-matched universe.

The 192 one-core fitting shards use 2.516 aggregate worker hours across both folds, with maximum shard elapsed time 78 seconds and peak worker resident memory 0.288 GiB. Merge-time moderation and empirical calibration add less than two minutes of serial work. This is substantially cheaper than repeatedly fitting the nested latent EC GLMM and shows that preserving primer-specific EC compatibility does not itself require the larger computational cost of the global random-effects model.

Table 1: Cell-type differential-splicing reproducibility across independent subject halves using the calibrated Tealeaf direct local-path test. “Tests” counts shared gene-by-cell-type-pair hypotheses, “Rep.” is the number of genes passing BH on p^{rep} , “held” is held-half nominal replication, and ρ is Spearman correlation of gene-pair $-\log_{10} p$. Every row uses the hypothesis universe shared by Tealeaf and that comparator in both halves. For SUPPA2, event p-values are combined by Simes within each gene-cell-type pair before applying the common two-fold procedure; the primer-aware row uses its event-eligible matched subset.

Comparator	Shared		Tealeaf			Comparator		
	tests	genes	Rep.	held	ρ	Rep.	held	ρ
LeafCutter	747	139	73	0.878	0.790	26	0.750	0.542
MAJIQ relaxed UMI	2,500	480	156	0.888	0.791	38	0.698	0.437
scQuint	2,600	507	160	0.875	0.813	56	0.775	0.458
SUPPA2 (matched split)	2,342	457	131	0.884	0.812	1	0.900	0.352
rMATS (native, historical)	2,564	496	133	0.907	0.814	0	0.000	0.530
SUPPA2 (primer aware; matched)	2,191	431	97	0.863	0.738	0	0.875	0.231

The matched-split SUPPA2 analysis uses the standard event catalogue on the production subject-fold manifests. Its event-level tests were reduced to gene-cell-type-pair hypotheses by Simes aggregation, and the resulting 2,342 hypotheses are shared with the Tealeaf direct-path results shown in the table. The earlier full-data and original event-universe split outputs remain in the comparator archive as separate audits.

The primer-aware matched row uses 2,191 gene-cell-type-pair hypotheses shared with Tealeaf in both folds and 431 shared genes. Tealeaf has 153 and 146 fold-level BH genes, 97 replicated genes, held-half nominal replication 0.863, and $\rho = 0.738$; primer-aware SUPPA2 has 4 and 5 fold-level BH genes, no replicated genes, held-half nominal replication 0.875, and $\rho = 0.231$. The same preparation also produces 645,555 full-data cell-type event-contrast tests over 8,997 events after production EC filtering, with 672 event-level BH calls. Its unrestricted fold audit contains 140,797 gene-contrast hypotheses over 3,716 genes and no fold-level BH calls after correction over that larger universe; its fold log-p correlation is 0.056. The unrestricted audit is retained as a sensitivity diagnostic, while the matched row is the comparator result. The complete tests, event offsets, and split audits are in `analyses/comparator_suppa_rnats/suppa2_primer_aware`.

Table 2 directly compares the fixed-total production scaling with the former path-dependent scaling. Set $\alpha_{\text{path}} = \zeta/2$ under the former rule, so the two penalties are identical for two-path blocks. Two-path blocks comprise 5,295 of 7,671 fold-0 candidates, while a block with $S_b > 2$ receives $S_b\zeta/2 > \zeta$ total concentration under the former rule. The curves are nearly indistinguishable around their optimum. Fixed-total scaling reaches peak fold discovery rates of 19.56% and 21.18%, compared with 19.55% and 20.95% under the path-dependent scaling. Both attain maximum replicated-gene counts of 78, 161, and 161. At $\zeta = 64$, fixed total gives 76, 161, and 161 replicated genes, versus 78, 161, and 161 at the matched $\alpha_{\text{path}} = 32$; their held-half replication rates are respectively 0.897/0.907/0.903 and 0.898/0.925/0.913. At $\zeta = 128$, the rates are 0.909/0.919/0.906 under fixed total and 0.908/0.919/0.907 under matched path-dependent scaling. Pooled null rejection remains near 0.05 and no fold-level null family makes a BH call at any fixed-total setting. Fixed total concentration therefore removes an arbitrary dependence on path count without a material discovery or replication penalty, and testing-level EB selects $\zeta = 32$ rather than choosing the maximum replicated-discovery count.

Table 2: Fixed-total versus current per-path concentration at matched two-path strength. Entries are fixed total / current scaling. ζ is the fixed total concentration and $\alpha_{\text{path}} = \zeta/2$. “BH rate” is the percentage of eligible block-by-cell-type-pair hypotheses passing fold-specific BH. Replicated genes use the complete Simes, two-fold conjunction, and BH procedure on the LeafCutter, MAJIQ Heterogen, and scQuint matched universes.

ζ	α_{path}	BH rate (%)		Replicated genes		
		fold 0	fold 1	LeafCutter	MAJIQ	scQuint
2	1	10.28 / 10.67	9.71 / 10.09	31 / 31	51 / 53	53 / 56
4	2	14.90 / 15.46	13.92 / 15.23	48 / 50	98 / 103	94 / 101
8	4	17.98 / 18.10	17.54 / 18.43	56 / 56	124 / 132	131 / 136
16	8	19.17 / 19.35	20.11 / 20.21	68 / 70	139 / 141	145 / 147
32	16	19.56 / 19.55	20.71 / 20.95	73 / 75	156 / 157	160 / 159
64	32	18.93 / 18.67	21.18 / 20.86	76 / 78	161 / 161	161 / 161
128	64	17.44 / 17.05	20.33 / 20.34	78 / 78	156 / 150	157 / 154
256	128	16.69 / 16.63	20.04 / 19.89	78 / 77	148 / 147	150 / 148

The smoothing sensitivity in Table 3 holds variance moderation, empirical calibration, gene aggregation, and the two-fold replication endpoint fixed while varying the former uniform per-path pseudocount. Without smoothing, local estimates can approach the simplex boundary and some hypotheses are lost. Replicated discovery increases through a maximum at $\alpha_{\text{path}} = 32$, then decreases at 64 and again at 128 for the MAJIQ and scQuint matched universes; the LeafCutter result plateaus at 78 replicated genes at 32–64 and decreases to 77 at 128. Fold-specific block calls peak earlier, at $\alpha_{\text{path}} = 16$ in half 0 and 16–32 in half 1, and also decline at higher precision. The turnover is visible in both intermediate block calls and the replication endpoint rather than being imposed by failed null calibration: every setting retains pooled sign-flip rejection near 0.05 and produces no fold-level BH calls in any of 64 null families. This grid motivates smoothing and verifies that excessive concentration eventually loses power, but it is not the selection rule; production uses fixed-total scaling and chooses ζ from the testing-level EB objective.

Table 3: Sensitivity of the pairwise subject-split benchmark to the uniform per-path pseudocount α_{path} . “Fold BH” gives block-by-cell-type-pair calls in halves 0 and 1. Replicated genes are reported on the respective LeafCutter, relaxed-coverage MAJIQ Heterogen, and scQuint matched universes after the complete Simes, conjunction, and BH procedure. The null column is the pooled rejection rate at nominal 0.05 in halves 0 and 1; none of the 64 fold-level null families makes a BH call at any setting. The unsmoothed fit has 7,295 and 4,911 eligible tests because of boundary instability; smoothed fits retain 7,517–7,523 and 5,048–5,058.

α_{path}	Fold BH 0 / 1	Replicated genes			Null rejection	
		LeafCutter	MAJIQ	scQuint	fold 0	fold 1
0	22 / 24	1	0	0	0.05014	0.05009
0.5	465 / 258	11	14	17	0.05005	0.05005
1	803 / 510	31	53	56	0.05004	0.05004
2	1,163 / 770	50	103	101	0.05004	0.05001
4	1,362 / 932	56	132	136	0.05004	0.05001
8	1,456 / 1,022	70	141	147	0.04998	0.05000
16	1,471 / 1,059	75	157	159	0.04999	0.04999
32	1,404 / 1,054	78	161	161	0.05003	0.05006
64	1,282 / 1,027	78	150	154	0.05004	0.05005
128	1,250 / 1,004	77	147	148	0.05002	0.04996

Primary omnibus and pairwise localization

The subject-blocked omnibus regression is the primary full-data cell-type test because it asks whether any represented cell type differs while producing one hypothesis per block. Pairwise tests are secondary, they localize which cell-type contrasts contribute to an omnibus event and provide the pair-specific hypotheses needed for comparator matching. Each family independently applies the same testing-level EB selection rather than sharing a concentration selected by the other family.

Full-data Tealeaf results

The full-data omnibus cell-type, secondary pairwise cell-type, and condition-within-cell-type families each apply the testing-level EB procedure independently. All three evidence curves select fixed-total concentration $\hat{\zeta} = 64$, rather than inheriting $\hat{\zeta} = 32$ from the subject-split comparator benchmark. The omnibus analysis tests 1,785 blocks and the condition analysis tests 14,833 block-by-cell-type hypotheses. Table 4 reports their observed and design-matched null results. Of 1,781 fitted cell-type blocks, the omnibus test calls 791 blocks from 572 genes. The condition family has nominal enrichment but no BH event.

Table 4: Full-data Tealeaf results after each analysis family independently applies testing-level EB; both primary families select fixed-total $\hat{\zeta} = 64$. Genes are counted among BH events and are descriptive rather than a separate gene-level FDR family.

Contrast	Tests	Eligible	Nominal	BH	BH genes	Minimum p	Null $< .05$	Null BH max
Cell type	1,785	1,781	894	791	572	9.53×10^{-5}	0.05015	0
Condition	14,833	14,833	1,243	0	0	4.71×10^{-5}	0.04994	0

The full cell-type result is consistent with the subject-split comparator benchmark rather than resting only on a large full-data discovery count. The two analyses estimate their concentrations independently, selecting 64 in full data and 32 in the split benchmark, and Tealeaf exceeds all three comparators in independently replicated genes on their matched universes. The complete null conjunction pipelines produce no replicated gene. The condition result shows that empirical calibration is not mechanically producing discoveries, its 1,243 nominal events do not reach the calibrated BH tail.

From prepared inputs, the selected omnibus, secondary pairwise, and condition fits use 0.488, 1.240, and 1.670 aggregate one-core worker hours, with peak worker memory of 0.500, 0.396, and 0.399 GiB. Fitting and calibrating all eight candidates costs 5.020, 11.211, and 12.801 worker hours. Selection therefore multiplies the one-concentration fitting cost, although the grid is embarrassingly parallel and is estimated once for the complete analysis family. The condition implementation reuses the REML biological variance fitted under its intercept-only null across label permutations, because that null is unchanged by the permutation.

The full selector is rerun independently inside each of the 32 matched null families. Selector-aware mean rejection at nominal 0.05 is 0.05043 for omnibus, 0.05000 for pairwise, and 0.04995 for condition, and no selected null family makes a BH call. All 32 pairwise null families select $\zeta = 2$; the omnibus and condition null selectors choose a range of values, so calibration audits the selection rule rather than only the observed $\zeta = 64$ fit. The older 64- and 128-replicate null-budget sensitivity used fixed add-two smoothing and is retained in the appendix as a calibration-budget check, not as a rerun of the EB selector.

Independent long-read replication

The independent long-read data can evaluate 1,401 of the 2,950 secondary pairwise Tealeaf discoveries from matched cell types. All tested paths map to source annotated transcripts for 1,338 tests from 369 genes, and 704 tests pass the primary pooled-depth rule. Their pooled long-read effects agree with Tealeaf for 405 of 704 tests, a 57.5% directional replication rate with 95% Wilson interval 53.8–61.1%, compared with the 50% sign-flip null ($p = 3.68 \times 10^{-5}$). Requiring a Tealeaf path-effect norm of at least 0.05 retains 403 tests and raises pooled replication to 59.8%, interval 54.9–64.5% ($p = 4.87 \times 10^{-5}$).

The stricter replicate-resolved rule retains 459 tests, of which 175 agree with Tealeaf in both long-read replicates, 38.1% with interval 33.8–42.7%. The conditional sign-flip null is 33.1%, and the orientation test gives $p = 0.00487$. At minimum Tealeaf effect norm 0.05, 119 of 276 tests agree in both replicates, 43.1% with interval 37.4–49.0%, compared with a 35.1% conditional null ($p = 0.000972$). Selecting only the largest eligible effect per gene gives similar rates, 57.1% pooled and 38.8% strict without an effect threshold, so repeated contrasts within genes do not explain the aggregate result. Excluding cross-region astrocyte and oligodendrocyte mappings gives 54.6% pooled replication among 480 tests and 34.3% strict replication among 309 tests, still above their conditional nulls. Expanding oligodendrocytes from mature MFOL/MOL annotations to every annotated stage changes either primary rate by less than 0.6 percentage points.

A pooled long-read UMI G-test passes BH and has the Tealeaf direction for 228 of 704 tests, 32.4%. This is a secondary read-level endpoint: the two source animals per region do not support a well-powered biological-replicate significance test for each event. The directional endpoints therefore measure whether independently generated long reads support the sign of selected Tealeaf effects, not a second genome-wide discovery analysis. Figure 2 shows the coverage and effect-size sensitivities.

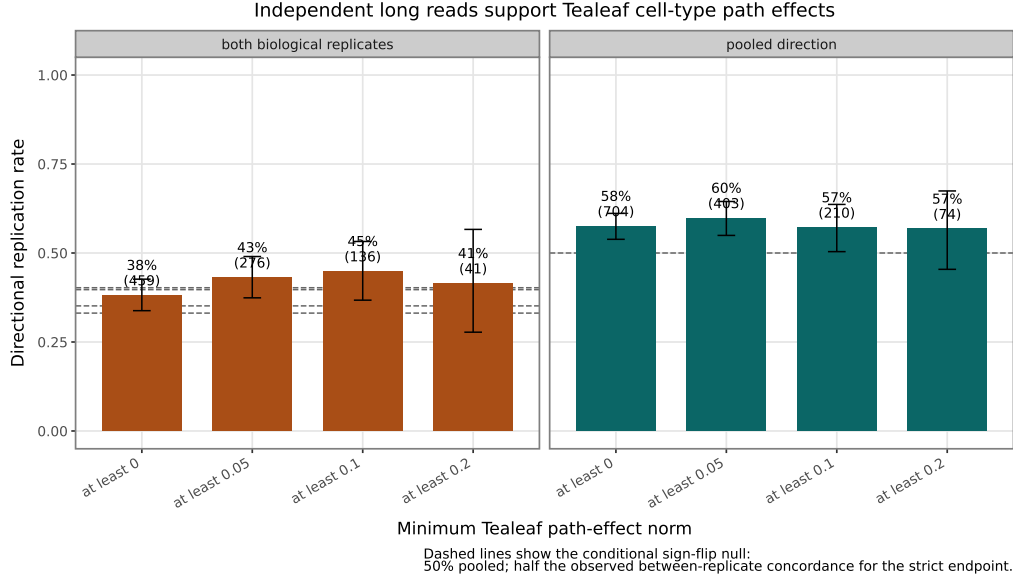


Figure 2: Directional replication of BH-significant secondary Tealeaf cell-type contrasts in independent P56 mouse-brain long reads from Joglekar et al. Bars use tests for which every path maps to a source annotated transcript and each cell type has at least 20 pooled transcript UMIs. The stricter endpoint also requires at least 10 UMIs per cell type in each source replicate. Parentheses give eligible tests, error bars are Wilson 95% intervals, and dashed lines are conditional sign-flip null rates. Tealeaf significance uses $\zeta_{\text{test}} = 64$, while displayed effect-size thresholds use the $\zeta_{\text{report}} = 1$ refit.

Junction methods differ sharply in conditional external replication, Table 5. LeafCutter and scQuint have higher pooled rates than Tealeaf, 87.5% and 83.7% versus 57.5%, and the advantage persists under the two-replicate endpoint. Relaxed-coverage MAJIQ is directionally consistent for 86 of 139 eligible LSVs, 61.9%, and 54 of 113 LSVs in both source replicates, 47.8%. The paired junction-CLR ablation is close to Tealeaf, 59.5% pooled and 39.8% in both replicates versus 57.5% and 38.1%. The event baselines are less concordant: full-data SUPPA2 agrees for 1,872 of 3,871 pooled calls, 48.4%, and for 802 of 2,590 calls in both source replicates, 31.0%; primer-aware SUPPA2 agrees for 1,738 of 3,822 pooled calls, 45.5%, and for 784 of 2,587 calls in both source replicates, 30.3%; paired-JCEC rMATS agrees for 1,778 of 3,859 pooled calls, 46.1%, and for 660 of 2,549 strict calls, 25.9%. The event rows use the top 500 ranked events per contrast rather than the BH-selected block or LSV universes and are therefore an event-level audit, not a matched power comparison. Retaining only the largest effect per gene–cell-type contrast changes the pooled rates to 87.3%, 62.4%, 59.2%, 56.2%, and 83.3% for LeafCutter, MAJIQ, paired CLR, Tealeaf, and scQuint, respectively, so repeated calls do not explain the junction-method ordering.

These are conditional validation rates rather than a matched power comparison. Complete source representation and the depth rule retain 704 Tealeaf calls, 139 MAJIQ LSVs, 56 LeafCutter clusters, and 43 scQuint groups. The event audit retains 3,871 full-data SUPPA2 events, 3,822 primer-aware SUPPA2 events, and 3,859 paired-JCEC rMATS events after the top-500-per-contrast and depth filters. The relaxed MAJIQ build tests 10,297 edges in the 28 mapped cell-type contrasts and calls 2,657 edges from 466 genes; restoring significant edges to multijunction LSVs, mapping every junction to the common short-read bundle, requiring complete source-transcript representation, and applying the long-read depth rule produces the 139 evaluable LSVs. The eligible LeafCutter

and scQuint sets are enriched for large, clean junction-composition shifts, with median short-read effect norms 0.190 and 0.619 compared with median path-composition norm 0.062 for Tealeaf; MAJIQ’s median common-estimator junction norm is 0.017, although the coordinates differ and these magnitudes are not directly interchangeable. Thus the external data do not support claiming that each selected Tealeaf call replicates better than junction-method calls. They instead show broader evaluable Tealeaf coverage, similar conditional replication to the paired CLR ablation, higher conditional replication for each junction method, and somewhat lower directional agreement for the event baselines. The subject-split benchmark remains the matched comparison of discovery power because it uses a shared gene–contrast universe in both halves.

Table 5: Directional replication of full-data discoveries in independent long reads. Every tested path, junction, or event must be represented in the source annotated-transcript matrix. Pooled depth is at least 20 UMIs per cell type, and the stricter endpoint also requires at least 10 UMIs per cell type in each biological replicate. Intervals are Wilson 95% intervals. SUPPA2, primer-aware SUPPA2, and paired-JCEC rMATS rows are top-500-per-contrast event audits and are not in the matched block/LSV discovery universe.

Method	Pooled eligible	Pooled agree	Pooled rate	Replicate eligible	Both agree	Both rate
Tealeaf	704	405	57.5% [53.8, 61.1]	459	175	38.1% [33.8, 42.7]
LeafCutter	56	49	87.5% [76.4, 93.8]	43	37	86.0% [72.7, 93.4]
MAJIQ Heterogen	139	86	61.9% [53.6, 69.5]	113	54	47.8% [38.8, 56.9]
scQuint	43	36	83.7% [70.0, 91.9]	32	26	81.2% [64.7, 91.1]
Paired junction CLR	131	78	59.5% [51.0, 67.6]	93	37	39.8% [30.4, 49.9]
SUPPA2 (full data)	3,871	1,872	48.4% [46.8, 49.9]	2,590	802	31.0% [29.2, 32.8]
SUPPA2 (primer aware)	3,822	1,738	45.5% [43.9, 47.1]	2,587	784	30.3% [28.6, 32.1]
rMATS paired JCEC [†]	3,859	1,778	46.1% [44.5, 47.6]	2,549	660	25.9% [24.2, 27.6]

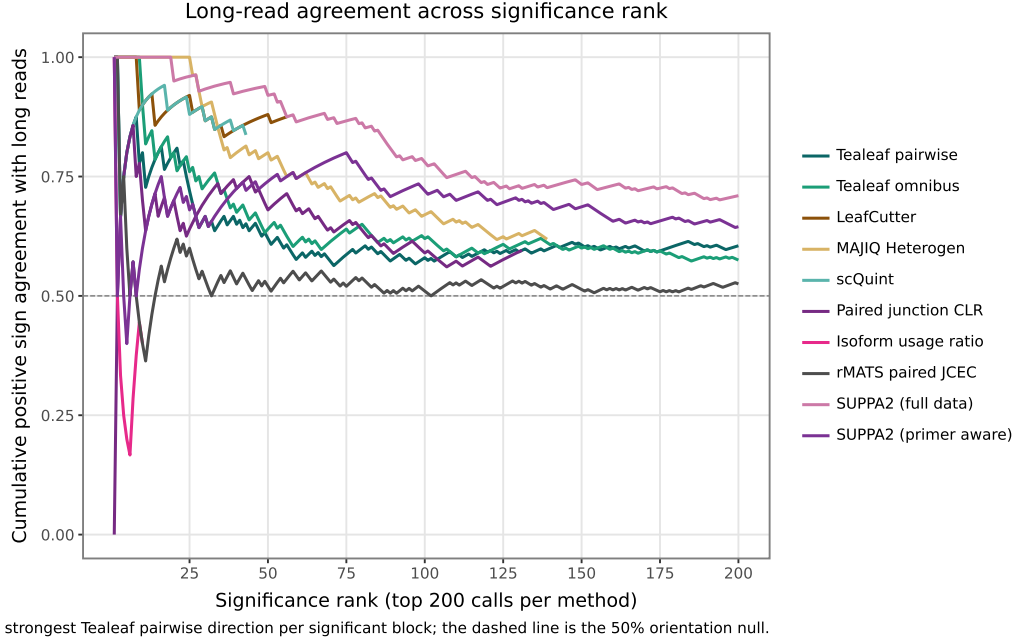


Figure 3: Cumulative directional agreement with independent long-read effects as a function of method-specific significance rank. The plot shows the top 200 long-read-evaluable calls per method, ordered by calibrated p-value, then continuous raw p-value and test statistic within ties. Calls require complete source representation, at least 20 pooled long-read UMIs per cell type, and a finite pooled direction; the x-axis is the integer rank among these evaluable associations rather than a method-specific fraction. Tealeaf omnibus points use the strongest Tealeaf pairwise direction per significant block. The isoform-usage-ratio baseline is shown only for the ten calls that overlap the retained Tealeaf long-read mapping table. Full-data SUPPA2, primer-aware SUPPA2, and paired-JCEC rMATS are included as event-level rank audits. The horizontal dashed line is the 50% orientation null.

The tie-aware curve changes the Tealeaf pairwise top-75 rate from 57.3% to 58.7% and the block-ranked omnibus curve reaches 64.0% (48 of 75). The pairwise curve remains close to its full eligible rate of 57.5% (405 of 704), so the original top-75 deficit was not only a plotting artifact. The calibrated empirical-null p-values have only 21 unique values among the top 200 Tealeaf calls; 69 of the first 75 share one p-value. The continuous tie-break makes the ordering reproducible, but the rank curve should not be interpreted as a finely resolved Tealeaf significance ranking. We then remapped the complete union of all eight fixed-total discovery sets to the independent source transcript matrix, rather than reusing the $\zeta = 64$ mapping intersection. Fixed-total concentrations give 65.3%, 68.0%, 65.3%, 61.3%, 56.0%, 58.7%, 60.0%, and 60.0% top-75 agreement after full remapping for $\zeta = 2, 4, 8, 16, 32, 64, 128, 256$, respectively. The corresponding pairwise discovery sets contain 1,594, 2,152, 2,575, 2,825, 2,922, 2,950, 2,919, and 2,871 calls, with 351, 491, 594, 661, 696, 704, 690, and 677 calls passing the common source-mapping and depth filters. The full results are recorded in `zeta_sensitivity.tsv` and `zeta_full_remap.tsv`.

We also ranked the three event baselines against the same independent source, using the top 500 tests per cell-type contrast to keep the event-unit comparison computationally bounded and then plotting the top 200 evaluable associations. The final rank table contains 142 of 200 positive agreements for full-data SUPPA2 (71.0%), 129 of 200 for primer-aware SUPPA2 (64.5%), and 105 of 200 for paired-JCEC rMATS (52.5%). Their top-500-per-contrast audits give 1,872 of 3,871,

1,738 of 3,822, and 1,778 of 3,859 pooled agreements, respectively, with 802 of 2,590, 784 of 2,587, and 660 of 2,549 calls agreeing in both source replicates. The higher top-rank agreement is not a genome-wide power claim because the original event-universe split audits are not matched to Tealeaf; the matched-split gene-pair aggregation in Table 1 is a separate analysis. These event-level results are not substituted for the full-data Tealeaf table above: they use event coordinates and event-specific PSI estimators, while the paired-JCEC rMATS values use paired PAIRADISE statistics and the common post-selection direction estimator.

Table 6: Event-based comparator agreement with the independent Tilgner long-read source. Calls are ranked within method after retaining at most 500 tests per cell-type contrast; the pooled and strict endpoints use the same 20-UMI and 10-UMI depth rules as Table 5.

Method	Pooled direction			Both source replicates		
	eligible	agree	rate	eligible	agree	rate
rMATS paired JCEC	3,859	1,778	46.1%	2,549	660	25.9%
SUPPA2 (full data)	3,871	1,872	48.4%	2,590	802	31.0%
SUPPA2 (primer aware)	3,822	1,738	45.5%	2,587	784	30.3%

The earlier unpaired rMATS result was nearly uninformative: 1,229,586 of 1,229,628 event p-values were exactly one, and no event survived BH. The corrected full-data paired-JCEC rMATS rerun tests 5,864,730 event-contrast hypotheses across 190 cell-type contrasts; 1,987,601 p-values are below one and 1,497 events pass the within-contrast JCEC FDR threshold. The FDR calls comprise 711 SE, 621 MXE, 102 A3SS, 44 A5SS, and 19 RI events. Thus the paired model materially restores a usable event-level test distribution, although its event multiplicity remains much larger than Tealeaf’s block hypotheses. The full-data SUPPA2 event audit contributes 667,224 tests over 14,752 events and no replicated genes in the original event-universe split audit, while the matched-split gene-cell-type-pair aggregation in Table 1 yields one replicated gene. Full-data SUPPA2 gives 48.4% pooled and 31.0% strict long-read directional replication. The primer-aware sensitivity gives 1,738 of 3,822 pooled agreements, 45.5%, and 784 of 2,587 strict agreements, 30.3%; its lower rate is consistent with a different event PSI estimator and is not a matched power claim. Together with the Tealeaf, LeafCutter, MAJIQ, and scQuint rows above, these results give both a matched split reproducibility comparison and a broader event-level audit. The split reproducibility row for rMATS above is retained as a historical unpaired audit; a new split paired rerun would be required to replace it with a directly matched conjunction analysis.

We also added an isoform-usage-ratio baseline. It fits the same subject-paired local compositions with a path-ratio likelihood, without the EC likelihood used by Tealeaf. After pooling the eight 50-permutation shards and applying the pooled empirical null, 89 pairwise calls pass FDR 0.05. Only ten of those calls overlap the current independent long-read mapping table, so their observed 40% directional agreement is descriptive and is not a fair replacement for the full comparator benchmark. A complete comparison requires remapping all ratio-baseline calls to the source transcript matrix.

Biological summaries use the 791 BH-significant blocks from the primary omnibus test, not the secondary pairwise localization family. Table 7 classifies each omnibus-significant block once from the supported paths in its fitted partition. A block adjacent to a terminal pseudo-anchor is called an alternative first or last *region*, because the paths may differ at several exons or transcript ends and do not necessarily isolate one promoter, terminal exon, or polyadenylation choice. Transcript-end regions dominate, but 139 distinct cassette-exon blocks are also discovered. The secondary pairwise analysis independently selects $\zeta = 64$, makes 2,950 calls across 716 distinct blocks, and is used only to localize which cell-type contrasts may drive an omnibus discovery and to match comparator hypotheses.

Table 7: Structural classes among the 791 BH-significant blocks from the primary subject-blocked omnibus cell-type test after testing-level EB selected $\zeta = 64$.

Event class	Blocks	Fraction
Alternative first region	462	58.4%
Cassette exon	139	17.6%
Alternative last region	94	11.9%
Whole-gene complex	41	5.2%
Compound internal	27	3.4%
Multi-exon skipping	14	1.8%
Mutually exclusive exons	9	1.1%
Complex internal	3	0.4%
Alternative acceptor	2	0.3%

Eight of nine literature-audited functional splice events pass BH in the primary omnibus family. The *Ntrk2* alternative-last-region block jointly differs across eight represented cell types and separates short kinase-deficient from long kinase-containing TrkB paths, including the TrkB.T1 family; astrocytes predominantly express TrkB.T1, and its loss impairs astrocyte morphogenesis and support of excitatory synaptogenesis. The *Grin1* block jointly differs across five cell types and isolates the 111-nt C1 cassette, whose retention motifs alter GluN1 surface trafficking. The *App* exon-15 cassette jointly differs across nine cell types; exclusion produces L-APP and changes polarized secretion and amyloidogenic processing. The six-path *Dlg4* alternative-first-region block jointly differs across six cell types and overlaps the functional PSD-95 alpha/beta isoform system, but the fitted partition does not isolate that pair of starts. Omnibus FDR is 0.000959 for *Ntrk2*, *App*, and *Dlg4*, and 0.00205 for *Grin1*.

Screening the remaining omnibus catalog identified four additional exact structural matches. The *Gria2* flip/flop and *Scn2a* 5N/5A pairs differ across eight and five represented cell types and each has FDR 0.000959; the former changes AMPA-receptor desensitization and steady-state current, while the latter alters Nav1.2 activation and the effects of disease variants. The 93-nt *Rbfox3* RRM cassette differs across four cell types and has FDR 0.00206; its exclusion disrupts Rbfox3 stability and splicing activity in the characterized systems. The 21-nt *Mef2d* beta microexon differs across four cell types and has FDR 0.0146; beta inclusion strengthens transcriptional activation in reporter assays. The audited *Enah* exon-11a block does not pass omnibus BH. These literature matches establish experimentally characterized consequences for the same local structures, not the direction, causal role, or mechanism of every observed cell-type contrast.

Table 8: Primary omnibus cell-type discoveries overlapping experimentally characterized splice isoforms. Mapping confidence concerns the structural correspondence between fitted paths and the published event.

Gene	Event	Mapping	Characterized consequence	Primary evidence
<i>Ntrk2</i>	Alternative last region	High	TrkB.T1 is enriched in astrocytes; its loss reduces astrocyte morphological maturation and impairs support of excitatory synaptogenesis.	Holt et al., 2019
<i>Grin1</i>	C1 cassette exon	High	C1 contains endoplasmic-reticulum retention motifs and changes surface trafficking of full-length GluN1.	Horak and Wentholt, 2009
<i>Dlg4</i>	Alternative first region	Moderate	The PSD-95 beta L27 domain binds CASK and Hrs; disrupting Hrs selectively blocks beta, but not palmitoylated alpha, synaptic clustering.	Chetkovich et al., 2002
<i>App</i>	Exon-15 cassette	High	Exon-15 exclusion produces L-APP, changes polarized APP secretion, and decreases beta-amyloid production relative to p3.	Hartmann et al., 1996
<i>Gria2</i>	Flip/flop exons	High	Flip-containing GluA2 channels desensitize more slowly and retain a larger steady-state current than flop-containing channels.	Koike et al., 2000
<i>Scn2a</i>	5N/5A exons	High	The alternatives alter Nav1.2 activation voltage and modify the effects of several early-onset epileptic-encephalopathy variants.	Thompson et al., 2020
<i>Rbfox3</i>	RRM cassette	High	Exclusion deletes 31 amino acids from the RNA recognition motif and compromises Rbfox3 stability and splicing activity.	Kim et al., 2013
<i>Mef2d</i>	Beta microexon	High	Beta inclusion increases MEF2D transcriptional activation in reporter assays.	Zhu et al., 2005

The complete omnibus discovery catalog, component annotations, coordinates, empirical p-values, FDR values, and literature-audit intersection are recorded in the repository’s local-path omnibus event-annotation directory. The separate local-path full-event annotation directory retains the secondary pairwise localization catalog.

Primer-separated read support

We first checked all significant omnibus discoveries against alignment-level evidence rather than selecting favorable examples. Of 791 BH-significant blocks, 750 have at least one explicit endpoint exon that can serve as a local depth reference; the remaining 41 are whole-gene complexes without a local endpoint anchor. Every aligned read was required to map uniquely and was deduplicated by cell barcode, UMI, genomic start, and CIGAR. Alternative donor and acceptor exons were split at every tested path boundary so that only their discriminating atomic segments contributed. For feature f in block b , cell type c , and primer p , let $\hat{u}_{fc} \in [0, 1]$ be the mean $\zeta_{\text{report}} = 1$ fitted proportion of paths containing f , let $s_{fcp} \in \mathbb{R}_+$ be pooled aligned depth for an exon segment or pooled UMI count for a junction, and let $a_{bcp} \in \mathbb{R}_+$ be mean aligned depth over the block’s one or two explicit endpoint exons. The raw endpoint-normalized evidence is $r_{fcp} = s_{fcp}/a_{bcp}$. This normalization asks whether a feature changes relative to local transcript abundance and prevents cell number or gene-expression differences from setting the comparison. Because exon depth and junction count remain on different scales and primer design changes their proportionality to path abundance, we compare cell-type ranks within each feature rather than their absolute magnitudes.

For each feature and primer we pooled signal and endpoint depth across subjects, retained at least three cell types with pooled endpoint depth at least 20 and fitted range at least 0.05, and computed the Spearman correlation between $\hat{u}_{fc}$ and r_{fcp} across cell types. Feature correlations were averaged within a block so that a block with many discriminating features did not dominate. Figure 4 shows positive median block-level agreement in every feature–primer stratum. For exon segments the median correlations are 0.18, 95% bootstrap interval 0.03–0.30, across 115 poly(dT) blocks and 0.12, interval 0.06–0.25, across 288 random-hexamer blocks. For junctions they are 0.27, interval 0.18–0.39, across 76 poly(dT) blocks and 0.26, interval 0.15–0.33, across 190 random-hexamer blocks. The fractions of blocks with positive agreement are 61%, 57%, 71%, and 66%, respectively, with one-sided sign-test p-values 0.0124, 0.00777, 0.000156, and 4.04×10^{-6} . Junction agreement is stronger, while the positive exon result shows that exonic observations also track the fitted changes. This is a descriptive same-read check, not independent replication or evidence used for statistical calibration, but it establishes that significant fitted changes usually order cell types in the direction visible in local raw coverage.

Figures 5–7 provide detailed alignment-level views of three strong, high-confidence omnibus discoveries. For each block, we first restrict to secondary pairwise contrasts with $\zeta_{\text{test}} = 64$ BH FDR below 0.05 and display the pair that maximizes the $\zeta_{\text{report}} = 1$ effect norm $\|\bar{\Delta}\|_2$, where $\Delta \in \mathbb{R}^{S-1}$ is a subject-specific paired difference in the $S - 1$ local-path isometric log-ratio coordinates and $S \in \mathbb{N}$ is the number of tested paths. The selected reporting-fit effect norms are 1.152 for cortical excitatory versus hippocampal pyramidal cells in *App*, 0.294 for cortical excitatory versus midbrain inhibitory cells in *Gria2*, and 1.498 for medium-spiny versus midbrain-inhibitory cells in *Grin1*. Each track pools unique-mapping, UMI-deduplicated spliced alignments within one cell type and priming strategy, then normalizes coverage and junction support per 1,000 primer-specific half-cells. Arc labels use the same units. The transcript models below each panel are the tested local paths, and long introns are compressed for display.

Both priming strategies support the annotated alternatives, while their absolute coverage profiles differ. This is expected because poly(dT) and random-hexamer priming have different positional exposure and is why Tealeaf retains separate primer likelihoods. Below each sashimi panel, grouped bars show the $\zeta_{\text{report}} = 1$ omnibus-refit mean proportion for each tested path across paired subjects, with standard-error bars. Using the omnibus reporting refit gives each subject–cell-type combination one estimate independent of which pair is displayed. These usage estimates jointly use both primer-specific likelihoods and all compatible observations, so they are indexed by cell

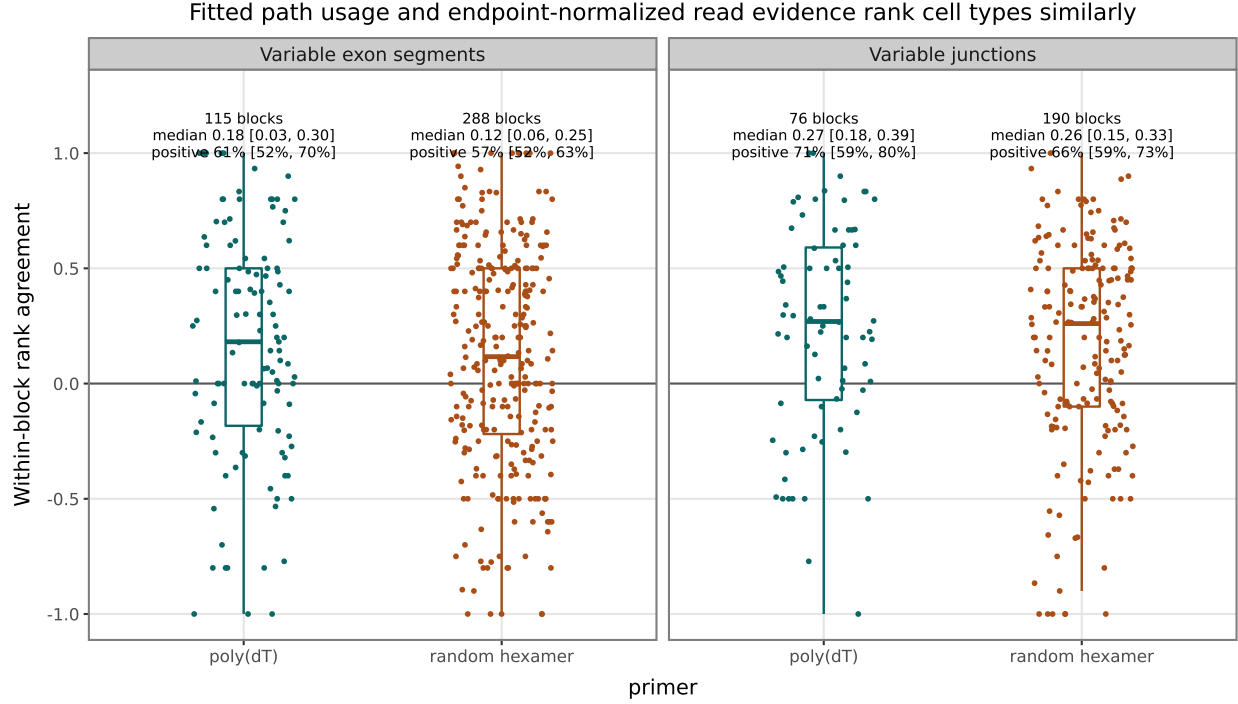


Figure 4: Agreement between weakly regularized fitted path usage and endpoint-normalized alignment evidence across significant omnibus blocks. Each point is one support-qualified block and averages within-feature Spearman correlations across cell types. Boxes show medians and interquartile ranges, annotations give block counts, bootstrap intervals for the median, and bootstrap intervals for the positive fraction, and the horizontal line marks no directional agreement. Significance is inherited from the production $\zeta_{\text{test}} = 64$ omnibus analysis; fitted inclusion uses the separate $\zeta_{\text{report}} = 1$ reporting refit and cannot alter discovery.

type and path rather than priming strategy; Path 1 and Path 2 follow the transcript-model order shown below the read tracks. The read tracks are descriptive checks rather than a reconstruction of the test because they use spliced genomic alignments only. The primary omnibus test also uses exonic, unspliced, and ambiguous EC observations through primer-specific compatibility maps and includes every represented cell type.

The plotted intervals contain 2,210 poly(dT) and 5,122 random-hexamer spliced UMIs from cortical excitatory cells and 186 and 478 from hippocampal pyramidal cells for *App*. Corresponding counts are 1,039/904 for cortical-excitatory and 167/189 for midbrain-inhibitory cells in *Gria2*, and 555/384 medium-spiny plus 148/330 midbrain-inhibitory UMIs for *Grin1*. These raw totals establish that the visible structures do not arise from a few alignments; normalization in the figures prevents unequal half-cell numbers from setting track scale.

We searched all 791 significant omnibus blocks for isoform switches. Hypothesis tests, calibrated p-values, and discovery selection retain the testing-level empirical-Bayes concentration $\zeta_{\text{test}} = 64$. Because that concentration is optimized for calibrated discovery rather than unbiased effect estimation, reported effect sizes and path usage are refitted after selection with the weak fixed-total concentration $\zeta_{\text{report}} = 1$. For each tested path, we computed the range of this weakly regularized mean usage across cell types and asked whether the identity of the most-used path changed. Ninety blocks change dominant path, but none contains a path whose cell-type means span from at most 0.1 to at least 0.9. Figures 8–10 therefore show the three largest unique-gene, non-whole-gene dominant-path changes rather than describing them as complete 0-to-1 switches. The largest path ranges are 0.344–0.965 for *Agtbp1* path 2, 0.258–0.836 for *Dlg4* path 3, and 0.171–0.739 for *Dclk1* path 4; all three omnibus FDR values are 0.000959.

The reporting concentration is a transparent stability convention and does not enter testing. All of $\zeta \in \{0.5, 1, 2\}$ retain 1,781 of 1,785 candidate fits. Across matched subject–cell-type–path estimates, $\zeta = 1$ versus $\zeta = 0.5$ has Pearson correlation 0.997 and mean absolute difference 0.021, while $\zeta = 1$ versus $\zeta = 2$ has correlation 0.995 and mean absolute difference 0.027. Among production-significant blocks, the individual path-usage 1st, 5th, 95th, and 99th percentiles are 0.003, 0.010, 0.959, and 0.986 at $\zeta = 1$, compared with 0.070, 0.129, 0.624, and 0.734 at $\zeta = 64$. Removing smoothing yields three apparent near-complete cell-type mean switches, but loses 99 candidate fits and puts the corresponding tail percentiles at zero and one. Two of those three apparent switches have their low-usage extreme supported by only one subject. Thus $\zeta_{\text{report}} = 1$ removes most testing-prior compression without adopting an unstable boundary estimator. The absence of a complete switch remains unresolved rather than a biological negative result.

Each figure contains separate poly(dT) and random-hexamer matrices and every cell type represented in that block’s omnibus fit. Columns begin with a $\zeta_{\text{report}} = 1$ mean path proportion and then follow that path’s variable exons and junctions in 5′-to-3′ RNA order. Exons and junctions present in every displayed path are omitted because they cannot distinguish path usage. Shared nonconstitutive features are repeated but keep gene-global E and J labels. Weakly regularized path usage and empirical exon and junction PSI share one fixed 0–1 color scale, and path columns retain percentage labels. For exon e , cell type c , and primer p , let $d_{ecp} \in \mathbb{R}_+$ be pooled mean aligned depth and let F_e contain the nearest path-shared exon on each available side. Variable-exon PSI is $\widehat{\text{PSI}}_{ecp}^{\text{exon}} = \min\{1, d_{ecp}/(|F_e|^{-1} \sum_{f \in F_e} d_{fcp})\}$; a zero anchor denominator is shown as missing. The path-shared exons used as depth references are not displayed. This is a local depth-ratio estimate, not an additional fitted model. For junction j , let $c_{jcp} \in \mathbb{N}_0$ be its pooled UMI count, d_j and a_j its donor and acceptor, and $\mathcal{J}_j = \{h : d_h = d_j \text{ or } a_h = a_j\}$ the set of all observed junctions sharing either splice site, including junctions whose other endpoint lies outside the displayed block. Junction PSI is $\widehat{\text{PSI}}_{jcp}^{\text{junction}} = c_{jcp}/\sum_{h \in \mathcal{J}_j} c_{hcp}$; a zero denominator is missing. Thus absolute depth and

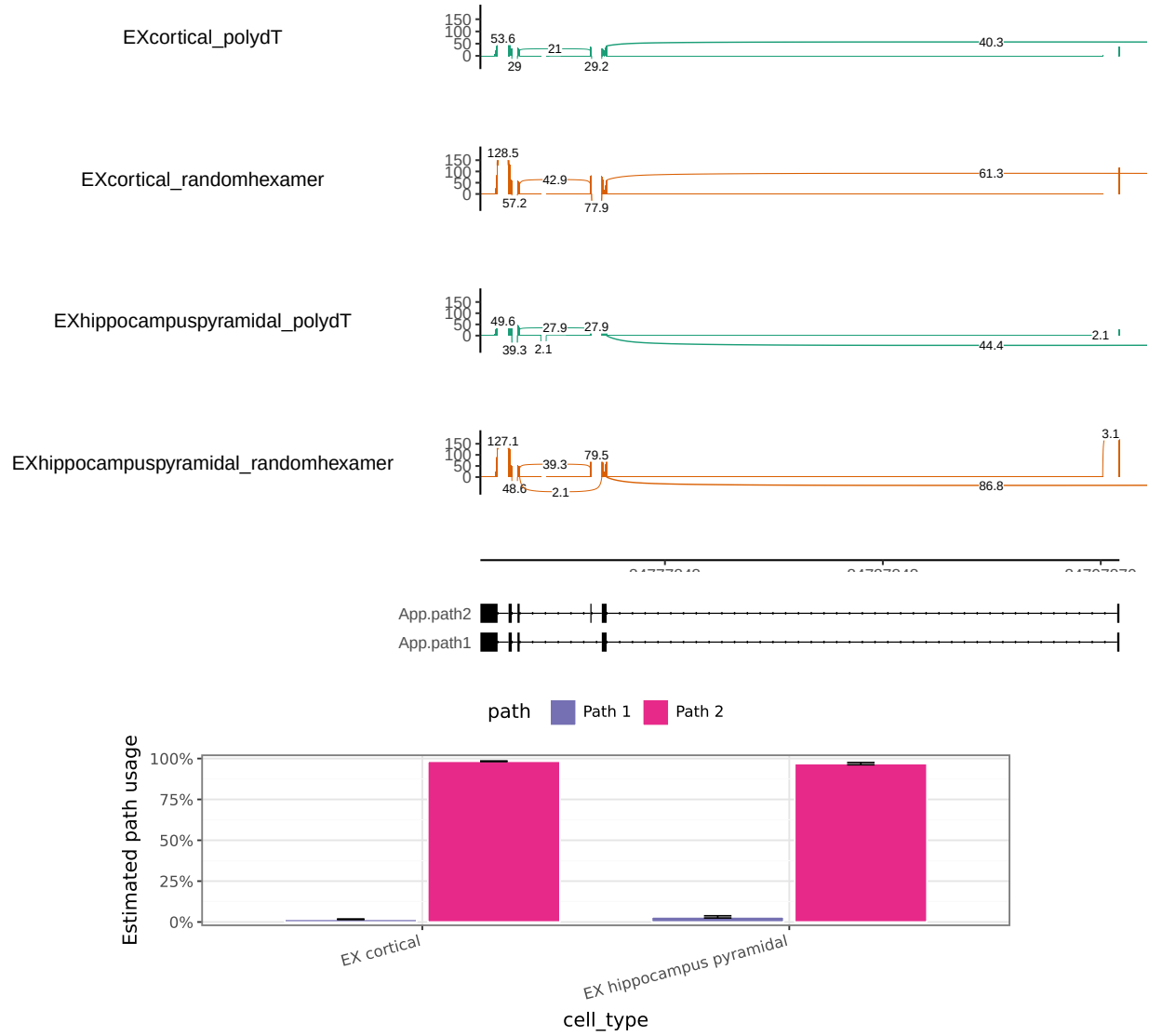


Figure 5: Primer-separated spliced-read support and fitted path usage for the *App* exon-15 cassette, block B3. EXcortical and EXhippocampuspyramidal denote cortical excitatory and hippocampal pyramidal cells. Teal and orange tracks are poly(dT) and random-hexamer half-cells. The lower panel shows $\zeta_{\text{report}} = 1$ mean path proportions with standard errors across four paired subjects.

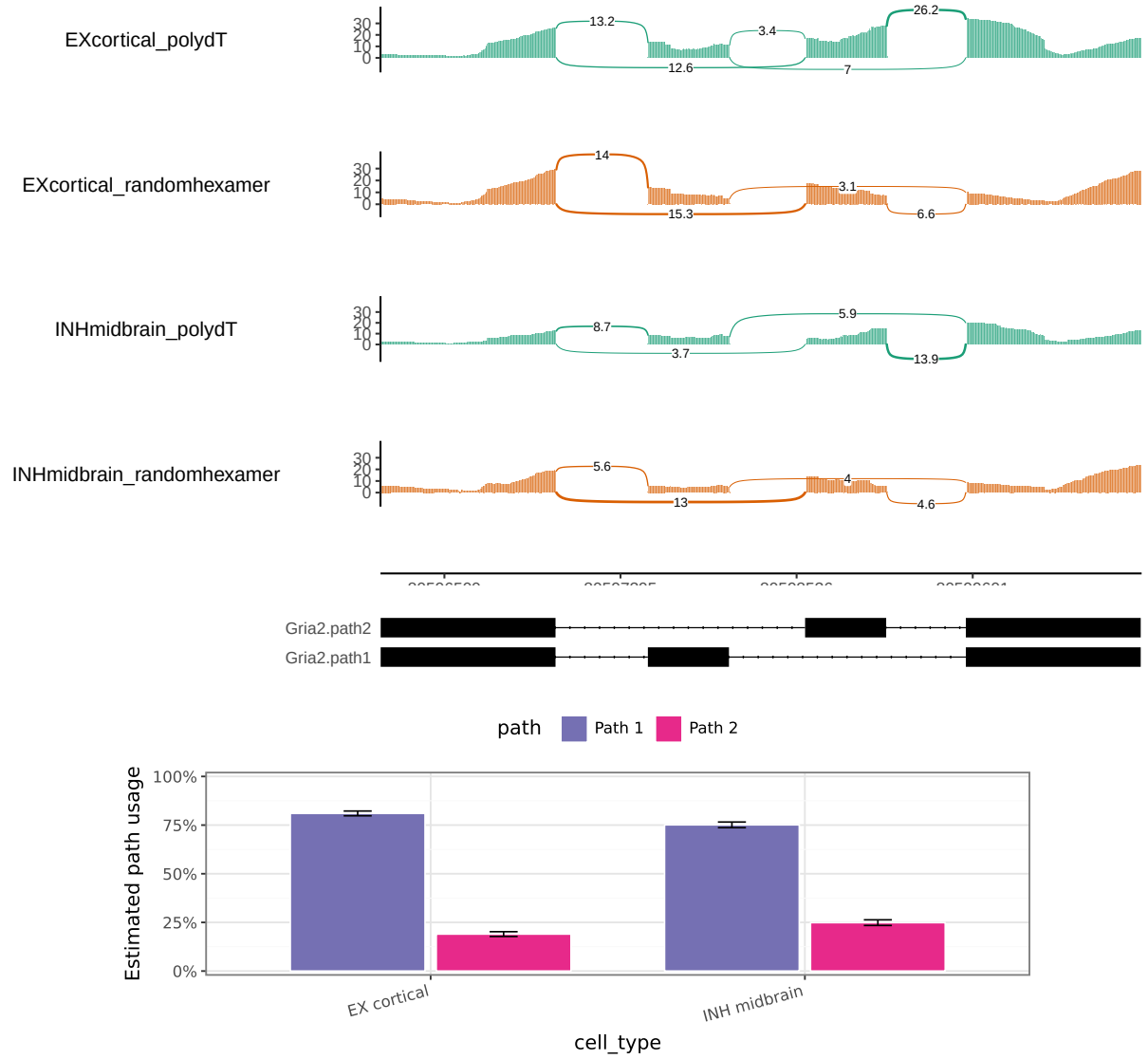


Figure 6: Primer-separated spliced-read support and fitted path usage for the mutually exclusive *Gria2* flip/flop exons, block B1. EXcortical and INHmidbrain denote cortical excitatory and midbrain inhibitory cells. Track labels and normalization follow Figure 5; the lower panel shows $\zeta_{\text{report}} = 1$ mean path proportions with standard errors across 39 paired subjects.

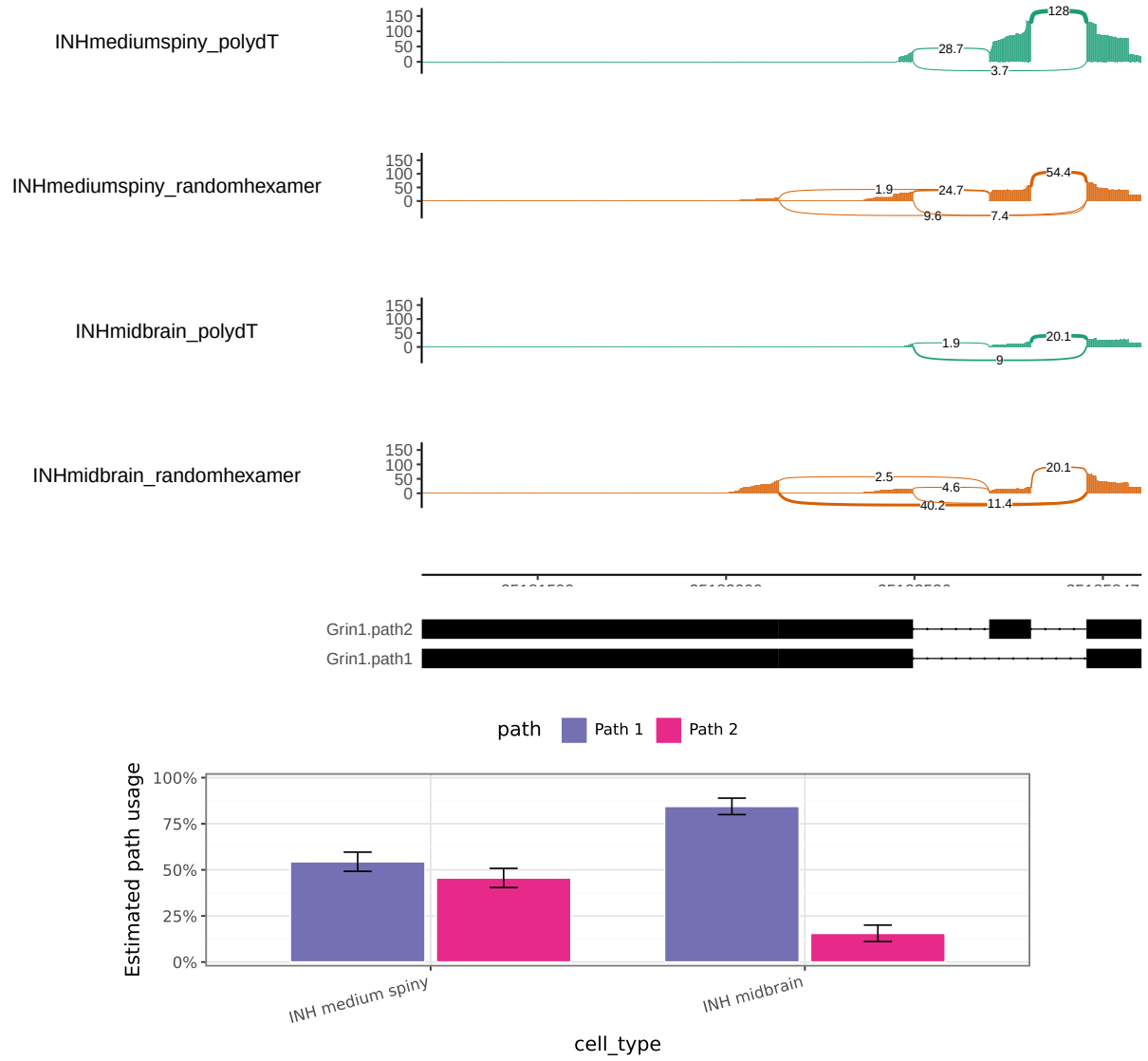


Figure 7: Primer-separated spliced-read support and fitted path usage for the *Grin1* C1 cassette, block B3. INHmediumspiny and INHmidbrain denote medium-spiny and midbrain inhibitory cells. Track labels and normalization follow Figure 5; the lower panel shows $\zeta_{\text{report}} = 1$ mean path proportions with standard errors across nine paired subjects.

unequal half-cell counts do not determine the PSI colors, while the two primer panels can reveal protocol-specific support.

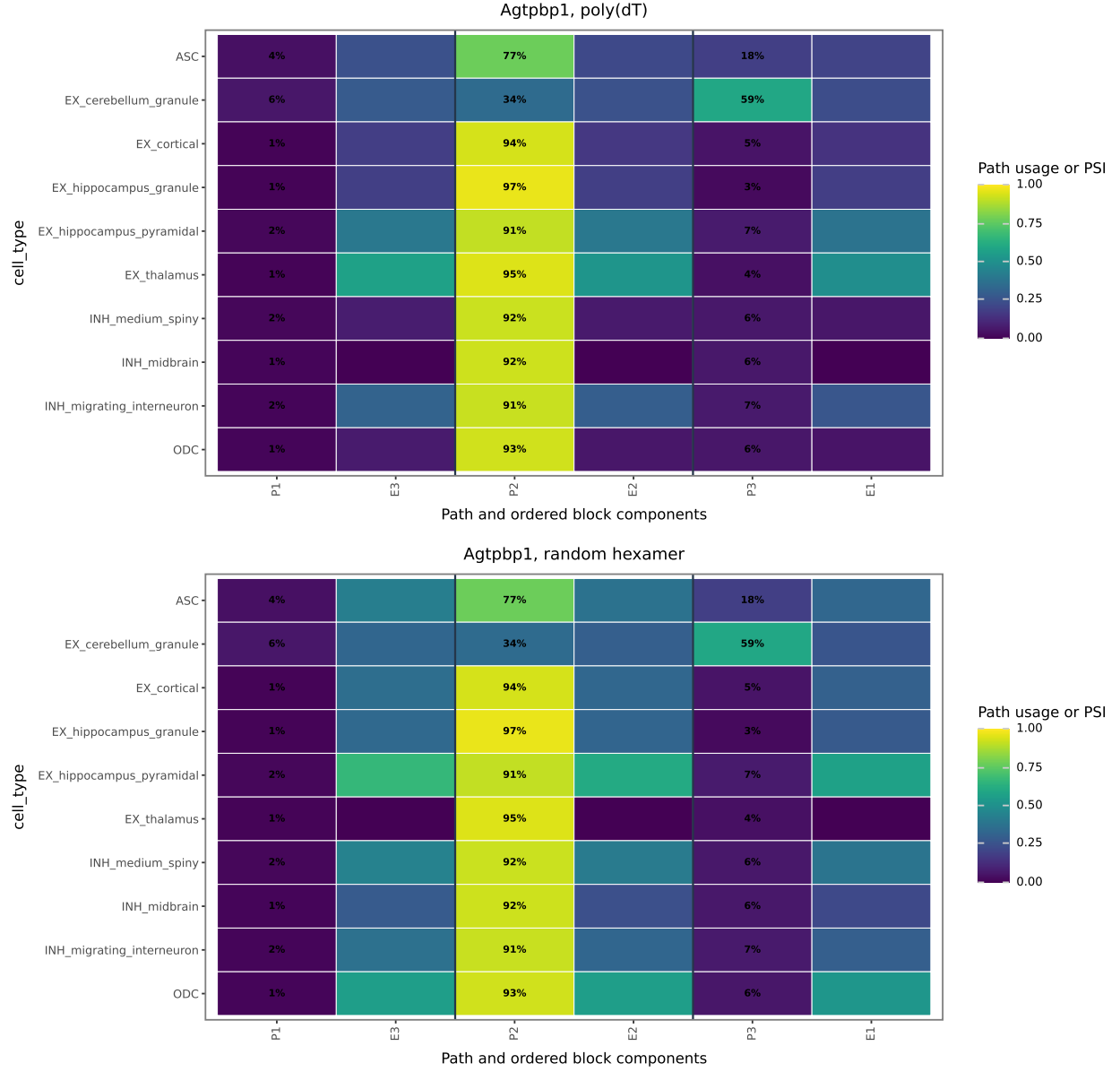


Figure 8: Strongest weakly regularized dominant-path change among production discoveries, the *Agtbbp1* alternative-first-region block. Path 2 rises from 0.344 in cerebellar-granule excitatory cells to 0.965 in hippocampal-granule excitatory cells. P columns show $\zeta_{\text{report}} = 1$ path usage; E and J columns show primer-specific exon and junction PSI.

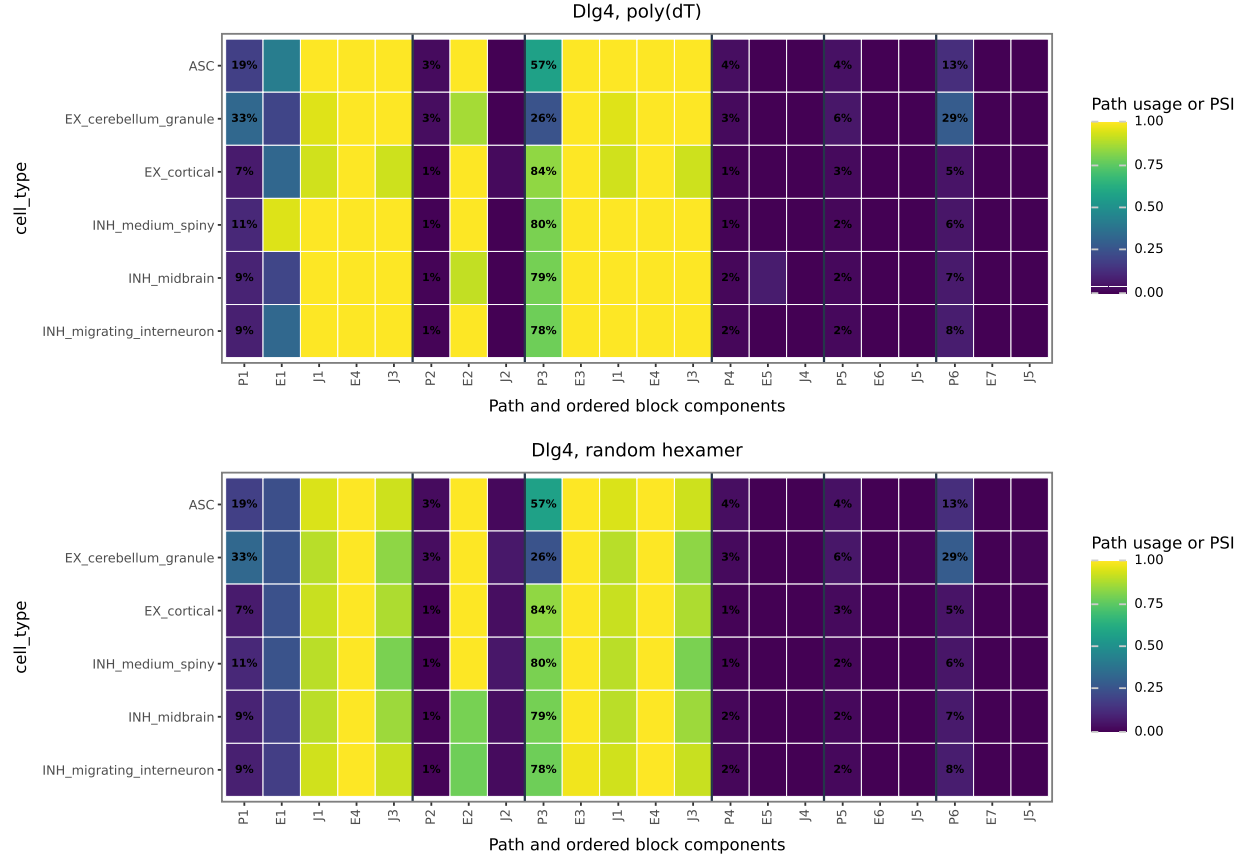
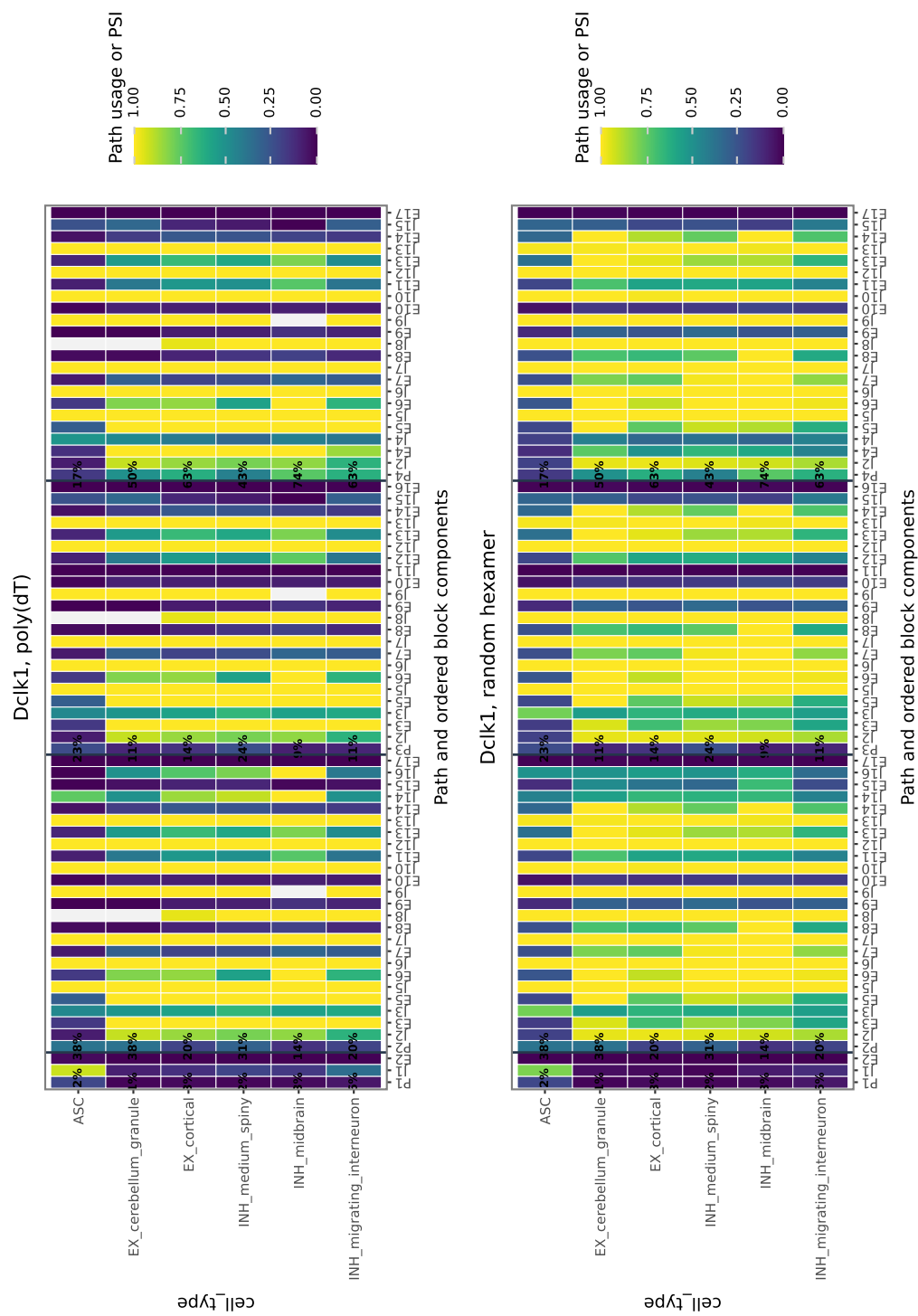


Figure 9: Second-largest weakly regularized dominant-path change, the six-path *Dlg4* alternative-first-region block. Path 3 rises from 0.258 in cerebellar-granule excitatory cells to 0.836 in cortical excitatory cells. Definitions and panel order follow Figure 8.



A Deprecated Full-Data EC GLMM

Let $r_g = T_g - 1 \in \mathbb{N}$ be the number of free reference-logit coordinates and let $\boldsymbol{\eta}_{gi} = (\boldsymbol{\eta}_{gi,-}^\top, 0)^\top \in \mathbb{R}^{T_g}$ append a zero reference-isoform logit. The primer-specific EC probability vector $\mathbf{q}_{gip}(\boldsymbol{\eta}_{gi}) \in \Delta^{K_{gp}-1}$ has components

$$q_{gipk}(\boldsymbol{\eta}_{gi}) = \frac{\sum_{t=1}^{T_g} A_{gpkt} \exp(\eta_{git})}{\sum_{h=1}^{K_{gp}} \sum_{t=1}^{T_g} A_{gpht} \exp(\eta_{git})}, \quad k \in \{1, \dots, K_{gp}\}.$$

The deprecated analysis fitted a latent global-isoform EC GLMM separately for every block. Let $m(i) \in \{1, \dots, M_b\}$ identify pseudobulk i 's mouse. Its multinomial form was $\mathbf{c}_{gip} \mid \boldsymbol{\eta}_{gi} \sim \text{Multinomial}\{n_{gip}, \mathbf{q}_{gip}(\boldsymbol{\eta}_{gi})\}$, with $\boldsymbol{\eta}_{gi,-} = \mathcal{F}_{bi}(\boldsymbol{\beta}_b) + \mathbf{u}_{g,m(i)}$ and $\mathbf{u}_{gm} \sim N(\mathbf{0}_{r_g}, \sigma_g^2 I_{r_g})$. The logistic-normal extension added $\mathbf{e}_{gi} \sim N(\mathbf{0}_{r_g}, \tau_g^2 I_{r_g})$ to each pseudobulk. Both primer likelihoods shared the same latent composition. The tested block coefficients used the pooled-baseline path-ILR Jacobian $J_b \in \mathbb{R}^{d_b \times r_g}$, its right inverse $H_b = J_b'(J_b J_b')^{-1} \in \mathbb{R}^{r_g \times d_b}$, and a nuisance basis $N_b \in \mathbb{R}^{r_g \times (r_g - d_b)}$ for $\ker(J_b)$. This gave a corrected tangent null while retaining global isoform nuisance directions.

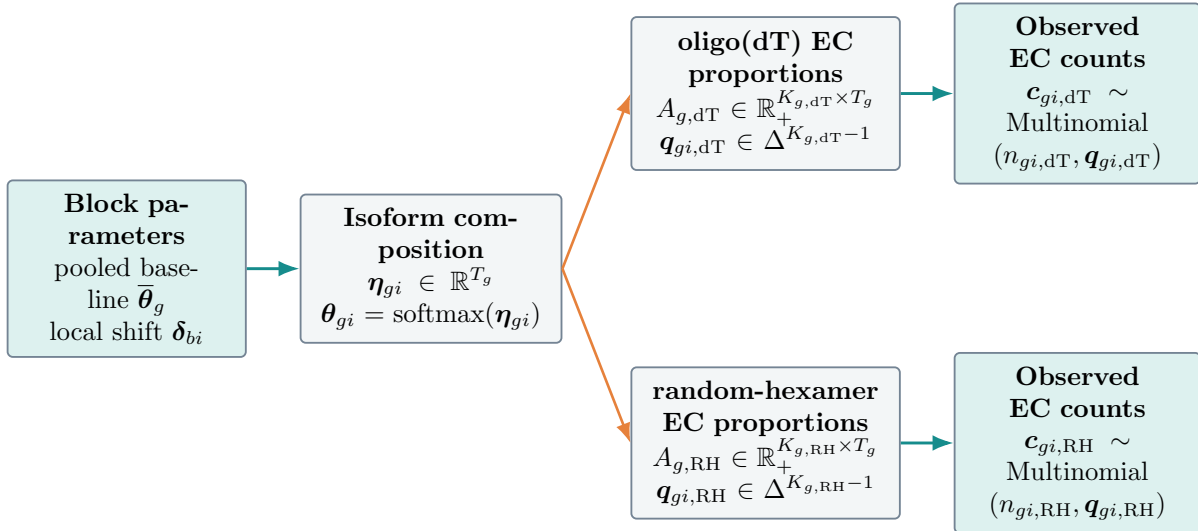


Figure 11: Generative flow for the deprecated global EC GLMM for gene $g = g(b)$ and pseudobulk i . The two primer maps share the same latent global isoform composition.

Two fitting options remain implemented. Full-covariance tilted variational inference factorizes across mice but retains unrestricted covariance within each mouse's latent vector; it uses an analytic tilted log-sum-exp bound and exact Gaussian KL term. Laplace fitting instead integrates around each mouse-specific posterior mode and uses implicit differentiation with analytic EC likelihood derivatives. The corrected full-data analysis calibrated nested variational objective gains using parametric null simulation, a cross-fitted generalized Pareto tail, and a leave-block-out empirical null CDF. GPD means generalized Pareto distribution. The corresponding derivations, algorithms, and computational comparisons are retained in the supporting sections below.

The corrected GLMM tested 4,825 cell-type partitions and 14,833 block-by-cell-type condition hypotheses. Both calibrated null families matched the audited nominal rates, but neither observed family contained a BH discovery.

Table 9: Deprecated corrected tangent-null full-data logistic-normal EC GLMM results.

Contrast	Tests	Nominal	BH	Minimum p	Null < .05	Null < .01	Null < .001
Cell type	4,825	522	0	0.00216	0.0500	0.0100	0.0010
Condition	14,833	922	0	0.000148	0.0500	0.0100	0.0010

B Exploratory and Supporting Analyses

Alternative smoothing targets and path-estimate uncertainty

Table 10 compares the production testing-level selector with discarded count-predictive objectives and the former fixed add-two rule.

The earlier count-predictive EB analysis is retained as a diagnostic. A baseline-centered spike-and-slab count model selected fixed total concentration 16, but applying only that scalar to uniform-centered testing was not a coherent transfer of one prior. Applying a common baseline-centered prior in both stages did not preserve power: an isotropic Gaussian prior on ILR perturbations selected precision 32 and produced only 23 and 11 fold BH calls, while a heavier-tailed symmetric Dirichlet prior on relative path fold changes selected concentration 64 and produced only 31 and 14. Both had pooled null rejection near 0.05 and no null-family BH calls. Shrinking every subject–cell-type composition toward one common biological center therefore attenuates the contrasts, regardless of local curvature or tail weight. Uniform-centered count-predictive evidence instead selected concentrations 2–4 and also lost replication. These failures motivate the production testing-level target, for which selection and testing use the same uniform-centered estimator and null/non-null objective.

Table 10: Smoothing selection sensitivities. Testing-level EB uses the same uniform-centered fixed-total estimator for selection and testing. Count-evidence rows transfer only the selected scalar and are retained as diagnostics. Replicated genes are reported on the LeafCutter, relaxed-coverage MAJIQ Heterogen, and scQuint matched universes.

Selection target	Scope	Scaling	Fold BH		Replicated genes		
			Half 0	Half 1	LeafCutter	MAJIQ	scQuint
Fixed $h = 2$	Global	Per path	1,163	770	50	103	101
Test mixture	Global	Fixed total	1,471	1,047	73	156	160
Baseline	Cross-fitted	Per path	1,358	1,022	70	137	142
Baseline	Cross-fitted	Fixed total	1,397	1,005	65	135	139
Baseline	Global	Per path	1,362	1,022	61	135	140
Baseline	Global	Fixed total	1,442	1,017	68	139	145
Uniform	Global	Per path	465	770	19	31	34
Uniform	Global	Fixed total	773	704	40	65	62

Second, if $I_{bmc} \in \mathbb{S}_+^{S_b-1}$ is the expected EC likelihood information and $C(\psi) = \text{diag}(\psi) - \psi\psi^\top$, the conditional Laplace covariance was estimated as

$$V_{bmc} = \left\{ I_{bmc} + \alpha_b S_b B_b^\top C(\hat{\psi}_{bmc}) B_b \right\}^{-1} \in \mathbb{S}_+^{S_b-1}.$$

For a paired contrast, $W_{bm} = V_{bm1} + V_{bm2} \in \mathbb{S}_+^{S_b-1}$ and the measurement-error model was $\hat{\mathbf{d}}_{bm} \mid \mathbf{d}_{bm} \sim N(\mathbf{d}_{bm}, \lambda W_{bm})$, $\mathbf{d}_{bm} \sim N(\boldsymbol{\mu}_b, \tau_b^2 I_{S_b-1})$, where $\lambda \geq 0$ is shared within cross-fitting strata and $\tau_b^2 \geq 0$ is fitted by REML. The analogous condition model scaled each mouse’s covariance before multivariate GLS.

At $\lambda = 1$, the fold-specific cell-type analyses made 684 and 524 BH calls, but independently replicated genes fell to 28, 10, and 11 against the three comparators. Cross-fitted REML profiles over $\lambda \in \{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 3\}$ selected zero in every split, full cell-type, and full condition stratum. The conditional EC covariance therefore did not explain held-out mouse-to-mouse error beyond the fitted biological variance and harmed the discovery tail when imposed at unit scale. Approximate uncertainty in $\hat{\delta}$, $\hat{y}$, and $\hat{d}$ remains available as a sensitivity but is not carried into the production test.

Pairwise Tealeaf sensitivity analyses

The corrected tangent-null Laplace EC GLMM was first rerun on the complete 25-UMI pairwise candidate manifests. Against LeafCutter, relaxed-coverage MAJIQ Heterogen, and scQuint it had 12, 19, and 18 replicated genes, compared with 25, 38, and 57, while its held-half nominal replication rates were 0.867, 0.828, and 0.828. The global latent model therefore retained stable signals but did not realize the expected local-path power advantage.

The direct local-path estimator without smoothing had unstable boundary estimates. Table 3 reports the final common-pipeline comparison of no smoothing and 0.5–128 pseudo-observations per path. The earlier native-test screens used to develop variance moderation are superseded by this comparison. Dropping nuisance isoforms and ECs with only nuisance support, combining globally and locally conditioned p-values, restricting to two-path blocks, and replacing Simes with Cauchy gene aggregation did not improve calibrated replication and were rejected. These analyses show that boundary stabilization and cross-block variance moderation, rather than discarding ambiguous EC information, account for the final power gain.

The concentration curve suggests an adaptive rule in which weakly supported effects receive stronger shrinkage while strong effects receive weaker shrinkage. We tested two calibrated diagnostics on the independent long-read validation. The first combined all eight calibrated concentration-specific p-values with a Cauchy statistic. The second used the zeta-2 p-value when it was below a pre-specified 0.01 threshold and otherwise used the zeta-32 p-value, followed by one BH correction; the same switch was applied to every signed-label null family. Both retained approximately 5% nominal null rejection and fewer than one BH call per null family on average. Neither improved pooled directional replication: Cauchy gave 430/740 (58.1%), the switch gave 449/772 (58.2%), and fixed zeta=4 and zeta=32 gave 290/491 (59.1%) and 410/696 (58.9%), respectively. MAJIQ Heterogen gave 86/139 (61.9%) on the same validation. Choosing the concentration from the observed effect magnitude can produce a larger apparent rate in this dataset, but that threshold was tuned on the validation data and is not a valid production procedure without incorporating the selector into null calibration. We therefore retain a single global testing concentration and use weak regularization only for post-selection effect reporting.

Depth-filtered split sensitivity

Filtering the pairwise split benchmark at a median gene depth of 100 modeled UMIs reduces the jointly converged Tealeaf fits to 1,101 block-pair tests from 202 genes in fold 0 and 564 tests from 94 genes in fold 1. The intersections with LeafCutter, relaxed-coverage MAJIQ, and scQuint then contain only 172, 364, and 337 gene-pair tests from 36, 63, and 61 genes. Tealeaf has 4, 11, and 11 replicated genes on these intersections, compared with 9, 17, and 11 for the respective comparators. Although the retained 77-test label-swap subset has a 0.0519 nominal rejection rate and no p-value below 0.01, its resolution is too low and its coverage loss is too severe for this filter to define the main genome-wide comparison. It is retained only as a depth sensitivity.

Lowering the inclusion threshold from 25 to 10 modeled gene UMIs and replacing the separate block tests with one unrestricted isoform-composition test per gene and cell-type pair expands the two split manifests to 22,453 tests from 2,503 genes and 16,753 tests from 2,238 genes. On the shared universes, this joint-gene sensitivity has 23 versus 29 replicated genes against LeafCutter, 66 versus 35 against relaxed-coverage MAJIQ, and 53 versus 104 against scQuint. Its held-half nominal replication is 0.780, 0.839, and 0.842, exceeding the corresponding comparator rates of 0.684, 0.659, and 0.776. The larger universe therefore recovers reproducible signal and exceeds MAJIQ’s discovery count, but it does not dominate the other comparators. More importantly, only 946 of 1,000 paired-label-swap fits jointly converge, 86, 21, and 6 have p-values below 0.05, 0.01, and 0.001, and three survive BH at 0.05. This low-depth joint test is too anti-conservative for production.

We also tested the requested gene model in which every annotated block contributes an independent fixed-effect subspace, with linear dependencies among overlapping block contrasts removed before fitting. At the 25-UMI screen, its shared-universe replicated-gene counts are 12, 19, and 18 against LeafCutter, relaxed MAJIQ, and scQuint, below both the respective comparator counts of 24, 38, and 57 and the unrestricted joint-gene sensitivity on the same screen. In its 1,000-test label-swap audit, 966 fits jointly converge, rejection rates are 0.0704, 0.0217, and 0.00414, and two tests survive BH. Independent block effects therefore reduce power without improving calibration over the production block tests, so this model is not retained as a fitting option.

Nonbootstrap Laplace sensitivity

Table 11 compares the analytic Laplace LRT and BIC Bayes-factor approximation with the completed mouse bootstrap analysis. A Laplace fit is retained only when the infinity norms of its scaled outer gradient and random-effect mode score are each at most 10^{-4} . Scaling the objective and gradient by the absolute initial objective changes neither the optimum nor the likelihood ratio, but prevents count depth from setting the numerical line-search scale.

Table 11: Nonbootstrap mouse cell-type sensitivity. BF counts use $\widehat{\text{BF}}_{10}^{\text{BIC}} > 10$; bootstrap calls use the primary GPD calibration.

Likelihood	Tests	Laplace conv.	LRT nominal	LRT BH	BF	Bootstrap BH
Multinomial	4,825	0.942	710	322	160	272
Logistic-normal	4,825	0.936	646	271	128	252

Restricting comparison to tests converged under both fitting strategies gives Laplace-versus-bootstrap statistic rank correlations 0.972 for multinomial and 0.964 for logistic-normal; p-value rank correlations are 0.937 and 0.922. The multinomial analyses share 247 BH calls, and the logistic-normal analyses share 204. Among these matched tests, the BF threshold selects 160 and 128 events, of which 142 and 116 are bootstrap BH calls. Applying the nominal chi-square reference to the existing variational-fit null simulations gives 0.05 rejection rates 0.0389 and 0.0374, indicating mild conservatism for that diagnostic rather than inflation.

The logistic-normal alternative estimates its observation-noise standard deviation at the lower numerical bound in 758 of 4,438 jointly converged fits. Because this nuisance variance is then on a boundary, its Wilks reference is less regular. The multinomial LRT is the cleaner nonbootstrap sensitivity, while the BF threshold supplies a smaller high-evidence subset; neither supersedes the empirically calibrated primary result without the mouse comparison above.

The independent human analysis gives the results in Table 12. The recommended multinomial

likelihood was rerun with closed-form pseudobulk likelihood derivatives, 12 posterior-mode Newton steps, and implicit mode differentiation on the first attempt, with 30 steps on continuation attempts; the logistic-normal table entry is the previous supporting sensitivity. The condition calls span 137 distinct blocks and 135 genes; 176 of 193 also exceed the BIC BF threshold. EX1, EX2, EX5, EX3, and EX4 contribute 62, 34, 21, 19, and 14 condition calls, respectively, with the remainder distributed across other neuronal and glial cell types.

Table 12: Nonbootstrap human block results from EC-count models. Condition hypotheses are block-by-cell-type tests. BF counts include only converged nested fits.

Contrast	Likelihood	Tests	Conv.	Nominal	LRT BH	BF
Cell type	Multinomial	11,597	0.947	2,557	1,463	239
Cell type	Logistic-normal	11,597	0.943	2,161	1,125	148
Condition	Multinomial	73,144	0.940	5,497	193	522

The condition nominal rate is 0.0799 among converged tests, whereas the cell-type multinomial rate is 0.2331, as expected for the stronger biological contrast. The BF and BH counts need not be ordered: $\text{BF} > 10$ is an uncorrected evidence threshold whose BIC penalty depends on tested dimension and subject count, while BH controls a genome-wide frequentist error rate.

Computational benchmarks

Table 15 compares aggregate worker time and peak resident memory for the broad cell-type and condition campaigns. Aggregate worker time is the sum of elapsed time over analysis tasks, so it is independent of queue delay and parallelism. Peak memory is the maximum for one worker. The accounting starts from each method’s prepared analysis input and includes model fitting and method-specific result construction, but excludes shared read processing. Tealeaf uses the same ordinary multinomial EC GLMM, closed-form pseudobulk likelihood derivatives, and Laplace LRT in both datasets. Each first attempt uses 12 posterior-mode Newton updates and implicit differentiation, failed-convergence fits receive a 30-update continuation, and compiled objectives are reused within bounded gene-and-shape groups. On a complete 72-test human condition shard, implicit differentiation reduced elapsed time from 79.7 to 37.1 minutes and peak memory from 6.88 to 6.66 GiB relative to differentiating through the Newton updates, with the same 69 jointly converged tests and median absolute statistic difference below 10^{-12} . Gene-group compilation reuse reduced a second shard benchmark from 37.9 to 37.1 minutes. Complete analytic-derivative cell-type and condition reruns used 51.0 mouse worker hours and 4.81 GiB peak worker memory, and 313.3 human worker hours and 8.25 GiB; each broad campaign is 3.3-fold slower than scQuint, compared with 7.8-fold before the derivative change. Human condition fitting uses finer shards than cell-type fitting because retaining compiled shapes across hundreds of heterogeneous condition tests in one process exceeds the worker memory budget; the mathematical model, initialization, optimizer, tolerances, and continuation are unchanged. The human run has 4.3-fold more tests than mouse and its median test has five rather than three isoforms, 26 rather than nine ECs, and five rather than two block paths; these larger likelihoods account for most of the remaining dataset difference. Lower-precision and early-stopping experiments did not preserve accuracy or reduce real-data elapsed time, so neither is used for the reported fits.

A matched optimization study retained L-BFGS as the production outer optimizer. Pure projected Adam at two learning rates failed the joint outer-gradient and posterior-mode convergence criteria on nearly every tested fit and changed some LRT statistics by hundreds; an Adam warm-up followed by L-BFGS recovered convergence but reached different local optima and changed statis-

tics by as much as 3.79. Muon is not used because the fixed-effect vector has no natural matrix axis, its apparent matrix shape would couple unrelated mice or depend on the arbitrary block-contrast basis. Momentum SGD, RMSProp, Lion, and related fixed-schedule first-order methods share Adam’s batching advantage but not L-BFGS’s line search or curvature history, so the failed Adam experiment provides no reason to replace the converged optimizer with them. Projecting a previously fitted alternative into later null models reduced iterations but also moved local optima. Running two independent L-BFGS workers reduced matched wall time by 38% for mouse and 24% for human, with greater peak memory and little reduction in aggregate CPU time.

Genome-wide shape census supports batched likelihood evaluation but not one globally padded tensor. Among 4,825 mouse and 11,597 human block tests, only 813 and 434 belong to an exact tensor-shape group of size at least two. Allowing pseudobulk and mouse counts to vary while holding the isoform, EC, and fixed-effect dimensions constant raises these counts to 3,726 and 7,971. Cost-sorted padded batches incur estimated likelihood-work inflation of 1.97-fold for mouse and 2.11-fold for human at batch size eight, increasing with batch size. On representative low-padding batches, a padded likelihood-gradient kernel was 2.5-fold faster than a `ragged_dot` kernel for mouse and 2.2-fold faster for human. On deliberately high-padding tails, the ragged kernel was 1.09-fold and 1.86-fold faster and compiled faster.

A full padded Laplace prototype gives every fit an independent L-BFGS history, projected gradient, active state, and Armijo line search while evaluating the separable objectives together. It preserves implicit differentiation of each posterior mode and reuse of a shared unrestricted alternative. In ordinary low-dimensional buckets, batches of eight were 5.48-fold faster for mouse and 6.43-fold faster for human than sequential scalar L-BFGS-B. All eight batched nested fits converged in each dataset, compared with seven mouse and eight human scalar fits. Among fits converged by both implementations, the maximum absolute LRT-statistic differences were 0.0090 and 0.00058. A four-fit high-dimensional mouse bucket was 1.53-fold faster after alternative reuse and gave identical statistics. A high-dimensional human bucket with ten isoforms, 18 null coefficients, 252 alternative coefficients, and roughly 900 pseudobulks was 0.62-fold as fast, meaning batching was 1.62-fold slower; the two commonly converged tests differed by 6.88 and 14.80 statistic units because the optimizers reached different local optima. Active masks preserve accepted fits but do not remove the arithmetic of inactive padded lanes from a compiled vectorized evaluation.

The initial selective policy was tested on every cell-type block, using cost-sorted batches of eight only for padding ratios at most two and nested models with at most 128 outer coefficients. The mouse hybrid reduced elapsed worker time only from 31.18 to 30.48 hours, while the human hybrid increased it from 98.39 to 125.21 hours. Peak worker memory fell from 3.97 to 1.40 GiB in mouse and from 6.55 to 4.19 GiB in human. The memory reduction did not compensate for the human slowdown or the movement between local optima. Production scalar fitting preserves bounded compilation reuse within a gene and covered pseudobulk set and initializes the first unrestricted alternative from its fitted null. The initial global shape-and-cost scheduler instead broke gene-local compilation reuse and fit deduplicated alternatives from their default initialization. The latter difference moved optima even on routes where both fits were scalar, while batching introduced additional trajectories. That initial hybrid was rejected without structured-null calibration because it failed the prior runtime and numerical-agreement gates.

A follow-up restored the production null-to-alternative initialization and made counts, EC maps, fixed-effect tensors, and cluster layouts dynamic arguments to a bounded global shape cache. On 20 real mouse tests, global caching reduced elapsed time by 10.3% and objective-evaluation time by 9.7% with identical statistics, at the cost of 11.0% greater peak memory, so this bounded cache remains available. Reusing each accepted L-BFGS iterate’s posterior mode with adaptive Newton termination halved optimizer time, but maximum statistic drift remained 0.319 at mode tolerance

10^{-5} and 0.187 at 10^{-8} . A plug-in raw-EC efficient-score screen was slower than the full nested LRT because forming the observed outer Hessian dominated its cost; seven of 20 fixed-effect Hessians were indefinite, and one extreme score disagreed with the LRT tail. The production-initialized GPU scheduler later reduced mouse worker time by 21.3% but increased human worker time by 36.9%, increased peak memory in both datasets, and retained local-optimum drift. Adaptive mode reuse, the score screen, and batched schedulers were therefore removed rather than exposed as analysis options.

The multinomial EC likelihood permits a cheaper exact derivative kernel than nested automatic differentiation. For one pseudobulk and primer, let $\mathbf{y} \in \mathbb{N}^E$ contain counts for E equivalence classes, let $M \in \{0, 1\}^{E \times K}$ map those classes to K isoforms, let $\mathbf{a} = \text{softmax}([\boldsymbol{\eta}^T, 0]^T) \in \mathbb{R}^K$ be isoform abundance for free logits $\boldsymbol{\eta} \in \mathbb{R}^{K-1}$, let $B = M \text{diag}(\mathbf{a}) \in \mathbb{R}^{E \times K}$, let $\mathbf{m} = B\mathbf{1}_K \in \mathbb{R}^E$, let $D = \mathbf{1}_E^T \mathbf{m}$, let $N = \mathbf{1}_E^T \mathbf{y}$, and let $\mathbf{c} = B^T \mathbf{1}_E \in \mathbb{R}^K$. Ignoring the multinomial coefficient, the log likelihood is

$$\ell = \mathbf{y}^T \log \mathbf{m} - N \log D.$$

Its full-logit score $\mathbf{g} \in \mathbb{R}^K$ and Hessian $H \in \mathbb{R}^{K \times K}$ are

$$\mathbf{g} = B^T(\mathbf{y} \oslash \mathbf{m}) - N\mathbf{c}/D, \quad H = \text{diag}(\mathbf{g}) - B^T \text{diag}(\mathbf{y} \oslash \mathbf{m}^2)B + N\mathbf{c}\mathbf{c}^T/D^2.$$

Deleting the reference isoform’s last matrix row and column gives derivatives with respect to $\boldsymbol{\eta}$, and derivatives sum over primers. Substituting these expressions for the pseudobulk-level autodiff score and Hessian leaves the Laplace objective, posterior-mode solve, implicit mode gradient, bounds, and L-BFGS convergence criteria unchanged while avoiding differentiation through an autodiff-generated Hessian.

On matched subsets of 12 mouse and 10 human tests, the analytic kernel reduced end-to-end time by factors of 1.49 and 1.48 and objective-evaluation time by factors of 1.24 and 1.48. Joint convergence was unchanged at 11 and 9 tests. Median absolute LRT differences were 2.4×10^{-11} and 2.3×10^{-13} , and maxima were 0.040 and 0.124. Peak memory fell from 2.35 to 1.33 GiB in mouse and from 4.81 to 4.32 GiB in human.

Table 13 gives the complete scalar reruns for both contrasts. Analytic derivatives and reduced repeated compilation lowered cell-type aggregate worker time, including merging, by factors of 2.48 in mouse and 1.78 in human and condition time by factors of 2.28 and 2.44. Mouse condition joint convergence and all three BH calls were identical. Human condition gained three joint convergences, and all 193 BH calls were identical. Median condition LRT drift was near floating-point precision, with maxima of 0.229 in mouse and 3.03 in human. Across both contrasts, broad worker time fell from 118.8 to 51.0 hours in mouse and from 729.4 to 313.3 hours in human. Broad peak memory changed from 4.74 to 4.81 GiB in mouse and from 6.83 to 8.25 GiB in human, where the peak occurs in the cell-type rather than condition run. The analytic derivative kernel with gene-grouped L-BFGS is therefore the production scalar implementation; condition discovery counts and the biological conclusions are unchanged.

Table 13: Full-data closed-form multinomial derivative audit. A and D denote analytic and nested-autodiff pseudobulk likelihood derivatives with otherwise matched scalar Laplace LRT fitting. Time is total worker hours and memory is peak GiB.

Dataset	Contrast	Tests	Conv. A/D	BH A/D	Shared	Jaccard	Median $ \Delta\text{LRT} $	Max $ \Delta\text{LRT} $	Time A/D	Memory A/D
Mouse	Cell type	4,825	4,546/4,546	322/323	322	0.997	3.64×10^{-12}	6.83	12.57/31.18	4.00/3.97
Mouse	Condition	14,833	13,779/13,779	3/3	3	1.000	2.27×10^{-13}	0.229	38.41/87.60	4.81/4.74
Human	Cell type	11,597	10,982/10,981	1,463/1,459	1,458	0.996	1.40×10^{-9}	26.68	55.17/98.39	8.25/6.55
Human	Condition	73,144	68,774/68,771	193/193	193	1.000	9.09×10^{-13}	3.03	258.10/631.05	6.16/6.83

A bounded trust-region Newton–CG alternative used exact Hessian–vector products through the same Laplace objective. On one representative mouse test, 20 trust iterations took 90.4 seconds, including 67.3 seconds in 261 Hessian–vector products, and neither nested fit converged; L-BFGS converged in 40.5 seconds. The corresponding human comparison took 151.4 versus 54.3 seconds, with 125.4 seconds in 251 Hessian–vector products and neither trust fit converged. Larger matched trust runs were still unfinished after more than three times the analytic L-BFGS benchmark duration. Exact second-order outer optimization is therefore rejected because differentiating the implicit mode and log-determinant terms is more expensive than the saved iterations.

The production-initialized genome-wide scheduler uses batches of 32, requires at least four equal-parameter fits, admits padding ratios at most three and at most 128 outer coefficients, and sends every other shape to the same scalar Laplace LRT. Failed batched fits continue from their batched parameters and, if necessary, restart from the production initialization. Table 14 gives the complete rerun. Mouse worker time fell 21.3%, from 31.18 to 24.55 hours, while peak memory rose from 3.97 to 5.31 GiB. Human worker time rose 36.9%, from 98.39 to 134.74 hours, and peak memory rose from 6.55 to 8.14 GiB. Only 1,344 of 11,597 human nulls met the batching gate; its 10,253 scalar nulls consumed 85.23 hours, so faster unrestricted alternatives could not recover the null-tail cost. The mouse and human discovery-set Jaccard indices against harmonized scalar production were 0.973 and 0.991. Median absolute LRT drift was small, 0.000131 and 0.000191, but maximum differences of 423.7 and 411.6 show that independently converged optimizers occasionally reached different local optima. A common production replacement is therefore rejected, and the reported results retain gene-grouped scalar fitting in both datasets.

Four multilevel mouse null families independently permuted the complete cell-type label vector within each subject while holding pseudobulk EC counts, nuisance columns, covered pseudobulks, and within-subject label counts fixed. Among 18,379 converged hybrid tests, rejection rates were 0.0983, 0.0385, and 0.0141 at nominal levels 0.05, 0.01, and 0.001, and every family produced 51–64 BH calls. An exact gene-grouped scalar rerun of the first permutation gave rates 0.1047, 0.0432, and 0.0148 and 62 BH calls, compared with 64 hybrid calls, 60 shared, and discovery Jaccard 0.909. Thus batching does not cause the null inflation, but the ordinary multilevel omnibus chi-square LRT is not calibrated and its BH counts in the matched method comparison must be treated as descriptive discovery yield rather than controlled FDR. This finding does not apply to the separately bootstrap-calibrated primary mouse analysis or the paired-label pairwise benchmark.

Table 14: Production-initialized full-data selective batched L-BFGS audit. H and S denote hybrid and scalar production fitting. Drift is evaluated among tests for which both procedures converged under both nested models. Time is elapsed worker hours and memory is peak GiB.

Dataset	Tests	Conv. H/S	BH H/S	Shared	Jaccard	Median $ \Delta\text{LRT} $	Max $ \Delta\text{LRT} $	Time H/S	Memory H/S
Mouse	4,825	4,597/4,546	324/323	319	0.973	0.000131	423.7	24.55/31.18	5.31/3.97
Human	11,597	11,004/10,981	1,460/1,459	1,453	0.991	0.000191	411.6	134.74/98.39	8.14/6.55

Adding the event-based baselines does not change the shared-universe comparison. The original event-universe SUPPA2 split audit has 4,204 genes represented in both halves, but no gene passes the two-fold conjunction BH threshold; its fold log-p correlation is 0.207. The matched-split SUPPA2 gene–cell-type-pair aggregation is the row in Table 1. The historical unpaired rMATs audit contributes two fold-level BH genes in each half but no replicated gene on its 496-gene shared universe, with held-half nominal replication 0.000 and fold log-p correlation 0.530. The new paired-JCEC rMATs run restores 1,497 full-data event FDR calls and is included in the event-level table and long-read comparison above, but those calls are not substituted for the historical split row because the paired split manifests were not rerun. Detailed tables, paired tests, and the event-catalog

audit are in `analyses/comparator_suppa_rmats/suppa2`.

Table 15: Analysis resource use for the matched comparator results. Time is aggregate worker hours and memory is peak GiB for one worker. Tealeaf totals include calibration or continuation retries used by the reported analyses.

Method	Mouse		Human	
	Time	Memory	Time	Memory
Tealeaf EC GLMM	51.0	4.81	313.3	8.25
LeafCutter	11.7	1.28	38.6	3.26
MAJIQ Heterogen	6.6	1.38	9.5	1.92
scQuint	15.3	3.99	94.1	4.76

Computational cost does not establish that one method is statistically better. The following independent-subject benchmark deliberately uses an in-house paired junction test on the same filtered junction groups as scQuint. It is a controlled comparison of statistical tests on a shared representation, not a comparison of the Tealeaf EC GLMM with the external methods.

Paired junction log-ratio baseline

This appendix baseline was introduced to match the pairwise hypotheses and junction-feature universe used by the external methods. It does not use EC counts, primer-specific compatibility maps, isoform ambiguity, splice blocks, or latent Gaussian EC fitting and is therefore not labeled as Tealeaf.

For one junction group, let $K \geq 2$ be its number of retained introns and let n be the number of subjects represented exactly once in each of two cell types. For subject $i \in \{1, \dots, n\}$, cell type $c \in \{0, 1\}$, and intron $k \in \{1, \dots, K\}$, let $y_{ick} \in \mathbb{N}_0$ be the pseudobulk junction count. Define $\mathbf{y}_{ic} = (y_{ic1}, \dots, y_{icK})' \in \mathbb{N}_0^K$, the centered log-ratio

$$\text{clr}(\mathbf{y}_{ic}) = \log(\mathbf{y}_{ic} + 1/2) - \frac{1}{K} \mathbf{1}'_K \log(\mathbf{y}_{ic} + 1/2) \in \mathbb{R}^K,$$

and the within-subject difference $\mathbf{d}_i = H'_K \{\text{clr}(\mathbf{y}_{i1}) - \text{clr}(\mathbf{y}_{i0})\} \in \mathbb{R}^{K-1}$. Here $H_K \in \mathbb{R}^{K \times (K-1)}$ is an orthonormal Helmert basis for the subspace orthogonal to $\mathbf{1}_K$; the result is invariant to the particular orthonormal basis. Pairing removes every additive subject effect shared by the two cell types, while the log-ratio removes junction-group depth.

The null is $H_0 : \mathbb{E}(\mathbf{d}_i) = \mathbf{0}_{K-1}$. For $K = 2$, the baseline uses the exact two-sided one-sample Student t test. For $K > 2$, with $q = K - 1$, sample mean $\bar{\mathbf{d}} \in \mathbb{R}^q$, and sample covariance $S_d \in \mathbb{R}^{q \times q}$, it computes

$$T^2 = n \bar{\mathbf{d}}' S_d^{-1} \bar{\mathbf{d}}, \quad \frac{n - q}{q(n - 1)} T^2 \sim F_{q, n - q} \quad \text{under } H_0.$$

The test requires $n \geq 8$, $n > q$, and full-rank S_d . The eight-subject threshold was fixed from permutation calibration before the observed split comparison. Junction groups use the scQuint filters: each intron must occur in at least three samples per arm, have global proportion at least 10^{-3} , and belong to a nonsingleton group.

Algorithm 1 Paired junction log-ratio test for one cell-type pair

Require: Count vectors $\mathbf{y}_{ic} \in \mathbb{N}_0^K$ and subject labels.

- 1: Keep subjects with exactly one observation from each cell type and nonzero junction-group depth in both observations.
 - 2: Apply the junction filters and require $n \geq 8$, $n > K - 1$.
 - 3: **for** each retained subject i **do**
 - 4: Compute $\mathbf{d}_i = H'_K\{\text{clr}(\mathbf{y}_{i1}) - \text{clr}(\mathbf{y}_{i0})\}$.
 - 5: **end for**
 - 6: **if** $K = 2$ **then**
 - 7: Return the two-sided one-sample t -test p-value.
 - 8: **else**
 - 9: Estimate S_d ; mark the feature ineligible if S_d is rank deficient.
 - 10: Return the exact Hotelling T^2 -to- F p-value.
 - 11: **end if**
-

Additional reproducibility calibration details

The benchmark divides biological subjects into two condition-balanced halves and reruns every cell-type analysis independently. Features are mapped to genes and feature p-values are combined within gene by Simes. Each baseline-comparator evaluation is restricted to genes and cell-type pairs eligible for both methods in both halves. For method m , gene g , and fold $h \in \{1, 2\}$, let $p_{mgh} \in [0, 1]$ be the Simes gene p-value. The conjunction p-value

$$p_{mg}^{\text{rep}} = \max(p_{mg1}, p_{mg2})$$

tests whether the gene has signal in both independent halves. BH at 0.05 is applied once to these conjunction p-values on the shared gene universe. Secondary endpoints are the fraction of nominal discoveries in one half with $p \leq 0.05$ in the other and Spearman correlation of $-\log_{10} p$ across all shared gene-pair tests.

Table 16 reports the appendix comparison. The paired CLR baseline has more independently replicated genes than each comparator on its pairwise shared universe, but this result isolates the paired testing procedure and cannot be attributed to EC ambiguity resolution, primer design, or the Tealeaf generative model.

Table 16: Appendix benchmark of cell-type differential-splicing reproducibility across independent subject halves. “Rep.” is the number of genes passing BH on p^{rep} ; “held” is held-half nominal replication; ρ is Spearman correlation of gene-pair $-\log_{10} p$. The paired CLR baseline and each comparator use a shared gene and cell-type-pair universe; comparator p-values are their native values.

Comparator	Shared genes	Paired CLR baseline			Comparator		
		Rep.	held	ρ	Rep.	held	ρ
LeafCutter	264	193	0.906	0.685	36	0.712	0.449
MAJIQ Heterogen	12	11	0.958	0.713	6	0.875	0.769
MAJIQ relaxed UMI	1,630	490	0.791	0.638	50	0.723	0.369
scQuint	2,164	646	0.788	0.642	154	0.819	0.435

Calibration uses 32 paired-label-swap families. Within each cell-type pair, subjects are swapped independently, conditional on at least one swapped and one unswapped subject; this excludes the

identity and complete-reversal configurations, both of which reproduce the observed two-sided contrast. Across 12,760 eligible paired CLR null tests, the rejection rates at 0.05, 0.01, 0.001, 10^{-4} are respectively 0.0509, 0.00964, 0.000627, 0.0000784. Reapplying BH within each permuted family yields two discoveries across 32 families. Thus the native baseline p-values are adequately calibrated at the selected threshold.

The comparator nulls are not uniformly calibrated: their pooled nominal 0.05 rates are 0.0620 for LeafCutter, 0.0837 for scQuint, and 0.0260 for MAJIQ. As a sensitivity analysis, each comparator p-value is mapped through its method-specific pooled empirical null CDF before repeating the same benchmark. The paired CLR baseline remains unchanged and has 193 versus 21 replicated genes for LeafCutter, 646 versus 134 for scQuint, and 11 versus 0 for MAJIQ. This shows that the baseline’s reproducibility advantage is not explained by favorable comparator miscalibration, but it remains evidence about the paired junction test rather than the EC GLMM. The 32-family calibration is intended to validate the body and moderate tail of each null, not to estimate genome-wide p-values smaller than its empirical resolution.

The stringent MAJIQ universe is small because its combined build and quantification defaults leave only 46 LSV edges from 13 genes; the cross-fold, common-pair intersection contains 12 genes. Lowering only the Heterogen experiment fraction increases feature-contrast tests but cannot add features absent from PSI coverage. Rebuilding with the relaxed UMI thresholds produces 20,954 and 18,914 fold-specific LSV features and a common comparison of 1,630 genes. Its 178,793 null edge tests reject at rates 0.0320, 0.00456, and 0.000151 at nominal levels 0.05, 0.01, and 0.001, with no within-family BH discoveries in 32 permutations. Empirical-null calibration corrects this conservatism. On the expanded universe, the paired CLR baseline has 490 replicated genes versus 50 for native MAJIQ and 89 for calibrated MAJIQ.

C Alternative Models and Supporting Derivations

The main text defines the two EC-count GLMMs, the direct paired local-path likelihood, and their implemented tests. This appendix records additional path-estimate models, covariance approximations, alternative EC likelihoods, and inference benchmarks used to understand or validate those analyses.

Paired-primer EC likelihood

For block b , pseudobulk i , and primer p , define $N_{bip} = \mathbf{1}'_{E_{bp}} \mathbf{c}_{bip} \in \mathbb{N}_0$, $\mathbf{q}_{bip} = A_{bp} \boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{E_{bp}}$, and $Z_{bip} = \mathbf{1}'_{E_{bp}} \mathbf{q}_{bip} \in \mathbb{R}_+$. Conditional on the primer-specific UMI total,

$$\mathbf{c}_{bip} \mid N_{bip}, \boldsymbol{\theta}_{bi} \sim \text{Multinomial}(N_{bip}, \mathbf{q}_{bip}/Z_{bip}).$$

The oligo(dT) and random-hexamer protocols use separate A_{bp} matrices and likelihood terms. Their different isoform exposures therefore model primer-specific positional coverage without forcing a common bias profile.

Why the former global basis lost local path power

The former block test lifted an unweighted Helmert path contrast to global isoform logits and defined its nuisance space by Euclidean orthogonality. Let $D_b \in \mathbb{R}^{T_g \times d_b}$ denote that centered lift and let $N_b^{\text{old}} \in \mathbb{R}^{r_g \times (r_g - d_b)}$ be the reference-logit image of its orthogonal complement. This construction

ensures an unweighted inner product of zero, but the aggregate path ILR has the pooled-baseline derivative J_b , whose entries depend on conditional isoform abundance within a path. In general,

$$J_b N_b^{\text{old}} \neq 0.$$

The old null could therefore change the aggregate local path balance that it was intended to hold fixed. This is a hypothesis-definition error rather than a missing-read problem. In the two subject halves, 1,081 of 1,353 and 653 of 859 eligible cassette tests have nonzero leakage. The median norms $\|J_b N_b^{\text{old}}\|_F / \|J_b H_b^{\text{old}}\|_F$ are 0.341 and 0.330, the 90th percentiles are both 0.523, and the maxima are 0.817 and 0.816. In fold 0, leakage has Spearman correlation 0.236 with the corrected test's gain in $-\log_{10} p$.

A second numerical issue arose because the unrestricted global alternative was cached for tests sharing a gene and covered pseudobulks. In the original cassette fits, a cached alternative had a worse optimized Laplace objective than a later block null for 232 of 1,302 and 137 of 820 jointly converged fits. The runner now detects a nesting violation, refines the shared alternative from the fitted block null, and retains the better converged optimum. This repair restores a valid nonnegative likelihood-ratio comparison but does not by itself produce the observed replication gain; changing the null geometry is necessary.

Pre-collapsed local-path comparator

The path-space comparator reduces the latent dimension before fitting but retains the raw EC count vectors. Let $T_g \in \mathbb{N}$ be the supported isoform count, $S_b \in \mathbb{N}$ the represented local-path count, and $U_b \in \mathbb{N}_0$ the number of isoforms retained only as normalization nuisances. It first estimates one label-independent pooled isoform mixture $\hat{\mathbf{w}}_g \in \Delta^{T_g-1}$ by maximizing the conditional EC log likelihood,

$$\hat{\mathbf{w}}_g = \arg \max_{\mathbf{w} \in \Delta^{T_g-1}} \sum_p \left\{ \sum_{e=1}^{E_{gp}} \bar{c}_{gpe} \log(A_{gp} \mathbf{w})_e - \bar{N}_{gp} \log(\mathbf{1}'_{E_{gp}} A_{gp} \mathbf{w}) \right\},$$

where $\bar{\mathbf{c}}_{gp} = \sum_i \mathbf{c}_{gip} \in \mathbb{N}_0^{E_{gp}}$ and $\bar{N}_{gp} = \mathbf{1}'_{E_{gp}} \bar{\mathbf{c}}_{gp} \in \mathbb{N}_0$. This pooling uses no cell-type or condition labels.

For represented path $s \in \{1, \dots, S_b\}$, let $\mathcal{I}_{bs} \subseteq \{1, \dots, T_g\}$ be its isoforms and define the fixed conditional mixture $\hat{\rho}_{bst} = \hat{w}_{gt} / \sum_{u \in \mathcal{I}_{bs}} \hat{w}_{gu}$ for $t \in \mathcal{I}_{bs}$. The collapsed primer map $B_{bp} \in \mathbb{R}_+^{E_{gp} \times (S_b + U_b)}$ has represented-path columns

$$(B_{bp})_{\cdot s} = \sum_{t \in \mathcal{I}_{bs}} \hat{\rho}_{bst} (A_{gp})_{\cdot t} \in \mathbb{R}_+^{E_{gp}},$$

while every nuisance isoform retains its original column separately. If $\phi_{bi} \in \mathbb{R}_+^{S_b + U_b}$ is the reduced latent abundance, the comparator replaces $A_{gp} \theta_{gi}$ by $B_{bp} \phi_{bi}$ and otherwise uses the same multinomial EC likelihood, subject random intercept, fixed-effect construction, Laplace fitting procedure, likelihood-ratio reference distribution, and BH correction as the isoform-space model.

Because counts are not assigned to paths before fitting, this construction retains the multinomial EC covariance $N_{bip} \{\text{diag}(\boldsymbol{\pi}_{bip}) - \boldsymbol{\pi}_{bip} \boldsymbol{\pi}_{bip}'\} \in \mathbb{S}_+^{E_{gp}}$ and propagates EC ambiguity through B_{bp} . It does not retain isoform-specific information within a path. The reduction is exact only when the conditional isoform mixture is constant across pseudobulks and the fixed mixture is correct; otherwise it is a lower-dimensional working model. Nuisance isoforms remain separate specifically to avoid imposing one mixture on heterogeneous gene-level isoforms outside the tested block.

This reduction is small in the fitted universes. Only 29.7% of mouse block tests and 12.8% of human block tests contain two represented isoforms that can be collapsed to the same path. The mean fractional latent-dimension reductions are 10.6% and 4.4%, respectively. Among multi-isoform mouse path-by-primer groups in the two subject folds, only 0.22% and 0.24% have identical compatibility columns, confirming that the reduction is generally a working-model approximation rather than an exact reparameterization.

Table 17 records the earlier multilevel and runtime ablation using the nested-autodiff Laplace implementation available at that time. Discovery is reported both on each method’s converged tests and on the common-converged universe. Independent-subject replication requires a block test to converge for both methods in both folds, combines block p-values to genes by Simes, and applies BH to the two-fold conjunction p-value. The human sensitivity analysis uses 359 matched completed shards containing 4,282 tests and is not a genome-wide human analysis. The small dimension reduction did not improve runtime because block-specific maps prevented reuse of the unrestricted gene alternative.

The cassette-specific audit isolates the hypothesis geometry that the earlier gene aggregation obscured. The exact local-path model has 732 event-by-cell-type-pair tests converged in both subject halves, 27 are nominal in both halves, and six pass BH on the conjunction p-value. All six BH events and 25 of the 27 both-nominal events agree in effect direction. On 711 tests also converged under the former global basis in both halves, the exact local model has 26 both-nominal and four conjunction-BH events, compared with 11 and two. Among 971 cassette paired-label swaps, its rejection rates are 0.0587, 0.0113, and 0.00206 at nominal levels 0.05, 0.01, and 0.001, with no BH calls. The former global model’s completed native-pairwise null audit, which was not cassette-restricted, gave 0.0745, 0.0186, and 0.00414. The local collapse therefore recovers reproducible cassette signal without evidence that the gain comes from a looser reference distribution.

Discarding ambiguous observations is not responsible for this gain. In a 100-cassette strict audit, 24 tests lack a pure EC for at least one path. Among the 76 feasible tests, pure path-identifying ECs contain a median 96.5 UMIs, 16 ECs, and 10.7% of crude gene UMI exposure. The strict likelihood gives 10 nominal tests, compared with 15 for the local-path likelihood on the same 76. Shared-exon and precursor-compatible reads are not automatically informative about cassette inclusion, but retaining the full EC vector lets every read contribute according to its primer-specific compatibility.

The tangent global-isoform model avoids fixing conditional isoform mixtures within paths and gives essentially the same cassette result, 28 both-nominal events, six conjunction-BH events, 26 of 28 signs agreeing, and effect Spearman correlation 0.722. Among 975 converged cassette paired-label swaps, its rejection rates are 0.0615, 0.0133, and 0.00205 at nominal levels 0.05, 0.01, and 0.001, with no BH calls. It remains the corrected hypothesis geometry for the full-data EC GLMM. The direct local-path estimator is used for the primary pairwise comparator benchmark because it is faster and its calibrated subject-paired test has higher replication.

Table 17: Isoform-space and pre-collapsed path-space EC GLMM comparison. Discovery rates use BH calls divided by eligible tests. Rep. is the gene count passing BH on the two-fold conjunction p-value, held is held-half nominal replication, and ρ is fold Spearman correlation. Time is aggregate worker hours and memory is peak GiB.

Endpoint	Isoform space	Path space
Mouse convergence	0.942	0.949
Mouse BH calls, method-specific	323	354
Mouse discovery rate, method-specific	0.0711	0.0773
Mouse BH calls, common converged	319	348
Mouse discovery rate, common converged	0.0710	0.0774
Replicated genes	37	39
Held-half replication	0.676	0.682
Fold rank correlation ρ	0.433	0.395
Mouse discovery worker hours	31.18	33.23
Mouse discovery peak GiB	3.97	4.04
Human subset convergence	0.949	0.949
Human subset BH calls	477	506
Human subset discovery rate	0.1174	0.1246
Human subset worker hours	33.69	36.55
Human subset peak GiB	6.39	6.65
Fold 0 worker hours	8.17	8.40
Fold 1 worker hours	7.60	8.26

Estimate-once path Wald ablation

This subsection records a rejected computational experiment. Its analytic reference failed the structured-null audit, so the estimate-once Wald implementation and its analysis launchers were removed; the result tables and findings remain for provenance.

The estimate-once ablation removes repeated raw-EC null and alternative GLMM fits. For each block b , it uses the same pooled, label-independent isoform baseline $\hat{\mathbf{w}}_g \in \Delta^{T_g-1}$ defined above. For pseudobulk $i \in \{1, \dots, n_b\}$, it estimates one path perturbation $\hat{\boldsymbol{\delta}}_{bi} \in \mathbb{R}^{d_b}$, where $d_b = S_b - 1 \in \mathbb{N}$, by maximizing the two-primer conditional EC likelihood while holding the within-path isoform mixture fixed. The resulting path composition is $\hat{\boldsymbol{\psi}}_{bi} \in \Delta^{S_b-1}$, its ILR response is $\hat{\mathbf{y}}_{bi} = Q'_b \log \hat{\boldsymbol{\psi}}_{bi} \in \mathbb{R}^{d_b}$, and the inverse conditional Fisher information gives $V_{bi} \in \mathbb{S}_+^{d_b}$. Thus the full off-diagonal covariance among path log ratios induced by the multinomial EC likelihood is retained; ECs are not assigned independently to paths.

The second stage uses the same nuisance and tested fixed-effect columns as the EC GLMM. Let $X_b \in \mathbb{R}^{n_b \times k_b}$ contain both sets of columns and $m(i) \in \{1, \dots, M_b\}$ map pseudobulk i to one of $M_b \in \mathbb{N}$ mice. Define $D_{bi} = X_{b,i} \otimes I_{d_b} \in \mathbb{R}^{d_b \times k_b d_b}$, $W_{bi} = V_{bi}^{-1} \in \mathbb{S}_+^{d_b}$, and $\boldsymbol{\beta}_b = \text{vec}(\mathbf{B}_b) \in \mathbb{R}^{k_b d_b}$, where $\mathbf{B}_b \in \mathbb{R}^{k_b \times d_b}$. The precision-weighted estimate is $\hat{\boldsymbol{\beta}}_b = A_b^{-1} \sum_i D'_{bi} W_{bi} \hat{\mathbf{y}}_{bi}$, where $A_b = \sum_i D'_{bi} W_{bi} D_{bi} \in \mathbb{S}_+^{k_b d_b}$. A pseudobulk whose EC information does not identify every tested path coordinate receives zero participation; a block is ineligible when the remaining pseudobulks do not span X_b or $M_b \leq k_b$.

Mouse dependence and biological variation enter through a clustered sandwich covariance rather than a fitted latent Gaussian. For mouse m , define its score $\mathbf{s}_{bm} = \sum_{i:m(i)=m} D'_{bi} W_{bi} (\hat{\mathbf{y}}_{bi} - D_{bi} \hat{\boldsymbol{\beta}}_b) \in \mathbb{R}^{k_b d_b}$. The coefficient covariance is $\hat{\Sigma}_b = c_b A_b^{-1} (\sum_m \mathbf{s}_{bm} \mathbf{s}'_{bm}) A_b^{-1} \in \mathbb{S}_+^{k_b d_b}$, where $c_b \in \mathbb{R}_+$ is the standard cluster and residual degrees-of-freedom correction. If the tested covariance has rank $q_b \in \mathbb{N}$, the reference distribution is $W_b/q_b \sim F_{q_b, M_b - k_b}$, which reduces the small-cluster tail inflation of the chi-square reference.

This is a two-stage working model, not an equivalent reparameterization of the EC GLMM. It does not propagate uncertainty in $\hat{\mathbf{w}}_g$, assumes the local likelihood is adequately Gaussian on the

ILR scale, freezes the observation covariance before estimating biological effects, and represents mouse dependence marginally rather than with a parametric random intercept. Its purpose is to measure the speed and empirical power available when raw EC ambiguity is resolved once rather than revisited inside every random-effect mode and optimizer iteration.

In the original nested-autodiff mouse cell-type ablation, the estimate-once test was eligible for 4,499 of 4,825 blocks and made 3,069 uncalibrated BH discoveries, a 68.2% discovery rate, compared with 323 of 4,546 for the isoform-space EC GLMM and 354 of 4,579 for the collapsed-path EC GLMM. On the 4,211 blocks eligible for all three methods, their discovery counts were 303, 335, and 2,876. Of the 303 isoform-space discoveries, 276 were also selected by estimate-once Wald, but its much larger discovery set gave a Jaccard index of 0.095. In the subject split, 948 blocks and 724 genes were testable by all three methods. Isoform-space EC GLMM, collapsed-path EC GLMM, and estimate-once Wald produced 35, 40, and 276 genes significant in both halves; their held-half nominal replication rates were 0.669, 0.685, and 0.751, and their fold rank correlations were 0.429, 0.393, and 0.426. The structured-null audit below shows that these Wald discovery and block-replication counts are inflated and must not be interpreted as evidence of additional power.

On the human cell-type analysis, 8,062 of 11,597 blocks were eligible and 5,906 passed uncalibrated BH, a 73.3% discovery rate, whereas the isoform-space EC GLMM was eligible for 11,003 blocks and selected 1,487, a 13.5% rate. On 7,685 common-eligible blocks, the methods selected 1,170 and 5,633; 873 calls overlapped and the discovery-set Jaccard index was 0.147. The human eligibility loss comes mainly from requiring the complete local-path ILR vector and covariance to be identifiable in each retained pseudobulk, while the high discovery rate reflects the same null failure seen in mouse.

Including the small correction reruns used to enforce positive denominator degrees of freedom, estimate-once Wald used 2.78 aggregate worker hours and 0.54 GiB peak memory for mouse cell type, versus 12.57 hours and 4.00 GiB for the analytic-derivative isoform-space EC GLMM. Human values were 14.18 hours and 3.58 GiB, versus 55.17 hours and 8.25 GiB. The full external-method campaigns in Table 15 used 6.61–15.28 mouse worker hours and 9.55–94.09 human worker hours. Those campaigns cover different feature units and broader reported effect families, so they are contextual rather than matched denominators; nevertheless, resolving each EC likelihood once moves runtime into the comparator range. Retaining a dense $d_b \times d_b$ covariance is therefore not the main cost. The primary EC GLMM remains slower because it repeatedly evaluates the raw EC likelihood and solves the latent Gaussian problem under both nested models.

The initial balanced two-level Gaussian null simulation did not reproduce the leverage and tested dimensions of the real design. The structured audit therefore holds every estimated $\hat{\mathbf{y}}_{bi}$, V_{bi} , nuisance column, and mouse membership fixed and independently shuffles the observed cell-type labels within each mouse. It also uses a restricted wild-cluster null, fitting the nuisance-only precision-weighted model and multiplying its complete multivariate residual vector by one Rademacher sign per mouse. Across 32 mouse null families, the CR1 analytic p-values rejected 65.8%, 58.3%, and 51.7% of label permutations at nominal 0.05, 0.01, and 0.001. Wild-cluster rates were 60.3%, 53.2%, and 46.8%. Each permuted family produced a mean 2,844 BH calls and each wild family a mean 2,581, compared with 3,069 in the observed data. Mapping observed analytic p-values through the matching null CDF within numerator-degree and cluster-degree strata left zero BH calls for either null construction.

A CR3 sensitivity premultiplies each whitened mouse score by the inverse cluster hat-matrix complement, rather than applying only the CR1 scalar correction. It reduced the observed mouse BH count from 3,069 to 1,906 but remained anti-conservative. Across 16 null families its nominal 0.05 rejection rates were 36.7% for label permutations and 33.8% for wild nulls, with mean BH counts 1,430 and 1,277. Stratified null calibration again left zero BH calls. The failure increases

with the joint tested dimension, but CR3 one-degree-of-freedom tests still reject roughly 12–24% under the structured null, so restricting to smaller blocks does not solve the problem.

An evenly spaced human sensitivity used 11 of 96 shards, 912 eligible observed blocks, and 16 null families. Label-permutation rejection was 75.3%, 72.3%, and 68.7% at nominal 0.05, 0.01, and 0.001; wild-null rejection was 72.3%, 68.8%, and 65.2%. The null families averaged 666 and 652 BH calls, and matched stratified calibration left zero. Thus the failure generalizes to human data and is stronger in its higher-dimensional tests.

Effect direction provides a separate check that does not depend on the analytic reference distribution. For each shared cell-type pair and local path, the two fold-specific ILR coefficient matrices were transformed to centered-log path effects. Across 985 blocks and 14,059 matched pair-by-path coordinates, signs agreed in 56.8% and the fold effect Spearman correlation was 0.151. Coordinates in blocks passing uncalibrated BH in both folds had 57.7% agreement and correlation 0.138, versus null sign agreement 51.0% for label permutations and 50.2% for wild draws. Requiring $|z| \geq 1.96$ in both folds raised agreement to 61.6%, but only 1,035 coordinates from 279 blocks remained and their correlation was 0.173. Under CR3, the corresponding strong-coordinate agreement was 80.6% among 217 coordinates from 69 blocks, but CR3 p-values themselves remain invalid. These results retain the runtime conclusion but reject estimate-once sandwich Wald as an inferential replacement for the EC GLMM.

Gaussian mixed-model LRT and score Bayes factors

This subsection likewise records a rejected inferential experiment. Neither the LRT tail nor fixed Bayes-factor thresholds were calibrated well enough to replace the EC GLMM, so these implementations were removed from the package and analysis runners.

Two alternatives retain the estimate-once path responses but replace the clustered sandwich calculation. Stack the usable responses as $\hat{\mathbf{y}}_b \in \mathbb{R}^{n_b d_b}$, let $Z_b \in \mathbb{R}^{n_b d_b \times a_b}$ be the nuisance design after taking its Kronecker product with I_{d_b} , and let $T_b \in \mathbb{R}^{n_b d_b \times q_b}$ be the corresponding tested design, where $a_b \in \mathbb{N}$ and $q_b \in \mathbb{N}$ are the nuisance and tested coefficient dimensions. The Gaussian random-intercept model is

$$\hat{\mathbf{y}}_b = Z_b \boldsymbol{\alpha}_b + T_b \boldsymbol{\gamma}_b + U_b \mathbf{u}_b + \boldsymbol{\epsilon}_b, \quad \mathbf{u}_b \sim N(\mathbf{0}, \sigma_{m,b}^2 I_{M_b d_b}), \quad \boldsymbol{\epsilon}_b \sim N\{\mathbf{0}, \text{blockdiag}_i(V_{bi} + \sigma_{e,b}^2 I_{d_b})\},$$

where $U_b \in \{0,1\}^{n_b d_b \times M_b d_b}$ maps responses to mouse-specific d_b -vectors, $\boldsymbol{\alpha}_b \in \mathbb{R}^{a_b}$, $\boldsymbol{\gamma}_b \in \mathbb{R}^{q_b}$, $\mathbf{u}_b \in \mathbb{R}^{M_b d_b}$, and $\boldsymbol{\epsilon}_b \in \mathbb{R}^{n_b d_b}$. Consequently, responses from the same mouse have cross-pseudobulk covariance $\sigma_{m,b}^2 I_{d_b}$, while $\sigma_{e,b}^2 \in \mathbb{R}_+$ absorbs biological variation not represented by the conditional EC covariance V_{bi} .

Let $X_{0b} = Z_b$, $X_{1b} = [Z_b \ T_b]$, and let $\Omega_b(\sigma_m^2, \sigma_e^2) \in \mathbb{S}_{++}^{n_b d_b}$ be the marginal covariance implied by the model. For model $h \in \{0,1\}$, maximum likelihood profiles its fixed effects as $\hat{\boldsymbol{\xi}}_{hb} = (X'_{hb} \Omega_b^{-1} X_{hb})^{-1} X'_{hb} \Omega_b^{-1} \hat{\mathbf{y}}_b$ and separately optimizes both nonnegative variance components using the profile objective

$$-2\ell_{hb} = \log |\Omega_b| + (\hat{\mathbf{y}}_b - X_{hb} \hat{\boldsymbol{\xi}}_{hb})' \Omega_b^{-1} (\hat{\mathbf{y}}_b - X_{hb} \hat{\boldsymbol{\xi}}_{hb}) + n_b d_b \log(2\pi).$$

This is ML rather than REML because the nested models have different fixed-effect dimensions. The likelihood-ratio statistic is $\Lambda_b = 2(\ell_{1b} - \ell_{0b}) \in \mathbb{R}_+$, with candidate reference $\chi_{q_b}^2$, and the cluster-count BIC approximation is $\log \text{BF}_{10,b}^{\text{BIC}} = \{\Lambda_b - q_b \log M_b\}/2$. Both variance components are refitted under the alternative, including boundary candidates with either or both components equal to zero.

The implementation does not form or factor a dense covariance for all pseudobulks from one mouse. Within mouse m , let $R_{bm} = \text{blockdiag}_{i:m(i)=m}(V_{bi} + \sigma_{e,b}^2 I_{d_b})$ and $K_{bm} = \mathbf{1} \otimes I_{d_b}$. It applies $(R_{bm} + \sigma_{m,b}^2 K_{bm} K_{bm}')^{-1}$ with the Woodbury identity and computes its determinant through the $d_b \times d_b$ matrix $I_{d_b} + \sigma_{m,b}^2 K_{bm}' R_{bm}^{-1} K_{bm}$. Runtime therefore scales with individual path-covariance factors and a path-dimensional correction rather than a cubic factorization in the number of pseudobulks per mouse.

The score Bayes factor requires only the fitted null covariance $\hat{\Omega}_{0b}$. Define the nuisance-residual precision $P_{0b} = \hat{\Omega}_{0b}^{-1} - \hat{\Omega}_{0b}^{-1} Z_b (Z_b' \hat{\Omega}_{0b}^{-1} Z_b)^{-1} Z_b' \hat{\Omega}_{0b}^{-1} \in \mathbb{S}_+^{n_b d_b}$, the efficient score $\mathbf{s}_b = T_b' P_{0b} \hat{\mathbf{y}}_b \in \mathbb{R}^{q_b}$, and its information $\mathcal{J}_b = T_b' P_{0b} T_b \in \mathbb{S}_{++}^{q_b}$. For a proper alternative prior $\boldsymbol{\gamma}_b \sim N(\mathbf{0}, W_{bs})$, the local Gaussian integral is

$$\log \text{BF}_{10,b}(s) = \frac{1}{2} \{ \mathbf{s}_b' (\mathcal{J}_b + W_{bs}^{-1})^{-1} \mathbf{s}_b - \log |I_{q_b} + W_{bs} \mathcal{J}_b| \}.$$

The first term rewards evidence away from the null and the determinant is the prior-volume penalty.

The production sensitivity averages equal prior mass over $s \in \{0.1, 0.25, 0.5, 1.0\}$, measured as the standard deviation of one latent cell-type effect in path-ILR units. If the tested cell-type design contains $L_b - 1$ reference contrasts, independent latent effects with variance s^2 induce the reference-invariant covariance $W_{bs} = s^2 \{(I_{L_b-1} + \mathbf{1}\mathbf{1}') \otimes I_{d_b}\} \in \mathbb{S}_{++}^{q_b}$. The reported mixture is $\log \text{BF}_{10,b}^{\text{mix}} = \text{logsumexp}_s \{ \log \text{BF}_{10,b}(s) \} - \log 4$. This avoids changing the evidence when a different reference cell type is chosen. It does not repair misspecification of the estimated-path Gaussian model, so both the LRT p-values and Bayes-factor thresholds are subjected to the same within-mouse label permutations and restricted wild-cluster nulls as the Wald ablation.

On the full mouse analysis, the Gaussian path model was eligible for 4,472 of 4,825 blocks and made 200 analytic BH discoveries, 4.47% of eligible blocks. The score-mixture Bayes factor exceeded 10 for 277 blocks and 100 for 163, while the BIC Bayes factor exceeded those thresholds for 83 and 51. On 4,232 blocks also eligible for the isoform-space EC GLMM, the EC model selected 305 and the Gaussian path LRT selected 195, with 105 shared calls and discovery-set Jaccard index 0.266. This is much less discrepant than the estimate-once sandwich Wald analysis, but nominal thresholds still require the structured-null check below.

All 96 mouse shards completed the 32 within-mouse label permutations. Their pooled LRT rejection rates were 4.16%, 1.51%, and 0.478% at nominal 0.05, 0.01, and 0.001, and one null family produced a mean 11.0 analytic BH calls. Thus the center is conservative but the deep tail is inflated. The restricted wild-cluster sensitivity covered 92 shards and gave pooled rates 2.24%, 0.638%, and 0.150%, with a mean 1.56 BH calls. Directly estimating the expected false-discovery count from each complete label-permutation family retained 12 LRT calls at empirical FDR 0.05, compared with 194 under the wild null; the conservative intersection therefore contains 12. The corresponding counts were 12 and 269 for the score-mixture Bayes factor and 27 and 99 for the BIC approximation. At the fixed threshold $\text{BF} \geq 10$, label permutations predict 28.3 null score-mixture calls among 4,472 tests versus 277 observed, and 7.94 null BIC calls versus 83 observed. Bayes factors therefore rank a small supported tail, especially after the BIC dimension penalty, but neither the score approximation nor the LRT supplies calibrated analytic thresholds. The production EC GLMM remains the primary inferential model; estimate-once Gaussian results are an ablation or must use design-matched empirical calibration.

Across 963 mouse block tests and 725 genes shared between both subject halves and the EC GLMM, the isoform-space model had 35 genes passing BH in both halves, held-half nominal replication 0.668, and fold log-p correlation 0.432. The Gaussian path LRT had 17 replicated genes, held-half replication 0.724, and fold correlation 0.716. Among 14,014 matched cell-type-pair by path coordinates, overall sign agreement was 0.573 and effect correlation was 0.148. Requiring $|z| \geq 1.96$

in both halves retained 171 coordinates from 54 blocks, with sign agreement 0.947 and correlation 0.932. The strong effects are more directionally stable than those selected by the sandwich Wald statistic, although this does not establish tail calibration.

The dense mouse run used 4.46 aggregate worker hours. Woodbury used 4.83 hours and therefore did not improve aggregate mouse time, although it reduced the slowest shard from 502 to 418 seconds. On an identical 20-block high-dimensional human estimate set, Woodbury reduced runtime from 193.6 to 168.0 seconds, 13.2%, and changed the largest LRT statistic by only 2.2×10^{-7} . The Woodbury implementation is retained because its memory and factorization dimension do not grow cubically with the number of pseudobulks per mouse.

An evenly spaced human sensitivity used 11 of 96 shards, with 935 eligible blocks among 1,329 candidates. The analytic LRT made 153 BH calls, the score-mixture Bayes factor exceeded 10 for 196 blocks, and the BIC Bayes factor exceeded 10 for 19. Across eight within-subject label-permutation families, LRT rejection was 6.08%, 3.37%, and 2.09% at nominal 0.05, 0.01, and 0.001. The first threshold is near nominal but the extreme tail remains inflated. Empirical tail-FDR estimates retained 37 LRT, 32 score-mixture, and 19 BIC-evidence calls at 0.05, but eight null families give a coarse sensitivity rather than a definitive genome-wide calibration.

The genome-wide hierarchical model supplies $\theta_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$. Let $s_b(t) \in \{1, \dots, S_b\}$ map isoform $t \in \mathcal{I}_b^{\text{sp}}$ to its path. The baseline spliced-isoform mass is the scalar $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \theta_{bit}^{(0)} \in \mathbb{R}_+$. For $t \in \mathcal{I}_b^{\text{sp}}$, local refinement uses

$$\theta_{bit}(\delta_{bi}) = \omega_{bi}^{\text{sp}} \frac{\theta_{bit}^{(0)} \exp\{(Q_b)_{s_b(t), \cdot} \delta_{bi}\}}{\sum_{u \in \mathcal{I}_b^{\text{sp}}} \theta_{biu}^{(0)} \exp\{(Q_b)_{s_b(u), \cdot} \delta_{bi}\}},$$

where $(Q_b)_{s, \cdot} \in \mathbb{R}^{1 \times d_b}$. Nuisance-isoform abundances are unchanged. This preserves total spliced mass and the relative isoform mixture within each path while estimating only $\delta_{bi} \in \mathbb{R}^{d_b}$.

Path response and inferential covariance

Each column of $C_b \in \{0, 1\}^{S_b \times T_g}$ corresponding to $t \in \mathcal{I}_b^{\text{sp}}$ contains one 1 in row $s_b(t)$; nuisance isoform columns are zero. Define

$$\mathbf{a}_{bi} = C_b \theta_{bi} \in \mathbb{R}_+^{S_b}, \quad \psi_{bi} = \frac{\mathbf{a}_{bi}}{\mathbf{1}_{S_b}' \mathbf{a}_{bi}} \in \Delta^{S_b-1}, \quad \mathbf{y}_{bi} = Q_b' \log \psi_{bi} \in \mathbb{R}^{d_b}.$$

The perturbation parameterization implies $\mathbf{y}_{bi} = \mathbf{y}_{bi}^{(0)} + \delta_{bi}$, where $\mathbf{y}_{bi}^{(0)} \in \mathbb{R}^{d_b}$ is computed from the baseline.

Let $I_{\delta, bi} \in \mathbb{S}_+^{d_b}$ be the expected Fisher information of the summed paired-primer log likelihood with respect to δ_{bi} . The conditional covariance is $V_{bi}^{\text{cond}} = I_{\delta, bi}^{-1} \in \mathbb{S}_+^{d_b}$ when the information is positive definite. It conditions on the fitted total spliced mass, the isoform mixture within each path, and every nuisance-isoform abundance. Its only free directions are the d_b path perturbations.

Full-isoform nuisance (profile) covariance

The profile calculation is a covariance sensitivity analysis, not a second estimate of path usage. It retains the same $\hat{\theta}_{bi} \in \mathbb{R}_+^{T_g}$ obtained from the constrained local fit, but asks how uncertain its path ILRs would be if all T_g isoform log-abundances were locally free. Equivalently, after a smooth reparameterization into path ILRs and $T_g - d_b$ nuisance coordinates, it approximates the path-ILR block of the inverse Fisher information after profiling the nuisance coordinates. No unconstrained

isoform likelihood is optimized, so *profile* below means this local expected-Fisher approximation rather than an exact profile-likelihood fit.

Set $\boldsymbol{\eta}_{bi} = \log \hat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}^{T_g}$, $D_{bi} = \text{diag}(\hat{\boldsymbol{\theta}}_{bi}) \in \mathbb{R}^{T_g \times T_g}$, $\mathbf{d}_{bp} = A'_{bp} \mathbf{1}_{E_{bp}} \in \mathbb{R}^{T_g}$, and $\hat{\mathbf{q}}_{bip} = A_{bp} \hat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{E_{bp}}$, $\hat{Z}_{bip} = \mathbf{1}'_{E_{bp}} \hat{\mathbf{q}}_{bip} \in \mathbb{R}_+$, and $\mathbf{r}_{0,bip} = (\mathbf{d}_{bp} \odot \hat{\boldsymbol{\theta}}_{bi}) / \hat{Z}_{bip} \in \mathbb{R}^{T_g}$. The primer-specific expected information is

$$I_{\eta,bip} = N_{bip} \left[\frac{D_{bi} A'_{bp} \text{diag}(\hat{\mathbf{q}}_{bip}^{-1}) A_{bp} D_{bi}}{\hat{Z}_{bip}} - \mathbf{r}_{0,bip} \mathbf{r}'_{0,bip} \right] \in \mathbb{S}_+^{T_g}.$$

The paired-primer information is $I_{\eta,bi} = \sum_{p=1}^P I_{\eta,bip} \in \mathbb{S}_+^{T_g}$.

The covariance must be propagated through the isoform-to-path map. Let $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \hat{\theta}_{bit} \in \mathbb{R}_+$ and $a_{bis} = \sum_{t: s_b(t)=s} \hat{\theta}_{bit} \in \mathbb{R}_+$. Define $R_{bi} = \partial \log \psi_{bi} / \partial \boldsymbol{\eta}'_{bi} \in \mathbb{R}^{S_b \times T_g}$ entrywise by

$$(R_{bi})_{st} = \begin{cases} \mathbb{I}\{s_b(t) = s\} \hat{\theta}_{bit} / a_{bis} - \hat{\theta}_{bit} / \omega_{bi}^{\text{sp}}, & t \in \mathcal{I}_b^{\text{sp}}, \\ 0, & t \notin \mathcal{I}_b^{\text{sp}}. \end{cases}$$

Thus

$$J_{bi} = \frac{\partial \mathbf{y}_{bi}}{\partial \boldsymbol{\eta}'_{bi}} = Q'_b R_{bi} \in \mathbb{R}^{d_b \times T_g}.$$

The multinomial likelihood necessarily has a global-scale null direction, and EC ambiguity can create additional null directions. Write the symmetric eigendecomposition

$$I_{\eta,bi} = U_{bi} \text{diag}(\boldsymbol{\lambda}_{bi}) U'_{bi},$$

where $U_{bi} \in \mathbb{R}^{T_g \times T_g}$ is orthogonal and $\boldsymbol{\lambda}_{bi} \in \mathbb{R}^{T_g}$. An eigenvalue is treated as positive when $\lambda_{bit} > 10^{-8} \max_u |\lambda_{biu}|$. Let $\mathcal{P}_{bi} \subseteq \{1, \dots, T_g\}$ index these positive eigenvalues, let $\mathcal{N}_{bi}$ index the remainder, and let $U_{bi,+} \in \mathbb{R}^{T_g \times |\mathcal{P}_{bi}|}$ and $U_{bi,0} \in \mathbb{R}^{T_g \times |\mathcal{N}_{bi}|}$ contain the corresponding eigenvectors.

The path response is identifiable only if its Jacobian annihilates every information-null direction. The implementation requires

$$\|J_{bi} U_{bi,0}\|_F \leq 10^{-9} \max\{\|J_{bi}\|_F, 1\}.$$

If this fails, V_{bi}^{prof} is set to an infinite matrix and the estimate is excluded from profile testing. Otherwise,

$$V_{bi}^{\text{prof}} = J_{bi} U_{bi,+} \text{diag}(\boldsymbol{\lambda}_{bi,+}^{-1}) U'_{bi,+} J'_{bi} = J_{bi} I_{\eta,bi}^+ J'_{bi} \in \mathbb{S}_+^{d_b}.$$

Unlike V_{bi}^{cond} , this covariance allows EC-supported changes in within-path isoform mixtures, spliced mass, and nuisance isoforms to absorb information that would otherwise identify a path contrast. It is therefore the more conservative sensitivity calculation when those nuisance directions are weakly identified.

Cell-resolved likelihood

For cell h in pseudobulk i , let $\mathbf{c}_{bihp} \in \mathbb{N}_0^{E_{bp}}$, $N_{bihp} = \mathbf{1}'_{E_{bp}} \mathbf{c}_{bihp} \in \mathbb{N}_0$, and $\boldsymbol{\theta}_{bih}^{(0)} \in \mathbb{R}_+^{T_g}$ be its EC counts, total, and baseline. Every cell shares the same pseudobulk perturbation $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$, but its isoform vector $\boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi}) \in \mathbb{R}_+^{T_g}$ is normalized using its own baseline. The objective is the scalar

$$\ell_{bi}(\boldsymbol{\delta}_{bi}) = \sum_{h=1}^{H_{bi}} \sum_{p=1}^P \log \text{Multinomial} \left(\mathbf{c}_{bihp}; N_{bihp}, \frac{A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})}{\mathbf{1}'_{E_{bp}} A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})} \right).$$

Thus cells contribute assignment information but do not become biological replicates. Let $U_{bih} \in \mathbb{R}_+$ be the cell's total UMI count and $g_{bih} \in [0, 1]$ its decoded proportion for the block's gene. The scalar weight is $w_{bih} = U_{bih}g_{bih}$, and the reported pseudobulk composition is

$$\psi_{bi} = \frac{\sum_{h=1}^{H_{bi}} w_{bih} \psi_{bih}}{\sum_{h=1}^{H_{bi}} w_{bih}} \in \Delta^{S_b-1}.$$

Cell- and primer-specific Fisher information matrices in $\mathbb{S}_+^{d_b}$ are summed, inverted, and propagated through this weighted average to obtain $V_{bi} \in \mathbb{S}_+^{d_b}$.

Effective counts and compositional regression

Treating all gene UMIs as known path assignments is overconfident when ECs are ambiguous. For each fitted ψ_{bi} , define the unit-multinomial ILR covariance

$$G_{bi} = Q'_b \text{diag}(\psi_{bi}^{-1}) Q_b \in \mathbb{S}_+^{d_b}.$$

The scalar effective count matches G_{bi}/n to the EC-derived V_{bi}^{cond} in Frobenius inner product $\langle U, W \rangle_F = \text{tr}(U'W)$, for $U, W \in \mathbb{R}^{d_b \times d_b}$:

$$n_{bi}^{\text{eff}} = \min \left\{ N_{bi}^{\text{gene}}, \frac{\langle G_{bi}, G_{bi} \rangle_F}{\langle G_{bi}, V_{bi}^{\text{cond}} \rangle_F} \right\} \in \mathbb{R}_+,$$

where $N_{bi}^{\text{gene}} \in \mathbb{N}_0$ is the gene UMI count. Fractional path counts are $\mathbf{x}_{bi} = n_{bi}^{\text{eff}} \psi_{bi} \in \mathbb{R}_+^{S_b}$, with $\mathbf{1}_{S_b}' \mathbf{x}_{bi} = n_{bi}^{\text{eff}}$.

For one block, stack $n = n_b \in \mathbb{N}$ observations in a design $X \in \mathbb{R}^{n \times k}$, where $k \in \mathbb{N}$, and coefficient matrix $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$. The mean composition $\boldsymbol{\mu}_{bi} \in \Delta^{S_b-1}$ is defined by reference-path logits

$$\log \frac{\mu_{bis}}{\mu_{biS_b}} = (X \mathbf{B}_b)_{is}, \quad s = 1, \dots, d_b.$$

The generalized Dirichlet-multinomial likelihood, extended to fractional counts through gamma functions, is

$$\mathbf{x}_{bi} \mid n_{bi}^{\text{eff}} \sim \text{DirichletMultinomial} \left(n_{bi}^{\text{eff}}, \kappa_b \boldsymbol{\mu}_{bi} \right),$$

where $\kappa_b \in \mathbb{R}_+ \cup \{\infty\}$ is a block-level concentration. Larger κ_b means less overdispersion; the endpoint $\kappa_b = \infty$ is exactly multinomial.

For nested designs $X_0 \in \mathbb{R}^{n \times k_0}$ and $X_1 \in \mathbb{R}^{n \times k_1}$, with $k_1 \geq k_0$ and the first $k_0, k_1 \in \mathbb{N}$, and the first k_0 columns of X_1 equal to X_0 , let $\widehat{\ell}_0, \widehat{\ell}_1 \in \mathbb{R}$ be maximized scalar log likelihoods. The statistic and nominal degrees of freedom are

$$\Lambda_b = 2(\widehat{\ell}_1 - \widehat{\ell}_0) \in \mathbb{R}_+, \quad \nu_b = (k_1 - k_0)d_b \in \mathbb{N}_0.$$

The fixed-concentration analysis estimates $\widehat{\kappa}_{0b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_0 and uses it under X_1 . The free-concentration sensitivity estimates $\widehat{\kappa}_{1b} \in \mathbb{R}_+ \cup \{\infty\}$ under X_1 . In the latter case both models contain one concentration nuisance parameter, so ν_b is unchanged. Both finite and exact multinomial endpoints are evaluated.

Gaussian mouse-replicate regression

For a condition test within one block and cell type, stack $\hat{\mathbf{y}}_{bi} \in \mathbb{R}^{d_b}$ over $n \in \mathbb{N}$ mice. Let $X_b \in \mathbb{R}^{n \times k}$, $k \in \mathbb{N}$, contain an intercept and condition indicators, $\mathbf{B}_b^{\text{GLS}} \in \mathbb{R}^{k \times d_b}$ be regression coefficients, and $\boldsymbol{\epsilon}_{bi} \in \mathbb{R}^{d_b}$ be residuals:

$$\hat{\mathbf{y}}_{bi} = (X_{b,i} \mathbf{B}_b^{\text{GLS}})' + \boldsymbol{\epsilon}_{bi}, \quad \text{Var}(\boldsymbol{\epsilon}_{bi}) = V_{bi} + \tau_b^2 I_{d_b} \in \mathbb{S}_+^{d_b}.$$

Here $X_{b,i} \in \mathbb{R}^{1 \times k}$, and $\tau_b^2 \in \mathbb{R}_+$ is residual between-mouse variance estimated by REML under the null. If $r_b \in \mathbb{N}$ non-reference condition columns are tested, their coefficients form $\boldsymbol{\beta}_b^{\text{test}} \in \mathbb{R}^{r_b d_b}$ with covariance $\Sigma_b^{\text{test}} \in \mathbb{S}_+^{r_b d_b}$. The Wald statistic is

$$W_b = (\boldsymbol{\beta}_b^{\text{test}})' (\Sigma_b^{\text{test}})^{-1} \boldsymbol{\beta}_b^{\text{test}} \in \mathbb{R}_+, \quad \nu_b^{\text{GLS}} = r_b d_b.$$

Paired cell-type test and permutations

For a cell-type test, let $M_b, L_b \in \mathbb{N}$ be the numbers of represented biological mice and cell types, respectively. The null design $X_{0b} \in \{0, 1\}^{n_b \times M_b}$ contains an intercept-coded set of mouse fixed effects. The alternative $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ appends treatment-coded cell-type indicators. Consequently $k_0 = M_b$, $k_1 = M_b + L_b - 1$, and $\nu_b = (L_b - 1)d_b$. The tested coefficient block is $\mathbf{B}_b^{\text{cell}} \in \mathbb{R}^{(L_b - 1) \times d_b}$, and

$$H_{0b} : \mathbf{B}_b^{\text{cell}} = 0 \quad \text{versus} \quad H_{1b} : \mathbf{B}_b^{\text{cell}} \neq 0.$$

Mouse effects absorb condition and every shift shared by cell types from the same mouse.

For $R = 500 \in \mathbb{N}$ permutations and $r \in \{1, \dots, R\}$, cell-type labels are shuffled only among pseudobulks from the same mouse, producing $X_{1b}^{(r)} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ and statistic $\Lambda_b^{(r)} \in \mathbb{R}_+$. Let $\varepsilon_b = 10^{-6} \max(1, |\Lambda_b|) \in \mathbb{R}_+$. The finite-permutation p-value is

$$p_b = \frac{1 + \sum_{r=1}^R \mathbb{I}\{\Lambda_b^{(r)} \geq \Lambda_b - \varepsilon_b\}}{R + 1} \in (0, 1], \quad R = 500.$$

The null coefficients and $\hat{\kappa}_{0b}$ are reused, but each permuted alternative is optimized. The tolerance counts numerically equal likelihood optima as ties. Benjamini–Hochberg correction is applied across canonical partitions with at least 30 pseudobulks.

Joint condition tests

ACAT combines $J \in \mathbb{N}$ component p-values $p_j \in (0, 1)$, $j = 1, \dots, J$, into the scalar

$$p_{\text{ACAT}} = \frac{1}{2} - \frac{1}{\pi} \arctan \left[\frac{1}{J} \sum_{j=1}^J \tan\{\pi(1/2 - p_j)\} \right] \in (0, 1).$$

One level combines cell-type-specific p-values for a path partition; a second combines distinct partitions within a gene.

The shared-effect Gaussian model for block b uses $M \in \mathbb{N}$ mice indexed by $m \in \{1, \dots, M\}$, and $L_m \in \mathbb{N}$ cell types for mouse m , indexed by $c \in \{1, \dots, L_m\}$. Its response is $\mathbf{y}_{bmc} \in \mathbb{R}^{d_b}$, design row $X_{bmc} \in \mathbb{R}^{1 \times k}$, where $k \in \mathbb{N}$, coefficient matrix $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$, mouse random effect $\mathbf{u}_{bm} \in \mathbb{R}^{d_b}$, and residual $\mathbf{e}_{bmc} \in \mathbb{R}^{d_b}$:

$$\mathbf{y}_{bmc} = (X_{bmc} \mathbf{B}_b)' + \mathbf{u}_{bm} + \mathbf{e}_{bmc}.$$

The random terms satisfy $E(\mathbf{u}_{bm}) = E(\mathbf{e}_{bmc}) = \mathbf{0}_{d_b}$, $\text{Var}(\mathbf{u}_{bm}) = \tau_{\text{mouse}}^2 I_{d_b}$, and $\text{Var}(\mathbf{e}_{bmc}) = V_{bmc} + \tau_{\text{resid}}^2 I_{d_b}$. They are independent across mice, and residuals are independent across cell types conditional on $\mathbf{u}_{bm}$. Hence, for $c \neq c'$,

$$\text{Cov}(\mathbf{y}_{bmc}, \mathbf{y}_{bmc'}) = \tau_{\text{mouse}}^2 I_{d_b}, \quad \text{Var}(\mathbf{y}_{bmc}) = V_{bmc} + (\tau_{\text{mouse}}^2 + \tau_{\text{resid}}^2) I_{d_b},$$

where $V_{bmc} \in \mathbb{S}_+^{d_b}$, and $\tau_{\text{mouse}}^2, \tau_{\text{resid}}^2 \in \mathbb{R}_+$ are shared-mouse and residual variance components estimated by two-component REML under the no-condition model. Condition labels are permuted once per mouse and applied to all its cell types.

Multinomial-logit GLMM

The formal generalized linear mixed model uses the same shared-condition design as the joint Gaussian model but retains the multinomial response. Let $\tilde{\mathbf{x}}_{bmc} \in \mathbb{N}_0^{S_b}$ be integer path counts with $\tilde{n}_{bmc} = \mathbf{1}_{S_b}' \tilde{\mathbf{x}}_{bmc} \in \mathbb{N}$. They are obtained from the fractional effective counts $\mathbf{x}_{bmc}$ by rounding n_{bmc}^{eff} to the nearest positive integer and distributing the remaining molecules to paths by largest fractional remainder. This makes the conditional response a proper multinomial while changing each effective total by at most one.

For design row $X_{bmc} \in \mathbb{R}^{1 \times k}$, fixed coefficients $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$, and mouse random intercept $\mathbf{u}_{bm} \in \mathbb{R}^{d_b}$, define

$$\boldsymbol{\xi}_{bmc} = X_{bmc} \mathbf{B}_b + \mathbf{u}_{bm}' \in \mathbb{R}^{1 \times d_b}, \quad \boldsymbol{\mu}_{bmc} = \text{softmax}\{(\boldsymbol{\xi}_{bmc}, 0)\} \in \Delta^{S_b-1}.$$

The final path is the reference category. The model is

$$\tilde{\mathbf{x}}_{bmc} \mid \mathbf{u}_{bm} \sim \text{Multinomial}(\tilde{n}_{bmc}, \boldsymbol{\mu}_{bmc}), \quad \mathbf{u}_{bm} \sim N(\mathbf{0}_{d_b}, \sigma_b^2 I_{d_b}),$$

where $\sigma_b^2 \in \mathbb{R}_+$ is re-estimated under both null and alternative. An isotropic random-effect covariance is used because some blocks have few mice; an unstructured $d_b \times d_b$ covariance would introduce $d_b(d_b + 1)/2$ variance parameters per block.

Let $\ell_{bm}(\mathbf{u}; \mathbf{B}_b, \sigma_b) \in \mathbb{R}$ be mouse m 's conditional multinomial log likelihood plus its Gaussian log density, and let

$$\hat{\mathbf{u}}_{bm} = \arg \max_{\mathbf{u} \in \mathbb{R}^{d_b}} \ell_{bm}(\mathbf{u}; \mathbf{B}_b, \sigma_b), \quad H_{bm} = - \frac{\partial^2 \ell_{bm}}{\partial \mathbf{u} \partial \mathbf{u}'} \bigg|_{\hat{\mathbf{u}}_{bm}} \in \mathbb{S}_+^{d_b}.$$

The Laplace log marginal likelihood, up to constants common to nested models, is

$$\ell_b^{\text{Lap}}(\mathbf{B}_b, \sigma_b) = \sum_{m=1}^M \left[\ell_{bm}(\hat{\mathbf{u}}_{bm}; \mathbf{B}_b, \sigma_b) + \frac{d_b}{2} \log(2\pi) - \frac{1}{2} \log |H_{bm}| \right] \in \mathbb{R}.$$

The implementation differentiates this approximation through 30 damped Newton updates for every $\hat{\mathbf{u}}_{bm}$. The null contains cell-type intercepts; the alternative appends r_b condition columns. The likelihood-ratio statistic has nominal degrees of freedom $r_b d_b$.

Additional EC likelihoods and inference approximations

The path-count GLMM above treats estimated effective path counts as its response. The following restatement of the observed-EC model supplies context for the additional Dirichlet–multinomial, Laplace, CAVI, Monte Carlo, and Rényi inference comparisons. For gene g , let $T_g \in \mathbb{N}$ be its

number of supported isoforms, $d_g = T_g - 1$, $I_g \in \mathbb{N}$ the number of retained pseudobulks, $M_g \in \mathbb{N}$ the number of mice, and $P = 2$ the primer count. Primer p has $K_{gp} \in \mathbb{N}$ gene-assigned ECs. Define the observed counts $\mathbf{c}_{gip} \in \mathbb{N}_0^{K_{gp}}$, total $n_{gip} = \mathbf{1}'_{K_{gp}} \mathbf{c}_{gip} \in \mathbb{N}_0$, and fixed compatibility or conditional-weight matrix $A_{gp} \in \mathbb{R}_+^{K_{gp} \times T_g}$. Unsupported annotated isoforms and ECs not uniquely assigned to g are excluded.

Let $X_g \in \mathbb{R}^{I_g \times q}$ be the fixed-effect design, $B_g \in \mathbb{R}^{q \times d_g}$ its coefficient matrix, and $u_{gm} \in \mathbb{R}^{d_g}$ mouse m 's random isoform-logit intercept. With mouse index $m(i) \in \{1, \dots, M_g\}$, the reference-isoform parameterization is

$$\eta_{gi} = \{X_{gi}B_g + u'_{g,m(i)}, 0\} \in \mathbb{R}^{T_g}, \quad u_{gm} \sim N(0, \sigma_g^2 I_{d_g}).$$

The EC probability vector is $\mathbf{q}_{gip}(\boldsymbol{\eta}_{gi})$ defined at the start of this appendix. Thus the two primers share B_g and u_{gm} , but retain distinct EC maps and positional-coverage weights. The conditional model is either

$$\mathbf{c}_{gip} \mid u_{g,m(i)} \sim \text{Multinomial}(n_{gip}, \mathbf{q}_{gip})$$

or

$$\mathbf{c}_{gip} \mid u_{g,m(i)} \sim \text{DirichletMultinomial}(n_{gip}, \kappa_g \mathbf{q}_{gip}), \quad \kappa_g \in \mathbb{R}_+.$$

The latter adds pseudobulk-level EC overdispersion. It is a direct Dirichlet–multinomial over EC categories, not a Dirichlet model over latent isoform allocations.

An alternative overdispersion model retains the multinomial observation but adds an isotropic observation-level logistic-normal effect. For pseudobulk i , let $e_{gi} \in \mathbb{R}^{d_g}$ and replace the linear predictor by

$$\eta_{gi} = \{X_{gi}B_g + u'_{g,m(i)} + e'_{gi}, 0\}, \quad e_{gi} \sim N(0, \tau_g^2 I_{d_g}), \quad \tau_g \in \mathbb{R}_+.$$

The same e_{gi} enters both primer likelihoods, because it perturbs the underlying isoform composition rather than each primer's EC probabilities independently. Consequently this model allows correlated extra-multinomial variation across the two primer halves. It is a logistic-normal analogue of the Dirichlet–multinomial, not an identical distribution: its dispersion is isotropic in reference-logit coordinates, whereas the Dirichlet–multinomial has one concentration on the EC simplex.

Laplace inference maximizes the joint conditional density over $u_g \in \mathbb{R}^{M_g \times d_g}$. Its negative Hessian satisfies $H_g \in \mathbb{S}_+^{M_g d_g}$, and the marginal approximation adds the correction $-\frac{1}{2} \log |H_g|$. The outer fit optimizes B_g , σ_g , and, for the overdispersed model, κ_g .

Let $\boldsymbol{\theta}_g \in \mathbb{R}^{p_g}$ be the outer parameter vector and $\hat{\mathbf{u}}_{gm} \in \mathbb{R}^{d_g}$ the cluster mode. The mode score is $\mathbf{s}_{gm} = \partial \ell_{gm} / \partial \mathbf{u}_{gm} \in \mathbb{R}^{d_g}$, and the positive mode Hessian is $H_{gm} = -\partial \mathbf{s}_{gm} / \partial \mathbf{u}_{gm} \in \mathbb{S}_{++}^{d_g}$. The fitted mode satisfies $\mathbf{s}_{gm}(\hat{\mathbf{u}}_{gm}, \boldsymbol{\theta}_g) = \mathbf{0}_{d_g}$. Rather than retaining and differentiating through every Newton update, the optimized implementation differentiates this stationarity equation implicitly,

$$\frac{\partial \hat{\mathbf{u}}_{gm}}{\partial \boldsymbol{\theta}_g} = H_{gm}^{-1} \frac{\partial \mathbf{s}_{gm}}{\partial \boldsymbol{\theta}_g} \in \mathbb{R}^{d_g \times p_g}.$$

For a reverse-mode cotangent $\mathbf{v}_{gm} \in \mathbb{R}^{d_g}$, it solves one $d_g \times d_g$ system with H_{gm} and applies the score vector–Jacobian product. This gives the derivative of the same Laplace objective at a converged mode without backpropagating through the 12 Newton updates. Fits that fail the joint outer-gradient and mode-score criteria receive a 30-update continuation. Compiled objectives are reused only among contiguous tests with the same gene, pseudobulk set, and tensor shape; counts and block-specific fixed-effect tensors are dynamic arguments, and the cache is released before the next group.

Nonbootstrap Laplace inference

The block test uses the same fixed-effect tensors as the primary variational fit. Write the null tensor as $Z_{0b} \in \mathbb{R}^{I_g \times d_g \times p_{0b}}$ and the alternative as $Z_{1b} \in \mathbb{R}^{I_g \times d_g \times p_{1b}}$, where the first p_{0b} slices of Z_{1b} equal Z_{0b} . The remaining $q_b = p_{1b} - p_{0b} = r_b d_b^{\text{path}}$ coefficients are exactly the tested block-path interactions. For coefficient vector $\beta_b \in \mathbb{R}^{p_{jb}}$, the fixed nonreference-isoform logits for pseudobulk i are

$$\eta_{gbi}^{\text{fix}} = Z_{jb,i,\cdot,\cdot} \beta_b \in \mathbb{R}^{d_g}, \quad j \in \{0, 1\}.$$

Thus the likelihood still contains every supported gene isoform and EC; only the tested fixed-effect subspace is block specific.

Let $\hat{\ell}_{0b}^{\text{Lap}}$ and $\hat{\ell}_{1b}^{\text{Lap}}$ be the maximized Laplace marginal log likelihoods, including re-estimation of all shared variance parameters. The nonbootstrap likelihood-ratio statistic and nominal reference distribution are

$$D_b = 2\{\hat{\ell}_{1b}^{\text{Lap}} - \hat{\ell}_{0b}^{\text{Lap}}\}, \quad D_b \sim \chi_{q_b}^2.$$

The alternative starts from the null estimate with its q_b added coefficients set to zero. A BIC approximation to the alternative-versus-null Bayes factor uses the number M_g of independent mice rather than the number of pseudobulks:

$$\log \widehat{\text{BF}}_{10,b}^{\text{BIC}} = \frac{1}{2}\{D_b - q_b \log M_g\} \in \mathbb{R}.$$

This is not a prior-integrated Bayes factor. Both calculations require only the null and alternative fits, compared with one null and one alternative fit for every parametric-bootstrap replicate. Their calibration can fail when the number of mice is small, a variance estimate lies on its boundary, or the Laplace outer optimizer is inaccurate. They are therefore compared with the completed mouse bootstrap analysis before being used as a replacement for empirical calibration.

For variational inference, use the diagonal Gaussian family

$$Q(u_g) = \prod_{m=1}^{M_g} \prod_{j=1}^{d_g} N(u_{gmj}; \mu_{gmj}, s_{gmj}^2), \quad \mu_g, s_g \in \mathbb{R}^{M_g \times d_g}, \quad s_{gmj} > 0.$$

For the multinomial likelihood, latent EC-to-isoform responsibilities $r_{gipkt} \in [0, 1]$, with $\sum_t r_{gipkt} = 1$ on the support of A_{gpkt} , give the optimized lower bound

$$E_Q \left[\log \sum_t A_{gpkt} e^{\eta_{git}} \right] \geq \log \sum_t A_{gpkt} e^{E_Q \eta_{git}}.$$

The normalizer needs an upper bound. For one observation and primer, let $m_{gipt} = E_Q \eta_{git} + \log \sum_k A_{gpkt}$, $v_{gipt} = \text{Var}_Q(\eta_{git})$, and $a_{gip} \in \Delta^{T_g-1}$. The tilted softmax bound of Knowles and Minka is

$$E_Q \log \sum_{t=1}^{T_g} e^{\eta_{git} + \log \sum_k A_{gpkt}} \leq \frac{1}{2} \sum_t a_{gipt}^2 v_{gipt} + \log \sum_t \exp \left\{ m_{gipt} + \frac{1 - 2a_{gipt}}{2} v_{gipt} \right\},$$

where the local coordinate satisfies

$$a_{gip} = \text{softmax}\{m_{gip} + \frac{1}{2}(1 - 2a_{gip}) \odot v_{gip}\}.$$

Alternating the analytic responsibilities, this local fixed point, and the global Gaussian parameters maximizes a tractable lower bound on the multinomial evidence. The cited method is non-conjugate

variational message passing rather than conjugate CAVI; here “tilted CAVI” refers only to this block-coordinate implementation.

The full-covariance extension factorizes only across mice. If mouse m has $q_{gm} \in \mathbb{N}$ observed pseudobulks, define its complete latent block

$$z_{gm} = (u'_{gm}, e'_{gi_1}, \dots, e'_{gi_{q_{gm}}})' \in \mathbb{R}^{L_{gm}}, \quad L_{gm} = (q_{gm} + 1)d_g,$$

where the e components are omitted in the model without observation noise. Its prior covariance is diagonal,

$$P_{gm} = \text{diag}(\sigma_g^2 I_{d_g}, \tau_g^2 I_{q_{gm}d_g}) \in \mathbb{S}_+^{L_{gm}},$$

but its posterior approximation is unrestricted within mouse,

$$Q(z_g) = \prod_{m=1}^{M_g} N(z_{gm}; \mu_{gm}, \Sigma_{gm}), \quad \mu_{gm} \in \mathbb{R}^{L_{gm}}, \quad \Sigma_{gm} \in \mathbb{S}_+^{L_{gm}}.$$

Thus full covariance describes posterior dependence among isoform logits and between a shared mouse effect and its pseudobulk residuals; it does not add an unstructured covariance parameter to the generative model.

For a block-specific cell-type test, the EC likelihood still contains all T_g supported isoforms of gene g ; the response is not reduced to path counts. Let $C_b^{\text{cell}} \in \mathbb{N}$ be the number of represented cell types, $Z_b \in \mathbb{R}^{I_g \times q_0}$ the one-hot condition design, and $W_b \in \mathbb{R}^{I_g \times (C_b^{\text{cell}} - 1)}$ the treatment-coded cell-type design. The pooled-baseline path Jacobian $J_b \in \mathbb{R}^{d_b \times d_g}$, its right inverse $H_b = J_b'(J_b J_b')^{-1} \in \mathbb{R}^{d_g \times d_b}$, and the kernel basis $N_b \in \mathbb{R}^{d_g \times (d_g - d_b)}$ are defined in the main methods. The free-logit predictor is

$$\eta_{gi,-} = Z_{bi} B_{g0} + \sum_{j=1}^{C_b^{\text{cell}} - 1} W_{bij} (N_b \alpha_{bj} + H_b \gamma_{bj}) + u_{g,m(i)} + e_{gi} \in \mathbb{R}^{d_g},$$

where $B_{g0} \in \mathbb{R}^{q_0 \times d_g}$ contains unrestricted condition effects, $\alpha_{bj} \in \mathbb{R}^{d_g - d_b}$ contains cell-type nuisance effects tangent to constant path usage, and $\gamma_{bj} \in \mathbb{R}^{d_b}$ is cell type j 's block-path contrast. The e_{gi} term is present only in the logistic-normal sensitivity model. The null estimates every α_{bj} but sets all $\gamma_{bj} = \mathbf{0}_{d_b}$; the alternative has $(C_b^{\text{cell}} - 1)d_b$ additional coefficients. The combined columns of N_b and H_b span all d_g isoform-logit directions, so the alternative permits unrestricted cell-type isoform composition while the null removes the local path derivative at the pooled baseline. This construction makes the EC GLMM a per-block test while preserving gene-wide ambiguity and nuisance mass.

For pseudobulk i , let $R_{gi} \in \{0, 1\}^{d_g \times L_{g,m(i)}}$ select and sum its mouse and pseudobulk effects. Its free-logit covariance is $R_{gi} \Sigma_{g,m(i)} R_{gi}' \in \mathbb{S}_+^{d_g}$; append a zero row and column for the reference isoform to obtain $V_{gi} \in \mathbb{S}_+^{T_g}$. The correlated tilted bound replaces the variance vector in the diagonal expression. For $m \in \mathbb{R}^{T_g}$, $V \in \mathbb{S}_+^{T_g}$, and $a \in \Delta^{T_g - 1}$,

$$E \log \sum_t e^{\eta_t} \leq \frac{1}{2} a' V a + \log \sum_t \exp\{m_t + \frac{1}{2} V_{tt} - (V a)_t\}, \quad a = \text{softmax}\{m + \frac{1}{2} \text{diag}(V) - V a\}.$$

The Gaussian KL term is analytic:

$$\text{KL}(Q_m \| P_m) = \frac{1}{2} [\text{tr}(P_m^{-1} \Sigma_m) + \mu_m' P_m^{-1} \mu_m - L_{gm} + \log |P_m| - \log |\Sigma_m|].$$

Each positive-definite Σ_m is represented by an unconstrained lower Cholesky factor with exponentiated diagonal. The full fit warm-starts from a diagonal variational posterior by copying its means and marginal variances and setting new off-diagonal entries to zero.

The direct EC Dirichlet–multinomial contains $\log \Gamma\{\kappa_g q_{gipk}(u) + c_{gipk}\}$, so the tilted softmax bound does not yield coordinate updates for it. Both likelihood families are therefore also fit with reparameterized Monte Carlo VI. If $w(u) = p(c, u)/Q(u)$, the optimized objective is the ELBO

$$\mathcal{L}_1 = E_Q \log w(u)$$

or the variational Rényi bound of order $\alpha \in (0, 1)$,

$$\mathcal{L}_\alpha = \frac{1}{1 - \alpha} \log E_Q \{w(u)^{1-\alpha}\}.$$

The comparison uses $\alpha = 0.5$, antithetic Gaussian draws, and the same diagonal or full Gaussian family. Conditional on B_g, σ_g, κ_g , both the likelihood and Q factor by mouse. Writing the corresponding ratio as $w_m(u_{gm})$, the implementation evaluates the equivalent expression

$$\mathcal{L}_\alpha = \sum_{m=1}^{M_g} \frac{1}{1 - \alpha} \log E_{Q_m} \{w_m(u_{gm})^{1-\alpha}\}.$$

This avoids the exponential importance-weight degeneracy caused by forming one product weight over all mice. Rényi VI is not called CAVI because its importance-weighted objective has no coordinate update under this model. For the full posterior with observation-level noise, directly optimizing a fixed set of Monte Carlo draws is ill-conditioned: the Cholesky factor can overfit sample correlations when L_{gm} is not small relative to the draw count. This variant is therefore fit with the analytic tilted objective; fresh antithetic draws evaluate its exact-likelihood ELBO and Rényi bound.

The block statistic is twice the optimized alternative-minus-null variational objective. Its reference distribution is empirical rather than χ^2 , because it compares optimized evidence bounds and can inherit approximation error. A synchronized label permutation is invalid here because the block null permits real cell-type nuisance effects tangent to constant path usage. Instead, a parametric bootstrap simulates mouse effects and optional pseudobulk effects from the fitted block null, generates paired-primer EC counts with the observed primer-specific pseudobulk totals, and refits both nested models. Each observed statistic is compared with a leave-block-out pool of bootstrap statistics matched by likelihood family and tested dimension $(C_b^{\text{cell}} - 1)d_b$. Tested dimensions represented by fewer than 20 blocks are combined into a rare-dimension stratum within the same likelihood. Matching gene-depth quartiles fragmented the empirical tail enough to limit BH resolution in the genome-wide analysis. Depth is therefore an independent calibration audit rather than a pooling variable. With null pool $\mathcal{N}_{-b}$,

$$p_b = \frac{1 + \sum_{z \in \mathcal{N}_{-b}} \mathbf{1}\{z \geq T_b\}}{1 + |\mathcal{N}_{-b}|}.$$

Every bootstrap statistic is calibrated by the same leave-block-out rule. Benjamini–Hochberg values are reported for a likelihood only if these calibrated null p -values reject between 2.5% and 7.5% at level 0.05, both in aggregate and separately within each gene-depth quartile.

Liang–Zeger multinomial GEE

GEE specifies only the marginal mean and first two moments. It can therefore use the fractional effective counts directly. Define $\mathbf{p}_{bmc} = \mathbf{x}_{bmc}/n_{bmc}^{\text{eff}} \in \Delta^{S_b-1}$, and let $\mathbf{p}_{bmc,-} \in \mathbb{R}^{d_b}$ omit the reference path. Its marginal mean uses the same reference-logit regression,

$$\boldsymbol{\mu}_{bmc} = \text{softmax}\{(X_{bmc}\mathbf{B}_b, 0)\}, \quad G_{bmc} = \text{diag}(\boldsymbol{\mu}_{bmc,-}) - \boldsymbol{\mu}_{bmc,-}\boldsymbol{\mu}_{bmc,-}' \in \mathbb{S}_+^{d_b}.$$

The working covariance of one proportion vector is $A_{bmc} = G_{bmc}/n_{bmc}^{\text{eff}} \in \mathbb{S}_+^{d_b}$. Let $L_{bmc} \in \mathbb{R}^{d_b \times d_b}$ be its symmetric square root, and let

$$D_{bmc} = \frac{\partial \boldsymbol{\mu}_{bmc,-}}{\partial \text{vec}(\mathbf{B}_b)'} = [X_{bmc,1}G_{bmc} \ \cdots \ X_{bmc,k}G_{bmc}] \in \mathbb{R}^{d_b \times kd_b}.$$

For mouse m , stack its $q_m \in \mathbb{N}$ observed cell types into $\mathbf{p}_{bm,-} \in \mathbb{R}^{q_m d_b}$, mean $\boldsymbol{\mu}_{bm,-} \in \mathbb{R}^{q_m d_b}$, and derivative $D_{bm} \in \mathbb{R}^{q_m d_b \times kd_b}$. With $L_{bm} = \text{blockdiag}_c(L_{bmc})$ and exchangeable $R_m(\alpha_b) \in \mathbb{R}^{q_m \times q_m}$, the Liang–Zeger working covariance is

$$W_{bm} = \phi_b L_{bm} \{R_m(\alpha_b) \otimes I_{d_b}\} L_{bm}' \in \mathbb{S}_+^{q_m d_b}.$$

The scale $\phi_b \in \mathbb{R}_+$ and working correlation $\alpha_b \in (-1/(q_{\max} - 1), 1)$ are updated from whitened Pearson residuals. Coefficients solve

$$\sum_{m=1}^M D_{bm}' W_{bm}^{-1} (\mathbf{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) = \mathbf{0}_{kd_b}.$$

If $\mathbf{U}_{bm} = D_{bm}' W_{bm}^{-1} (\mathbf{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) \in \mathbb{R}^{kd_b}$ and $\mathcal{H}_b = \sum_m D_{bm}' W_{bm}^{-1} D_{bm} \in \mathbb{S}_+^{kd_b}$, the canonical cluster-robust covariance is

$$\widehat{\text{Var}}\{\text{vec}(\hat{\mathbf{B}}_b)\} = \mathcal{H}_b^+ \left(\sum_{m=1}^M \mathbf{U}_{bm} \mathbf{U}_{bm}' \right) \mathcal{H}_b^+ \in \mathbb{S}_+^{kd_b}.$$

The condition coefficients are tested by a robust Wald statistic. No small-sample sandwich correction is imposed; synchronized mouse-level permutations assess whether the canonical large- M reference is adequate.

Matched power benchmark

Power is evaluated only for partitions represented by exactly two conditions. For replicate r , condition labels are first permuted among mice. Let $z_{bmc}^{(r)} \in \{0, 1\}$ be the resulting non-reference condition indicator and let $\mathbf{v}_b \in \mathbb{R}^{d_b}$ be a reproducible random unit vector. For ILR effect $\gamma \in \{0.25, 0.5\}$, define

$$\mathbf{y}_{bmc}^{(r,\gamma)} = \mathbf{y}_{bmc} + \gamma z_{bmc}^{(r)} \mathbf{v}_b, \quad \boldsymbol{\psi}_{bmc}^{(r,\gamma)} = \text{softmax}\{Q_b \mathbf{y}_{bmc}^{(r,\gamma)}\}.$$

The shifted effective counts are $n_{bmc}^{\text{eff}} \boldsymbol{\psi}_{bmc}^{(r,\gamma)}$. Thus every method sees the same imposed ILR effect, observed effective depth, cell-type pattern, and mouse clustering. For each simulated alternative, null-calibrated power uses the finite-permutation rank of its statistic among synchronized null statistics for the same block and method. This event-matched calibration preserves dimension, depth, and cluster structure.

D Supporting Algorithms

All algorithms operate independently over canonical blocks unless a joint gene-level combination is stated. “Fit” below means bounded L-BFGS with analytic gradients and a fourfold iteration-limit retry after a warm start from the first endpoint.

Algorithm 2 Construct splice blocks and path maps

Require: K_g annotated spliced isoforms with exon chains; retained set $\mathcal{I}_g$, $|\mathcal{I}_g| = T_g$

Ensure: Blocks b , maps $C_b \in \{0, 1\}^{S_b \times T_g}$, and bases $Q_b \in \mathbb{R}^{S_b \times d_b}$

- 1: Split the gene at every annotated exon boundary into atomic intervals.
 - 2: Merge adjacent intervals exonic in every annotated isoform into constitutive anchors.
 - 3: Form internal blocks between consecutive anchors and terminal blocks using pseudo-anchors.
 - 4: **for all** candidate blocks b **do**
 - 5: Record each annotated isoform’s ordered intervening exon chain.
 - 6: Assign identical chains to one path and restrict to $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$.
 - 7: Remap represented path labels to $\{1, \dots, S_b\}$.
 - 8: **if** $S_b < 2$ **then**
 - 9: discard b
 - 10: **else**
 - 11: Set $(C_b)_{s_b(t), t} = 1$ for $t \in \mathcal{I}_b^{\text{sp}}$; leave nuisance columns zero.
 - 12: Construct the Helmert basis Q_b .
 - 13: **end if**
 - 14: **end for**
 - 15: Canonicalize the unordered isoform partition and retain one representative of equivalent blocks.
-

Algorithm 3 Fit one pseudobulk’s local path composition

Require: Counts $c_{bip} \in \mathbb{N}_0^{E_{bp}}$, designs $A_{bp} \in \mathbb{R}_+^{E_{bp} \times T_g}$, baseline $\theta_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$, C_b , and Q_b

Ensure: $\hat{\theta}_{bi}$, $\hat{\psi}_{bi}$, $\hat{y}_{bi}$, and V_{bi}^{cond}

- 1: Initialize $\delta_{bi} \leftarrow \mathbf{0}_{d_b}$.
 - 2: **if** $S_b = 2$ **then**
 - 3: Evaluate the fixed log-ratio grid and retain its best point.
 - 4: **end if**
 - 5: **function** OBJECTIVE(δ)
 - 6: Form $\theta_{bi}(\delta)$, preserving spliced mass and within-path isoform mixtures.
 - 7: **return** the negative sum of the $P = 2$ multinomial log likelihoods and its analytic gradient.
 - 8: **end function**
 - 9: Minimize OBJECTIVE(δ) by bounded L-BFGS on $[-20, 20]^{d_b}$.
 - 10: Set $\hat{\theta}_{bi} \leftarrow \theta_{bi}(\hat{\delta}_{bi})$.
 - 11: Set $\hat{\psi}_{bi} \leftarrow C_b \hat{\theta}_{bi} / (\mathbf{1}_{S_b}' C_b \hat{\theta}_{bi})$.
 - 12: Set $\hat{y}_{bi} \leftarrow Q_b' \log \hat{\psi}_{bi}$.
 - 13: Compute $I_{\delta, bi}$ by summing primer expected information.
 - 14: Apply the eigenvalue/null-space check to $(I_{\delta, bi}, I_{d_b})$; if identifiable, set $V_{bi}^{\text{cond}} = I_{\delta, bi}^{-1}$, otherwise set it to infinity.
-

Algorithm 4 Compute full-isoform nuisance (profile) covariance

Require: $\hat{\theta}_{bi} \in \mathbb{R}_+^{T_g}$, A_{bp} , N_{bip} , C_b , Q_b , relative threshold $r_{\text{eig}} = 10^{-8}$, and null tolerance $a_{\text{null}} = 10^{-9}$

Ensure: $V_{bi}^{\text{prof}} \in \mathbb{S}_+^{d_b}$ and an identifiability flag

- 1: **for** $p = 1, \dots, P$ **do**
 - 2: Compute $\hat{q}_{bip}$, $\hat{Z}_{bip}$, and $I_{\eta,bip} \in \mathbb{S}_+^{T_g}$.
 - 3: **end for**
 - 4: Set $I_{\eta,bi} \leftarrow \sum_p I_{\eta,bip}$.
 - 5: Compute path masses a_{bis} , spliced mass ω_{bi}^{sp} , and $R_{bi} = \partial \log \psi_{bi} / \partial \eta'_{bi}$.
 - 6: Set $J_{bi} \leftarrow Q'_b R_{bi}$.
 - 7: Eigendecompose $I_{\eta,bi} = U_{bi} \text{diag}(\lambda_{bi}) U'_{bi}$.
 - 8: Put t in $\mathcal{P}_{bi}$ when $\lambda_{bit} > r_{\text{eig}} \max_u |\lambda_{biu}|$, and put all other indices in $\mathcal{N}_{bi}$.
 - 9: **if** $\|J_{bi} U_{bi,0}\|_F > a_{\text{null}} \max\{\|J_{bi}\|_F, 1\}$ **then**
 - 10: Set $V_{bi}^{\text{prof}} \leftarrow \infty_{d_b \times d_b}$ and mark the response unidentified.
 - 11: **else**
 - 12: Set $V_{bi}^{\text{prof}} \leftarrow J_{bi} U_{bi,+} \text{diag}(\lambda_{bi,+}^{-1}) U'_{bi,+} J'_{bi}$.
 - 13: Mark the response identifiable.
 - 14: **end if**
-

Algorithm 5 Fit a cell-resolved pseudobulk path composition

Require: Cell EC vectors c_{bihp} , baselines $\theta_{bih}^{(0)}$, weights w_{bih} , and A_{bp} , C_b , Q_b

Ensure: One pseudobulk response $\hat{y}_{bi}$ and covariance V_{bi}

- 1: Allocate one shared $\delta_{bi} \in \mathbb{R}^{d_b}$.
 - 2: **for all** cells h and primers p **do**
 - 3: Perturb the cell's baseline, normalize its EC probabilities, and add its multinomial log likelihood and gradient.
 - 4: **end for**
 - 5: Optimize the summed objective over δ_{bi} .
 - 6: Compute each fitted $\hat{\psi}_{bih}$ and their normalized w_{bih} -weighted mean $\hat{\psi}_{bi}$.
 - 7: Sum cell-by-primer Fisher matrices and propagate their inverse through the weighted composition response to obtain V_{bi} .
 - 8: Set $\hat{y}_{bi} = Q'_b \log \hat{\psi}_{bi}$.
-

Algorithm 6 Gaussian condition or paired cell-type test

Require: Responses $\hat{\mathbf{y}}_{bi}$, covariances V_{bi} , mouse, condition, and cell-type labels; test type

Ensure: One Wald p-value for block b

- 1: **if** testing condition within a cell type **then**
 - 2: Retain conditions represented by at least three mice.
 - 3: Build full design $X_b \in \mathbb{R}^{n \times k}$ and reduced design $X_{0b} \in \mathbb{R}^{n \times (k-r_b)}$.
 - 4: **else**
 - 5: Build mouse-effect design $X_{0b} \in \{0, 1\}^{n_b \times M_b}$.
 - 6: Append $L_b - 1$ cell-type columns to obtain $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$.
 - 7: **end if**
 - 8: Require full column rank in both designs.
 - 9: Estimate τ_b^2 by REML under the reduced design using $V_{bi} + \tau_b^2 I_{d_b}$.
 - 10: Hold $\hat{\tau}_b^2$ fixed and fit the full design by GLS.
 - 11: Stack the tested coefficients and compute their Wald statistic W_b .
 - 12: Compare W_b with the applicable chi-squared reference.
 - 13: Calibrate by mouse-level condition permutations or within-mouse cell-type permutations, as appropriate.
-

Algorithm 7 Paired compositional cell-type test

Require: $\hat{\psi}_{bi}$, V_{bi}^{cond} , mouse and cell-type labels, and $R = 500$

Ensure: One exact permutation p-value p_b per canonical block

- 1: Keep converged, identifiable estimates satisfying the path floor and the three-mouse-per-cell-type requirement.
 - 2: Convert each $(\hat{\psi}_{bi}, V_{bi}^{\text{cond}})$ to n_{bi}^{eff} and $\mathbf{x}_{bi}$.
 - 3: Build X_{0b} from mouse effects and X_{1b} by appending cell-type effects.
 - 4: Fit the null coefficients and $\hat{\kappa}_{0b}$.
 - 5: Fit the observed alternative and compute Λ_b .
 - 6: **if** using fixed concentration **then**
 - 7: Set $\kappa_{1b} \leftarrow \hat{\kappa}_{0b}$.
 - 8: **else**
 - 9: Optimize $\hat{\kappa}_{1b}$ and compare the finite optimum with the exact multinomial endpoint.
 - 10: **end if**
 - 11: **for** $r = 1, \dots, R$ **do**
 - 12: Shuffle cell-type labels within each mouse to form $X_{1b}^{(r)}$.
 - 13: Reuse the null fit, refit the alternative, and store $\Lambda_b^{(r)}$.
 - 14: **end for**
 - 15: Compute the tolerance-aware finite-permutation rank p_b .
 - 16: Require $n_b \geq 30$, then apply BH across canonical blocks.
-

Algorithm 8 Liang–Zeger multinomial GEE condition test

Require: Fractional counts $\mathbf{x}_{bmc}$, design X_b , mouse labels, and tested condition columns

Ensure: Robust Wald statistic and p-value

- 1: Initialize $\mathbf{B}_b$ from independent multinomial regression and set $\alpha_b \leftarrow 0$, $\phi_b \leftarrow 1$.
 - 2: **repeat**
 - 3: Compute $\boldsymbol{\mu}_{bmc}$, G_{bmc} , A_{bmc} , and D_{bmc} for every observation.
 - 4: Whiten residuals by $A_{bmc}^{-1/2}$.
 - 5: Update ϕ_b from squared Pearson residuals and α_b from within-mouse Pearson cross-products.
 - 6: Construct every W_{bm} , quasi-score $\mathbf{U}_{bm}$, and sensitivity matrix $\mathcal{H}_b$.
 - 7: Update $\text{vec}(\mathbf{B}_b) \leftarrow \text{vec}(\mathbf{B}_b) + \mathcal{H}_b^+ \sum_m \mathbf{U}_{bm}$.
 - 8: **until** the maximum coefficient update is at most 10^{-8}
 - 9: Form the cluster sandwich covariance $\mathcal{H}_b^+ (\sum_m \mathbf{U}_{bm} \mathbf{U}_{bm}') \mathcal{H}_b^+$.
 - 10: Compute the robust Wald test for all condition coefficients.
 - 11: Permute condition once per mouse and repeat the complete GEE fit.
-

Algorithm 9 Laplace multinomial GLMM condition test

Require: Effective counts, nested designs, mouse labels, and random-effect dimension d_b

Ensure: Likelihood-ratio statistic and p-value

- 1: Integerize each effective count vector by largest remainder.
 - 2: **for all** null and alternative models **do**
 - 3: Initialize fixed coefficients from multinomial regression and initialize σ_b .
 - 4: **repeat**
 - 5: **for all** mice m **do**
 - 6: Find $\hat{\mathbf{u}}_{bm}$ with 30 damped Newton updates.
 - 7: Compute random-effect Hessian H_{bm} and its log determinant.
 - 8: **end for**
 - 9: Sum the Laplace log marginal likelihood and differentiate through the Newton updates.
 - 10: Update $\mathbf{B}_b$ and $\log \sigma_b$ by bounded L-BFGS.
 - 11: **until** the outer gradient tolerance or iteration limit is reached
 - 12: **end for**
 - 13: Compute twice the null-to-alternative Laplace log-likelihood gain.
 - 14: Compare with $\chi_{r_b d_b}^2$.
 - 15: For permutations, reuse the fitted null and refit each permuted alternative.
-

Algorithm 10 Nonbootstrap paired-EC Laplace block test

Require: Paired EC counts, compatibility matrices, nested tensors Z_{0b} , Z_{1b} , and mouse indices

Ensure: LRT p-value and BIC Bayes-factor approximation

- 1: Fit the null by maximizing its Laplace marginal log likelihood.
 - 2: Append q_b zero coefficients to the fitted null parameters.
 - 3: Warm-start and fit the alternative, re-estimating shared variance parameters.
 - 4: Set $D_b \leftarrow 2 \max\{0, \hat{\ell}_{1b}^{\text{Lap}} - \hat{\ell}_{0b}^{\text{Lap}}\}$.
 - 5: Set $p_b \leftarrow \Pr\{\chi_{q_b}^2 \geq D_b\}$.
 - 6: Set $\log \widehat{\text{BF}}_{10,b}^{\text{BIC}} \leftarrow (D_b - q_b \log M_g)/2$.
-

Algorithm 11 Inference comparison for the paired EC-count GLMM

Require: Paired counts $\mathbf{c}_{gip}$, maps A_{gp} , nested fixed-effect designs, mouse indices, likelihood family, and Rényi order $\alpha = 0.5$

Ensure: Null and alternative evidence approximations and coefficient estimates

- 1: **for all** null and alternative designs **do**
 - 2: Fit the conditional posterior mode of $u_g \in \mathbb{R}^{M_g \times d_g}$ by damped Newton updates.
 - 3: Optimize the Laplace marginal likelihood over B_g , $\log \sigma_g$, and optional $\log \kappa_g$.
 - 4: **if** the likelihood is multinomial **then**
 - 5: **repeat**
 - 6: Update every latent EC-to-isoform responsibility from $A_{gpkt} \exp(E_Q \eta_{git})$.
 - 7: Iterate every tilted coordinate a_{gip} to its softmax fixed point.
 - 8: Update $B_g, \mu_g, \log s_g, \log \sigma_g$ against the bounded ELBO; damp the global update when the bound decreases.
 - 9: **until** the ELBO gradient tolerance or iteration limit is reached
 - 10: **end if**
 - 11: Draw fixed antithetic standard-normal samples and reparameterize $u_g = \mu_g + s_g \odot \epsilon$.
 - 12: Optimize the exact-likelihood Monte Carlo ELBO $\mathcal{L}_1$.
 - 13: Warm-start from the ELBO fit and optimize $\mathcal{L}_{0.5}$; report its effective importance-sample size.
 - 14: **end for**
 - 15: Compare each method's null-to-alternative gain on its own objective; do not compare raw objective values across approximation families.
-

Algorithm 12 Block-specific paired EC-count GLMM cell-type test

Require: Block b , paired gene EC counts $\mathbf{c}_{gip}$, maps A_{gp} , condition design Z_b , cell-type design W_b , mouse indices, and R parametric-bootstrap replicates

Ensure: Empirically calibrated block p-value

- 1: Fit the label-independent pooled isoform baseline, construct the path-ILR Jacobian J_b , set $H_b = J'_b(J_b J'_b)^{-1}$, and obtain N_b from $\ker(J_b)$.
 - 2: Form the null fixed-effect tensor from unrestricted condition-by-isoform coefficients $Z_b \otimes I_{d_g}$ and cell-type nuisance columns $W_{bij}(N_b)_{:s}$.
 - 3: Fit the null full-covariance posterior and variance parameters.
 - 4: Append the $(C_b^{\text{cell}} - 1)d_b$ columns $W_{bij}(H_b)_{:s}$, warm-start all null parameters, and fit the observed alternative.
 - 5: Set $T_b \leftarrow 2(\mathcal{L}_{1b} - \mathcal{L}_{0b})$, using the same variational objective for both nested fits.
 - 6: **for** $r = 1, \dots, R$ **do**
 - 7: Draw mouse effects and optional pseudobulk effects from the fitted null; compute primer-specific EC probabilities.
 - 8: Generate multinomial or Dirichlet–multinomial EC counts while preserving every observed primer-specific pseudobulk total.
 - 9: Refit the null, warm-starting from the observed null, then append the block-effect columns and fit the nested alternative.
 - 10: **end for**
 - 11: Pool converged bootstrap statistics by likelihood, tested dimension, and depth stratum, excluding block b 's replicates.
 - 12: Compute the finite-pool rank p-value and assess the same rank calculation on every null replicate before applying BH.
-

Algorithm 13 Matched semi-synthetic power benchmark

Require: Two-condition partitions, effect sizes γ , and replicate count R_{power}

Ensure: Asymptotic and null-calibrated power for each method

- 1: **for all** eligible blocks b **do**
 - 2: Draw and fix a unit ILR direction $\mathbf{v}_b$.
 - 3: **for** $r = 1, \dots, R_{\text{power}}$ **do**
 - 4: Permute condition labels among mice.
 - 5: **for all** nonzero effect sizes γ **do**
 - 6: Set $\mathbf{y}_{bmc}^{(r,\gamma)} = \mathbf{y}_{bmc} + \gamma z_{bmc}^{(r)} \mathbf{v}_b$.
 - 7: Map shifted ILRs to compositions and effective path counts.
 - 8: Fit GEE, GLMM, clustered GLS, and Dirichlet–multinomial tests.
 - 9: **end for**
 - 10: **end for**
 - 11: **end for**
 - 12: Rank each alternative statistic against null statistics for its block and method, using the finite-permutation correction.
 - 13: Report asymptotic and event-matched permutation rejection at 0.05.
-

Algorithm 14 Pairwise and dominant-path sensitivity tests

Require: Path estimates and conditional covariances from Algorithm 3**Ensure:** Calibrated sensitivity tables

- 1: **for all** cell-type pairs **do**
 - 2: Retain M mice containing both types.
 - 3: Use $X_0 \in \mathbb{R}^{2M \times M}$ for mouse effects and $X_1 \in \mathbb{R}^{2M \times (M+1)}$ for mouse plus cell type.
 - 4: Permute the two labels within mouse and correct across all block-by-pair hypotheses.
 - 5: **end for**
 - 6: **for all** blocks b **do**
 - 7: Rank paths by $\sum_i N_{bi}^{\text{gene}} \psi_{bis}$ and select $\mathcal{S}_b$, the two largest, without labels.
 - 8: Construct selector E_b , two-path basis $Q_b^{(2)}$, and $\mathcal{R}_b = (Q_b^{(2)})' E_b' Q_b$.
 - 9: Set $V_{bi}^{(2)} = \mathcal{R}_b V_{bi} \mathcal{R}_b'$ and renormalize the two retained proportions.
 - 10: Run Algorithm 7 and record path-floor sensitivity sweeps.
 - 11: **end for**
-

Algorithm 15 Joint condition tests

Require: Calibrated block-by-cell-type p-values or ILR responses**Ensure:** Partition- or gene-level joint p-values

- 1: For ACAT, combine valid cell-type p-values within a canonical partition; optionally combine partition p-values within a gene.
 - 2: As a negative control, fit one shared condition effect while treating cell-type pseudobulks as independent.
 - 3: For clustered GLS, construct diagonal covariance blocks $V_{bmc} + (\tau_{\text{mouse}}^2 + \tau_{\text{resid}}^2) I_{d_b}$ and same-mouse off-diagonal blocks $\tau_{\text{mouse}}^2 I_{d_b}$.
 - 4: Estimate τ_{mouse}^2 and τ_{resid}^2 by REML under the null.
 - 5: Fit the shared condition coefficient by GLS.
 - 6: **for all** mouse-level condition permutations **do**
 - 7: Apply one permuted condition to all cell types from that mouse and refit the model.
 - 8: **end for**
 - 9: Apply BH to the canonical-partition family; keep negative-control and calibrated results separate.
-

E Supporting Implementation Details

Splice-block construction

GTF exons are represented by zero-based, half-open intervals. Exon boundaries define atomic intervals; intervals exonic in every annotated spliced isoform are merged into constitutive anchors. Consecutive anchors define a block, and an isoform path is its ordered exon-interval chain between them. Terminal pseudo-anchors include alternative first and last regions. A gene without a shared constitutive region receives one whole-gene block.

Blocks with one path are discarded. Simple two-path blocks include skipped exons, mutually exclusive exons, alternative donors and acceptors, and retained introns. The implementation also supports nested and multi-path blocks. Precursor isoforms remain nuisance isoforms in the EC model but are not assigned to spliced paths.

Different genomic blocks can induce exactly the same partition of annotated isoforms, for example when several exons always occur together. Such blocks have identical statistical responses and must not count as repeated evidence. A canonical partition is the unordered collection of isoform sets assigned to paths. Equivalent blocks are collapsed before FDR or joint testing; all member interval identifiers remain in the output.

Pseudobulks and local fitting

Cell isoform proportions from the hierarchical decoder are weighted by cell UMI totals and aggregated by mouse, condition, and cell type. Pseudobulks must contain at least 20 cells and 10^5 total UMIs. A block is fit when its gene has at least 25 gene-assigned UMIs in at least two qualifying pseudobulks. Only ECs whose compatible isoforms map uniquely to that gene enter the local likelihood.

Bounded L-BFGS optimizes the paired multinomial likelihood, with $\|\delta_{bi}\|_\infty \leq 20$ to prevent floating-point underflow, where $\|\cdot\|_\infty$ is the maximum absolute component. Two-path fits are initialized from a log-ratio grid because the paired-primer normalizers can make the objective non-concave.

For the cell-resolved variant, sparse cell-by-EC matrices are retained and one local isoform matrix is decoded at a time. The optimizer dimension is still only the number of paths minus one. Cell-specific Fisher contributions are accumulated analytically, so the method does not fit one free splicing parameter per cell.

Identifiability and reliability

Finite Fisher covariance is not sufficient near a simplex boundary. Parametric bootstrap identified two reliability requirements:

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \hat{\psi}_{bis} \geq 0.15.$$

Thus every tested log-ratio direction has effective information of at least eight and every fitted path has at least 15% usage. All estimates are written, together with convergence, identifiability, and reliability flags; only reliable estimates enter GLS. Here $\lambda_{\max}(V_{bi}) \in \mathbb{R}_+$ is the largest eigenvalue of the $d_b \times d_b$ covariance, and $\hat{\psi}_{bis} \in [0, 1]$ is path s 's fitted proportion.

F Supporting Results and Benchmarks

Condition-test scope

The primary analysis used the 25-gene-UMI cutoff and blocks with at most eight represented paths. It fit 10,100 blocks and 359,713 pseudobulk-by-block path estimates; 359,711 fits converged. After reliability filtering, there were 829 conditional and 506 profile tests.

These 1,335 fits represent 829 distinct block-by-cell-type hypotheses, covering 527 distinct splice blocks in eight cell types. The profile analysis is available for a subset of the conditional hypotheses. The possible conditions were 5xFAD, 5xFIRE, 5xFIRE PBS, 5xFIRE transplant, FIRE, and WT. For the 829 conditional hypotheses, 275 compared two conditions, 155 compared three, 153 compared four, 130 compared five, and 116 compared all six. The included set varies because every condition in a test must have at least three independent mouse pseudobulks for that block and cell type.

Condition-test results

No block was significant at 5% FDR in either covariance analysis. At the unadjusted $p < 0.05$ level:

- the conditional analysis identified 24 block-by-cell-type hypotheses, covering 21 distinct blocks;
- the profile analysis identified 20 block-by-cell-type hypotheses, covering 17 distinct blocks;
- 20 hypotheses were nominally significant in both analyses.

The union is 24 nominal block-by-cell-type hypotheses covering 21 blocks, but the corrected number of significant hypotheses and blocks is zero. Thus there is no FDR-significant evidence for condition-dependent splicing within cell type. The two covariance variants should not be added as if they were separate discoveries.

Formal mouse-clustered models

The shared-condition analysis was repeated with the multinomial-logit GLMM and Liang–Zeger GEE defined above. Clustered Gaussian ILR GLS and the unclustered Dirichlet–multinomial were refit to the same canonical partitions as comparators. Each block used 50 synchronized mouse-level null permutations. For 88 two-condition blocks, 50 semi-synthetic replicates were generated at each ILR effect size; 4,369 of 4,400 replicate designs had full rank. “Perm. nominal” below is the number of observed partitions with an event-matched permutation p-value at most 0.05. Table 18 summarizes the complete analysis.

Table 18: Calibration and power comparison. Null rejection uses asymptotic p-values on permuted labels; power uses event-matched permutation ranks.

Method	Tests	Conv.	Null < .05	Perm. nominal	Power .25	Power .50
Clustered GLS	359	359	0.025	17	0.171	0.322
Dirichlet–multinomial	359	359	0.138	18	0.188	0.364
Liang–Zeger GEE	359	355	0.415	8	0.139	0.268
Laplace GLMM	359	359	0.108	19	0.199	0.390

The canonical GEE sandwich reference is unusable at the available mouse counts: 41.5% of permuted-null p-values were below 0.05, and its apparent 116 asymptotic FDR discoveries are artifacts of this inflation. The GLMM reference was closer but still anti-conservative at 10.8%, while clustered GLS was conservative at 2.5%. Event-matched permutation calibration is therefore required for all comparisons. On that scale, the GLMM had the highest power, improving over clustered GLS by 2.8 and 6.9 percentage points at ILR effects 0.25 and 0.5. GEE was less powerful than clustered GLS after calibration. No method produced a permutation-calibrated 5% FDR discovery in the observed condition analysis.

EC-count GLMM inference benchmark

The EC-count methods were first compared in 96 independent simulations per likelihood and effect size. Each replicate had 12 mice, two observations per mouse, two isoforms, two distinct three-EC primer maps, 150 molecules per primer and observation, and mouse random-effect standard deviation 0.4. Dirichlet–multinomial simulations used concentration 30. The null threshold for each method was its own 95th percentile evidence gain at effect zero; power was evaluated at an

isoform logit effect of 0.5. This calibration is required because Laplace likelihoods, tilted ELBOs, and Rényi bounds are not numerically interchangeable objectives.

Table 19: Ambiguous-EC simulation comparison. RMSE is for the condition logit coefficient. Time is median seconds per nested null/alternative fit at effect 0.5. ESS is the median alternative mouse-level importance ESS out of 128.

Likelihood	Approximation	Null rejection	Power	RMSE	Time	ESS
Multinomial	Laplace	0.052	0.542	0.223	9.87	–
Multinomial	Tilted ELBO	0.052	0.542	0.223	1.87	–
Multinomial	Rényi $\alpha = 0.5$	0.052	0.542	0.223	1.58	127.67
Dirichlet–multinomial	Laplace	0.052	0.417	0.236	21.41	–
Dirichlet–multinomial	Monte Carlo ELBO	0.052	0.417	0.236	3.06	127.51
Dirichlet–multinomial	Rényi $\alpha = 0.5$	0.052	0.417	0.236	3.35	127.66

Within each likelihood family, all three approximations had the same empirically calibrated power to the reported precision, and their coefficient bias differed by less than 2×10^{-4} . The tilted bound’s null cutoff was 4.70, compared with 4.60 for multinomial Laplace and Rényi VI, but this small scale difference did not change rejection. Variational fitting was five- to seven-fold faster than Laplace. A single global importance weight over all mice had ESS 1/128 and collapsed the fitted random-effect variance; the mouse-factorized Rényi objective corrected this failure and produced ESS above 127.5/128 for both likelihoods.

A second simulation used three isoforms and otherwise retained the same 96 null and 96 effect replicates. Full-covariance and diagonal VI were nearly indistinguishable. Tilted evidence gains had Spearman correlation 0.9999 and median absolute difference 0.073 between posterior families; Rényi gains had correlation 0.99996 and median absolute difference 0.044. Every VI variant had calibrated power 0.438 and condition-coefficient RMSE 0.237–0.238. The full-covariance tilted fit took a median 2.43 seconds per nested effect fit versus 2.00 seconds for diagonal tilted VI. Thus, in a controlled multi-isoform setting, omitted posterior covariance was not a material source of error.

The same conclusion held for the direct EC Dirichlet–multinomial. In 96 three-isoform null and effect simulations, diagonal and full ELBO gains had correlation 0.9997 and median absolute difference 0.150; diagonal and full Rényi gains had correlation 0.9999 and median difference 0.092. Both posterior families had calibrated power 0.219. In the 65-gene real-data analysis, diagonal and full ELBO gains had correlations 0.998 for condition and 0.9998 for cell type, with median absolute differences 0.0035 and 0.0050. Fixed-draw Rényi optimization was again less stable than ELBO optimization.

The genome-wide paired-primer comparison retained 65 genes with mean depth at least 100 EC UMIs per pseudobulk; this was 65 of 66 depth-eligible genes, with one 3,905-EC outlier excluded by the 128-EC computational cap. All retained genes had at most six EC-supported isoforms. The authoritative run initialized each Gaussian variational posterior from the corresponding Laplace means and covariance diagonal, then initialized Rényi VI from the ELBO fit. Across the 65 condition models, convergence was 98.5% for both Laplace likelihoods, 95.4% for both ELBO variants, 92.3% for multinomial Rényi VI, and 98.5% for Dirichlet–multinomial Rényi VI. Cell-type convergence ranged from 93.8% to 100%. There were no fit exceptions.

Approximation agreement depended on isoform dimension. Among the 38 two-isoform genes, multinomial Laplace and Rényi evidence gains had Spearman correlation 0.974 for condition and 0.9996 for cell type; Laplace and tilted ELBO correlations were 0.969 and 0.9995. All 16 two-

isoform condition genes above the method-specific nominal threshold under multinomial Laplace were also above it under Rényi VI. For the 20 three-isoform genes, the analogous gain correlations fell to 0.49 and 0.59. The mean-field posterior has only a diagonal covariance across isoform random effects, whereas Laplace retains their within-mouse posterior covariance. The dimension-dependent divergence therefore motivated a full-covariance extension, but by itself did not distinguish covariance error from objective or optimizer differences. One-dimensional simulation thresholds were not applied to multi-isoform or cell-type omnibus tests.

The tilted surrogate must also be separated from objective evaluation. For two-isoform condition models, median nested gains were 5.54 for the tilted training bound, 4.95 for an exact-likelihood Monte Carlo ELBO evaluated at the same posterior, 4.93 for its Rényi evaluation, 4.93 for Laplace, and 4.95 for separately optimized Rényi VI. For cell type, the corresponding medians were 89.06, 74.38, 74.36, 72.12, and 74.36. Exact ELBO and Rényi gains on the same posterior had Spearman correlations 0.998 and 1.000 for condition and cell type. The tilted bound becomes looser under the larger alternative and therefore overstates nested evidence gains; it is retained as a fitting surrogate, not used as a likelihood-ratio statistic.

Full-covariance results did not support posterior covariance as the cause of the earlier multi-isoform discrepancy. On the real condition analysis, fresh-sample ELBO gains from diagonal and full tilted posteriors had Spearman correlation 0.991 across 65 genes and median absolute difference 0.002. Among the 20 three-isoform genes, the correlation was 0.938 and median difference was 0.024. The corresponding cell-type values were 1.000 overall and within the three-isoform subset, with median three-isoform difference 0.023. These results, together with the three-isoform simulation, point instead to differences in the Laplace objective or its outer optimization. Separately optimizing a fixed-draw full-covariance Rényi objective was less stable than fitting the tilted objective and evaluating both exact objectives with fresh draws; the latter is the reported comparison.

The observation-level logistic-normal model was validated with 96 null and 96 effect simulations using three isoforms, true residual standard deviation 0.6, and the same ambiguous paired-primer maps. Tilted full-covariance VI estimated median residual standard deviation 0.559 under the effect and had calibrated power 0.354. Dirichlet–multinomial Laplace had power 0.365, while the misspecified ordinary multinomial had power 0.313. Logistic-normal Laplace had power 0.146 despite estimating residual scale 0.472, indicating that its high-dimensional marginal approximation was not competitive with the tilted fit.

In the real data, tilted logistic-normal fits converged under both nested models for 80.0% of condition genes and 86.2% of cell-type genes. The alternative residual standard deviation had median 0.040 for condition and 0.029 for cell type; the respective interquartile ranges were 0.007–0.239 and 0.006–0.234. Relative to the multinomial full posterior, fresh ELBO condition gains changed by a median absolute 0.335 and retained rank correlation 0.809. Cell-type gains changed by 10.68 and retained correlation 0.959, consistent with extra-multinomial variation having more impact on the larger cell-type omnibus statistic. Because residual-scale estimates often approached zero and convergence was weaker, this model is retained as an overdispersion sensitivity analysis rather than replacing the primary test.

The initial block-specific EC analysis inherited 406 coverage-eligible partitions from the path-count analysis and tested 399 after EC identifiability filtering. This was unnecessarily restrictive because it required reliable path estimates before fitting the EC likelihood. The genome-wide EC screen instead includes one EC count vector for every pseudobulk–primer combination in the likelihood, retains cell types represented in at least three mice, and requires at least two represented cell types. The strict rule additionally requires at least 25 gene UMIs and 30 covered pseudobulks; the permissive rule requires at least 10 gene UMIs. These rules retain 1,926 and 5,650 blocks, respectively. Five parametric-bootstrap replicates per block provide pooled null distributions matched

by tested dimension. Table 20 reports the calibrated tests.

Table 20: Parametric-bootstrap-calibrated cell-type tests using gene EC counts. The depth column gives the range of null rejection rates across independent gene-depth quartiles. MN is multinomial, LN is logistic-normal multinomial, and DM is Dirichlet–multinomial.

Coverage	Likelihood	Tests	Conv.	Nominal	FDR	Null < .05	Depth range
Strict	MN	1,926	1.000	415	174	0.0496	0.0480–0.0505
Strict	LN	1,926	0.991	377	157	0.0498	0.0439–0.0541
Strict	DM	1,926	0.988	145	0	0.0491	0.0454–0.0582
Permissive	MN	5,650	0.999	997	294	0.0497	0.0475–0.0519
Permissive	LN	5,650	0.984	974	193	0.0497	0.0428–0.0541
Permissive	DM	5,650	0.989	421	0	0.0498	0.0456–0.0516

The permissive rule increases absolute discoveries from 174 to 294 for MN and from 157 to 193 for LN, while testing 2.9 times as many blocks. MN and LN share 152 strict and 192 permissive discoveries. DM has no genome-wide FDR discoveries under either rule despite acceptable calibration. On the 1,926 shared hypotheses, applying BH only to that common family gives 316 MN, 265 LN, and 40 DM discoveries under the permissive fits, versus 174, 157, and zero under the strict fits. Thus lower-count pseudobulks add signal, but much of that gain is spent on the larger genome-wide hypothesis family. The corresponding strict-versus-permissive statistic rank correlations are 0.781, 0.764, and 0.760, so changing the cutoff also changes individual fits rather than only the BH denominator.

Expanded EC likelihood and coverage comparison

The annotation-level block equivalence relation is insufficient after EC screening: removing unsupported isoforms can make two distinct annotated blocks induce exactly the same partition of the remaining isoforms. The final hypothesis family therefore retains one representative for each tuple of gene, EC-supported isoforms, canonical block-path partition, covered observations, and tested factor levels. This reduces the permissive cell-type family from 5,650 to 4,825 tests and the condition family from 17,685 to 14,833 block-by-cell-type tests over 5,608 representative blocks. This collapse is performed before calibration and BH correction.

The initial empirical calibration compared each test only with its degrees-of-freedom-specific bootstrap pool. After supported-partition deduplication, its minimum attainable p-values were approximately 4.5×10^{-4} , whereas BH requires resolution near 10^{-5} at the first ranks. Its zero logistic-normal calls were therefore a bootstrap-tail resolution artifact, not disappearance of the cell-type signal.

The former resolved-tail procedure replaced this initial calibration. Its nonredundant deprecated results are reported in Table 9; the earlier annotation-level counts in Table 20 are retained only to document why supported-partition deduplication and resolved-tail calibration were required.

Cell-type effects within mouse

The initial Gaussian analysis tested 369 conditional and 218 profile partitions. It found 22 and five FDR-significant partitions, respectively, with all profile discoveries contained in the conditional set. After canonicalizing annotation intervals that encode the same isoform-to-path partition, these 22 intervals represented 14 distinct partitions.

The compositional analysis used converged, conditionally identifiable estimates and required every modeled path proportion to be at least 5%. This produced 646 nonredundant tests. A

minimum of 30 pseudobulk observations per partition, chosen as a label-effect-independent coverage requirement, retained 406 hypotheses for FDR correction. Among these, 149 had exact permutation $p < 0.05$, and **101 splice-path partitions in 95 genes passed 5% FDR**. Thirteen of the 14 nonredundant Gaussian discoveries were recovered; the remaining partition had exact $p = 0.032$ but FDR 0.098. Significant compositional results had median maximum cell-type logit effect 0.84 and median effective path count 21.7.

The observation-count threshold is not a narrow boundary effect: thresholds from 29 through 32 pseudobulks each gave 101 discoveries, while 33 gave 99. All 646 observed fits converged. The median number of valid permutations was 500; the minimum was 476 after excluding failed permutation refits. Asymptotic Dirichlet–multinomial p-values were anti-conservative, with 15.5% of permutation-null values below 0.05, which is why only exact per-event permutation ranks are used for the primary result.

Two sensitivity analyses did not improve the event-level result. Testing 3,708 individual cell-type pairs gave 97 FDR-significant pair hypotheses but only 32 distinct partitions after correction across all pairs. Restricting each block to its two molecule-weighted dominant paths increased coverage; with a 1% path floor it gave 88 discoveries, and removing the floor gave 91, under a cross-fitted empirical null. These analyses support the full-path exact omnibus test as the primary cell-type analysis.

Allowing the alternative to re-estimate κ also left the result unchanged: it found the same 101 partitions in the same 95 genes, with no event entering or leaving the 5% FDR set. Exact p-values from fixed- and free-concentration fits had Spearman correlation 0.9995, and both had 148 nominal events. The null optimum was multinomial for 499 of 646 tests, and the alternative optimum was multinomial for 580. Among 66 tests with finite concentration in both models, the median alternative/null concentration ratio was 1.54, consistent with cell-type mean effects explaining some apparent overdispersion. Exact permutation calibration absorbs this nuisance-parameter flexibility, so freeing κ provides no net power gain here.

Calibration

Across the 25-UMI analysis, 2.7% of permuted-label Wald p-values were below 0.05 for both conditional and profile covariance, indicating conservative null calibration. A parametric bootstrap of 3,000 reliable local fits gave mean squared standardized error 1.10 and 94.5% coverage for nominal 95% intervals. Independent Gaussian simulations of REML–GLS gave type-I error between 2.7% and 5.0% across the tested biological-variance settings.

Numerical tests additionally verify skipped-exon block construction, identifiable covariance after collapsing indistinguishable isoforms, rejection of information-null path contrasts, the analytic binomial Fisher variance, and recovery of imposed GLS effects.

Coverage-cutoff sensitivity

UMIs	blocks	tests	conditional null $p < .05$	profile null $p < .05$	bootstrap coverage
25	10100	1335	0.027	0.027	0.945
50	5896	755	0.031	0.030	0.936
100	2820	283	0.029	0.025	0.946
200	1034	69	0.028	0.026	0.948
500	195	10	0.030	0.050	0.946

No cutoff produced an FDR-significant block. The 500-UMI calibration estimate is based on only 100 permutation tests.

For reliable two-condition, two-path conditional tests, mean asymptotic power for a 0.5 ILR effect was 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The final two values are selected from only eight and two eligible tests. Among 103 hypotheses shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449. Lower coverage cutoffs therefore increase the number of testable blocks without degrading calibrated precision among shared tests. The 25-UMI cutoff is the default.

Does cell resolution increase condition-test power?

The cell-resolved likelihood was run on the 527 blocks testable in the 25-UMI primary analysis. All 41,242 block-by-pseudobulk fits converged; 21,624 passed the reliability rules, compared with 21,010 pseudobulk conditional estimates on the same blocks. This produced 836 condition tests, compared with 829 for pseudobulk fitting. No cell-resolved test passed 5% FDR. Condition-label permutations were conservative, with 2.49% of null p-values below 0.05.

An exact matched comparison used the same reliable samples for both methods in 807 block-by-cell-type tests. Cell resolution reduced the median local covariance trace by 1.6%, and the median path-response RMS difference was 0.033 ILR units. P-values were strongly concordant (Spearman rank correlation 0.958); 24 pseudobulk and 25 cell-resolved tests were nominally significant.

For the 259 matched two-condition, two-path tests, mean asymptotic power for a 0.5 ILR effect was 0.504 for both methods. The median ratio of cell-resolved to pseudobulk condition-coefficient standard error was 0.995. Retaining cell likelihood terms therefore did not materially increase power in this dataset. The aggregated EC counts and decoder baseline preserve nearly all relevant local information within a cell type, while residual between-mouse variation dominates the small reduction in inferential variance. Pseudobulk fitting remains the default; the cell-resolved method is available for datasets with stronger within-pseudobulk isoform heterogeneity.

Compositional and joint-test assessment

Collapsing equivalent isoform partitions reduced the 829 conditional block-by-cell-type fits to 505 nonredundant tests over 317 partitions. The median effective path count was 32 molecules, or 62% of gene UMIs; only 0.6% of estimates reached the gene-UMI cap.

On the same 505 tests, conditional GLS had 12 nominal results and the Dirichlet–multinomial had 18. Neither had a 5% FDR discovery. The Dirichlet–multinomial was conservatively calibrated: 3.76% of 10,100 permuted-label p-values were below 0.05. In contrast, pure multinomial regression gave 197 nominal and 155 FDR results, but 39.1% of its permutation p-values were below 0.05. UMI sampling alone therefore cannot represent the large between-mouse compositional variation. Output tables mark pure multinomial and naive shared-effect fits as invalid for inference while retaining them as calibration diagnostics.

ACAT across cell types produced 8 nominal GLS and 12 nominal Dirichlet–multinomial partitions among 317 tests, with no FDR discoveries. Combining distinct partitions within genes gave 7 and 11 nominal genes among 276 tests, again with no FDR discoveries.

Naive shared-effect pooling illustrates the pseudoreplication risk. It gave 43 nominal and one FDR Dirichlet–multinomial partition, but synchronized mouse permutations rejected 13.2% of nulls at 0.05. The apparent discovery had asymptotic $p = 6.4 \times 10^{-5}$ but permutation $p = 0.143$. Naive shared GLS was also inflated, with 8.4% null rejection.

The mouse-clustered shared GLS restored conservative calibration: 2.19% of 7,169 synchronized permutation p-values were below 0.05. It produced 11 nominal partitions among 359 tests and no FDR discoveries; its smallest p-value was 0.0126 with FDR 0.86. Gene-level ACAT gave 9 nominal genes among 313 and no FDR discoveries. Thus the calibrated gain is limited to the increase from 12 to 18 nominal within-cell-type tests under the Dirichlet-multinomial. None of the evaluated strategies creates a genome-wide condition discovery in this dataset.